A critical Poisson law in a dyadic block

Starting points of long constant stretches of an extended Rademacher random completely multiplicative function

Brice Pouly*

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Abstract

Let f be an extended Rademacher random multiplicative function, i.e. a completely multiplicative function whose values at the primes are independent Rademacher variables. We count, in the dyadic block $[N, 2N)$, the starting points of maximal constant stretches of length at least L. Complete multiplicativity creates deterministic long-range copies, so no sparse dependency graph is available a priori. In a dyadic block, the dominant dilation is removed from the bulk and the remaining couplings form an arithmetic family. An exact affine formulation and an absolute homogeneous two-block estimate control its total mass. After conditioning on the small primes, outside a family of total event probability $o(1)$, each remaining start has conditional probability exactly 2^{-L} and shared large-prime coordinates form an exact sparse dependency graph. Chen–Stein then gives, uniformly for $|L - \log_2 N| \leq C$,

$$d_{\mathrm{TV}}\big(\mathcal{L}(Z_{N,L}), \mathrm{Pois}(N2^{-L})\big) \ll_C (\log \log N)^{-2}.$$

The arithmetic estimate leaves an $\exp(-c\sqrt{\log N}/\log \log N)$ margin, while the conditioning step sets the displayed rate. The same mechanism yields deterministic masks, the spatial Poisson point process and geometric excess-length marks. Bounded-ratio gluing gives the critical Poisson law for interior starts in $[1, M]$, without a rate, and the finite-prefix longest-run law follows by coupling. The border–interior joint in the moderate tail is not addressed.

*Independent author. Research carried out with substantial assistance from artificial-intelligence systems. The manuscript is accompanied by a Lean 4 formalization of the internal deductions from seven explicit literature interfaces; see section 18 and the declarations at the end of the article. ORCID: 0009-0008-8491-2467.

Résumé. Soit f une fonction aléatoire complètement multiplicative du modèle de Rademacher étendu, dont les valeurs aux nombres premiers sont des variables de Rademacher indépendantes. Nous comptons, dans le bloc dyadique $[N, 2N)$, les points de départ des plages constantes maximales de longueur au moins L. La complète multiplicativité crée des copies déterministes à longue distance, de sorte qu'aucun graphe de dépendance clairsemé n'est disponible a priori. Dans un bloc dyadique, la dilatation dominante est écartée du cœur du bloc et les couplages restants forment une famille arithmétique. Une formulation affine exacte et une estimation homogène absolue à deux blocs contrôlent sa masse totale. Après conditionnement sur les petits premiers, hors d'une famille de contribution probabiliste totale $o(1)$, chaque départ restant a une probabilité conditionnelle exactement égale à 2^{-L} et les grands premiers partagés forment un graphe de dépendance exact et clairsemé. Stein–Chen donne alors, uniformément pour $|L - \log_2 N| \le C$,

$$d_{\mathrm{TV}}\big(\mathcal{L}(Z_{N,L}), \mathrm{Pois}(N2^{-L})\big) \ll_C (\log\log N)^{-2}.$$

L'estimation arithmétique laisse une marge $\exp(-c\sqrt{\log N}/\log\log N)$, tandis que le conditionnement fixe le taux affiché. Le même mécanisme fournit les masques déterministes, le processus ponctuel spatial de Poisson et les marques géométriques d'excès de longueur. Le recollement à rapport borné donne, sans taux, la loi critique pour les départs intérieurs de $[1, M]$, puis la loi du plus long run du préfixe fini par couplage. Le joint bord–intérieur en queue modérée n'est pas traité.

Keywords. Random multiplicative function; constant stretches; consecutive values; Poisson approximation; Poisson point process; Chen–Stein method; Pell equations; Runge's method.
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1 Introduction and main results

A completely multiplicative function forgets nothing: $f(2x) = f(2)f(x)$, so the even positions of the interval $[2x, 2x+2L]$ copy, up to the factor $f(2)$, the signs of $[x, x+L]$ — a deterministic copy at distance x which does not decay, defeats the usual mixing arguments, and seems to close off to random multiplicative functions the dependency-graph version of the Chen–Stein method, which requires a *sparse* graph. Yet at the extreme scale $L \asymp \log_2 N$, where a maximal constant stretch of length L has probability of order N^{-1}, a Poisson approximation is the natural statement.

The starting point of this work is that, in a dyadic block $[N, 2N)$, this obstruction becomes *amenable to counting*. The exact pair $(x, 2x)$ never lives inside it; the couplings between windows that remain — rational ratios a/b, including the ratio 2 itself at the edges of the block, or a shared large prime factor — form an arithmetic family whose total mass is bounded by the central sections of this text. Conditioning on the signs of the small primes then makes the dependency graph exact, the count of shared primes makes it sparse, and the arithmetic mass controls the pair term: the Chen–Stein method applies exactly where the dependency structure seemed to rule it out.

We consider the Rademacher model, defined by signs $f(p)$ independent and uniform on $\{-1, +1\}$ at the primes, extended by complete multiplicativity. For $x \ge 2$ and $L \ge 1$, set

$$J_{x,L} = \mathbf{1}_{\{f(x-1)=-f(x),\ f(x)=f(x+1)=\cdots=f(x+L-1)\}}.$$

Thus $J_{x,L} = 1$ means that a maximal constant stretch of length at least L starts exactly at x. The condition $f(x-1) = -f(x)$ pins the left edge of the maximal stretch; no condition is imposed at $x + L$, so length excesses remain allowed. In the dyadic block

$$I_N = \{N, N+1, \dots, 2N-1\},$$

we study

$$Z_{N,L} = \sum_{x \in I_N} J_{x,L}, \qquad \lambda_N = N2^{-L}, \qquad B = L + 1.$$

Throughout the article, $N \to \infty$ along the integers. For $z \in \mathbb{R}$ we write $(z)_2 = z(z-1)$. We use the convention

$$d_{\mathrm{TV}}(\mu, \nu) = \sup_A |\mu(A) - \nu(A)|.$$

Theorem 1.1 (Critical Poisson law in a dyadic block)**.** *Fix $C > 0$. Uniformly over integers $L = L(N)$ such that*

$$|L - \log_2 N| \le C,$$

we have

$$\boxed{d_{\mathrm{TV}}(\mathcal{L}(Z_{N,L}), \mathrm{Pois}(\lambda_N)) \ll_C \frac{1}{(\log\log N)^2}}. \tag{1.1}$$

The rate is proved in Corollary 13.10. It deserves to be displayed because it locates the difficulty: the dominant term comes from the conditioning on the small primes, while the two-block arithmetic estimate (1.4), which occupies Sections 4 to 11, leaves a margin of $\exp(-c\sqrt{\log N}/\log\log N)$. Remark 13.11 argues that $(\log\log N)^{-3}$ is a floor for this method (a heuristic lower bound: the edge count is only bounded from above). In particular,

$$\mathbb{P}(Z_{N,L} = 0) = e^{-\lambda_N} + o_C(1), \qquad \mathbb{P}(Z_{N,L} \ge 1) = 1 - e^{-\lambda_N} + o_C(1).$$

To state the point-process extensions, write ℓ_x for the length of the maximal stretch starting at x on the event $J_{x,L} = 1$, and set

$$\Xi_{N,L} = \sum_{x \in I_N} J_{x,L}\, \delta_{x/N}, \qquad \widehat{\Xi}_{N,L} = \sum_{x \in I_N} J_{x,L}\, \delta_{(x/N, \ell_x - L)}.$$

The second process is supported on $[1,2] \times \mathbb{N}_0$, where $\mathbb{N}_0$ is discrete; the space of locally finite measures carries the vague topology.

Theorem 1.2 (Masks and point processes)**.** *Under the hypotheses of Theorem 1.1, the following assertions are uniform for $|L - \log_2 N| \le C$.*

(i) For every deterministic set $A_N \subseteq I_N$, if

$$Z_{N,L}(A_N) = \sum_{x \in A_N} J_{x,L}, \qquad \lambda_N(A_N) = |A_N| 2^{-L},$$

then

$$\sup_{A_N \subseteq I_N} d_{\mathrm{TV}}(\mathcal{L}(Z_{N,L}(A_N)), \mathrm{Pois}(\lambda_N(A_N))) = o_C(1).$$

(ii) Along any subsequence with $\lambda_N \to \lambda \in (0, \infty)$,

$$\Xi_{N,L} \Longrightarrow \mathrm{PPP}(\lambda\, dt)$$

vaguely on $[1,2]$.

(iii) Along the same subsequence,

$$\widehat{\Xi}_{N,L} \Longrightarrow \mathrm{PPP}(\lambda\, dt \otimes \nu), \qquad \nu(\{e\}) = 2^{-(e+1)}, \quad e \in \mathbb{N}_0,$$

vaguely on $[1,2] \times \mathbb{N}_0$.

Remark 1.3 (Lattice effect)**.** Since L is an integer, the parameter $\lambda_N = N 2^{-L}$ need not have a limit when $L = \log_2 N + O(1)$. The theorem is therefore a uniform approximation by a Poisson law with oscillating parameter. Along any subsequence with $\lambda_N \to \lambda \in (0, \infty)$, one obtains convergence to $\mathrm{Pois}(\lambda)$. This lattice effect is already classical for longest stretches in an independent Bernoulli sequence [14, 16, 2].

Theorem 1.4 (Moments and the absolute homogeneous sum)**.** *In the same critical window,*

$$\mathbb{E}Z_{N,L} = \lambda_N + O_C\left(N^{-1/2+o_C(1)}\right), \tag{1.2}$$

$$\mathbb{E}[(Z_{N,L})_2] = \lambda_N^2 + o_C(1), \tag{1.3}$$

and, if $\rho(x,y)$ denotes the dimension of the homogeneous space of relations between two strictly separated starts,

$$\boxed{\sum_{\substack{x,y\in I_N\\ |x-y|>L}} (2^{\rho(x,y)} - 1) = o_C(N^2).} \tag{1.4}$$

In particular, $\mathrm{Var}(Z_{N,L}) = \lambda_N + o_C(1)$*.*

These one-block laws glue together. Uniformly for $|L - \log_2 M| \le C$, with $Z_M = \sum_{2\le x<M} J_{x,L}$ and $\Lambda_M = \mathbb{E}Z_M = (1 + o_C(1))M2^{-L}$,

$$d_{\mathrm{TV}}(\mathcal{L}(Z_M), \mathrm{Pois}(\Lambda_M)) \longrightarrow 0 \qquad (M \to \infty)$$

(Theorem 16.2): the critical Poisson law holds globally, and unconditionally, for the interior starts of $[1, M]$. The proof truncates the deep blocks (proposition 15.5: intermediate-prime pivots, Laishram–Shorey large primes, Pell pairs and the Balasubramanian–Shorey maximum), and proposition 16.1 then extends the two-block sum to a bounded-ratio range. The longest-run law in the whole finite prefix $(f(1), \ldots, f(M))$ follows (Corollary 16.4): $\mathbb{P}(R_M < L) = \exp(-M2^{-L}) + o_C(1)$, the exact multiplicative analogue of the Erdős–Rényi law; the one-block rate does not propagate to the global law (Remark 16.3).

1.1 Prior work, terminology, and novelty

Random multiplicative functions go back to Wintner's model [36]; subsequent work includes short intervals [6], moments and multiplicative chaos [19], and central-limit phenomena [31]. We call *extended Rademacher model* the completely multiplicative function determined by independent signs at the primes. This terminology distinguishes it from the classical Rademacher model, which is multiplicative but vanishes on non-squarefree integers.

In an independent Bernoulli sequence, the logarithmic scale of the longest run goes back to Erdős and Rényi [10]. Földes, Gordon–Schilling–Waterman and Arratia–Gordon–Waterman sharpened the critical distribution and the lattice effect [14, 16, 2]. For counts of runs and rare patterns, Godbole obtains Poisson approximations in total variation in the critical regime via Chen–Stein [15]. Novak treats longest stretches for stationary m-dependent sequences [28]. More generally, under dependence, clustering may lead to a compound Poisson process rather than a simple one [20, 12]. In our definition, the left-edge condition eliminates the clusters caused by mere overlaps of a single stretch; the new difficulty is to exclude clusters produced by multiplicative relations between distant windows.

Najnudel [26] proves exact independence of consecutive windows of *fixed* length k, for a large enough starting point, when the values at the primes are uniform on a finite group of roots of unity; the case $q = 2$ covers our model. His statement does not, however, provide for $q = 2$ a control of the threshold $n_0(k, 2)$ usable uniformly when $k \asymp \log N$. In the Steinhaus model, Najnudel recalls after Turk [33] a quantitative threshold compatible with growing length; it concerns ordinary multiplicative independence, not the independence modulo squares required for signs. The dichotomy between the two models is that of two populations of integers: in Steinhaus, only exact multiplicative relations matter, and the integers involved are smooth, hence very rare; in Rademacher, squares are invisible, and the relevant population — integers whose squarefree part is smooth — already contains all perfect squares. It is this richer population — $N^{1/2+o(1)}$ against $N^{o(1)}$ — that makes Rademacher the hard case. One must control simultaneously about N rare events and the relations between all pairs of windows.

Relations modulo squares between elements of an interval have their own Diophantine literature: products of integers in short intervals, perfect powers in a block, and prescribed classes modulo squares [11, 30, 4, 4, 17]. These works anticipate part of the arithmetic framework, without treating the count of starts or the Poisson limit. On the random-multiplicative side, Klurman–Shkredov–Xu [22], Chinis–Shala [8] and Wang–Xu [35] relate central-regime laws to the rarity of relations between polynomial values. De la Bretèche–Wang–Xu [9] study the Rademacher and extended Rademacher models, square products of polynomial values, Evertse–Silverman and Pell–Fermat. This last overlap is methodological: their polynomials are fixed and the statistic is central, whereas here the number of constraints grows like $L \asymp \log N$ and the statistic is extreme.

To our knowledge, among the works consulted up to July 24, 2026, no publication treats exactly the total-variation approximation of the number of starting points of maximal constant stretches in a dyadic block, for this model and in the window $L = \log_2 N + O(1)$. This bounded bibliographic search is not a proof of priority. The claimed contribution concerns this precise combination of the dependent multiplicative model, the growing length, the maximality, and the uniform control of dependencies between windows; it does not claim a first Poisson law for runs in general.

Work	Model and regime	Relation to the present article
I.i.d. runs [15, 2]	independent Bernoulli; run extremes at logarithmic length	direct probabilistic analogue, without arithmetic dependence
Najnudel–Turk [26, 33]	completely multiplicative; independence at fixed length ($q = 2$), growing threshold in Steinhaus	same sign model, no joint extreme statistic
Block products [11, 4]	square relations between consecutive integers, no limit law	precedence of the modulo-squares framework
de la Bretèche–Wang–Xu [9]	extended Rademacher; polynomial values, central-regime laws	methodological overlap (Evertse–Silverman, Pell)

For the Poisson approximation we use the dependency-graph theorem of Arratia–Goldstein–Gordon [1]; see also the treatise [5] and the modern formulation of Chen–Röllin [7, Theorem 4.1]. The coding tools used in the defect and aligned-core arguments rest on the Hamming bound [24]. Our approximation at infinity is a self-contained quantitative specialization of Runge's method [29]; Walsh [34] gives general quantitative versions, but we do not attribute the specialized upper bound below to him. The usual estimates for arithmetic functions and Euler products are used in the standard form of [32]. The Pell count is proved below from the factorization of ideals and the unit group of a real quadratic field; [25] is cited only as classical context.

1.2 Conceptual mechanism and architecture of the proof

The argument is organized around the absolute homogeneous two-block estimate (1.4). This statement is stronger than the second-moment asymptotic and serves as the interface between the two halves of the proof: the arithmetic sections establish it, and the probabilistic sections convert it into the pair estimate required by Chen–Stein.

Dyadic localization. Complete multiplicativity produces exact copies such as $f(2x) = f(2)f(x)$. In a single dyadic block, the pair $(x, 2x)$ does not occur in the bulk. The surviving couplings are not discarded as a generic "dependence error": they are encoded by rational channels and by shared large-prime coordinates, and are then counted according to their arithmetic structure.

Arithmetic half. The events $J_{x,L}$ first receive an exact affine formulation. Homogeneous relations are pairs of even subsets whose product is a square. Private occurrences provide pivots, while the rare configurations without a large pivot are controlled by Runge's method and the Hamming bound. For two separated blocks, the relation space is decomposed into a systematic rational code and a residual quotient described by the large-prime graph. CRT and interpolation close the moderate sectors; Runge excludes the aligned deep cores; and determinants, small-kernel estimates and Pell rigidity close the non-aligned terminal matching. The result is (1.4), with an exponential margin.

Probabilistic half. We condition on the signs of the primes

$$Y = \lfloor B^2 \log B \rfloor.$$

Outside a family of starts whose total event probability is $o(1)$, projection onto the primes $p > Y$ has rank exactly L, so every conditional marginal is exactly 2^{-L}. Two good starts are then dependent only when they share a prime coordinate above Y; this gives an exact dependency graph with $o(N^2)$ edges. The absolute homogeneous sum controls its Chen–Stein pair term. The arithmetic part therefore leaves an exponential margin, whereas the conditioning step sets the final rate $(\log \log N)^{-2}$.

The same interface is stable under deterministic masks and under the affine, mixed-length formulation needed for the spatial and marked point processes. The global interior law requires two additional operations: truncation of the deep dyadic blocks and extension of the two-block estimate to bounded ratios.

1.3 Scope

Theorem 1.1 and Theorem 1.2 close, within a single dyadic block, the scalar count, deterministic masks, the spatial process and the excess-length marked process. The interior gluing over $[1, M]$ is proved (proposition 15.5, proposition 16.1, Theorem 16.2); independence from the border window near 1 is not established, but it is not needed at the critical scale, where a coupling gives the finite-prefix law (Corollary 16.4); the quantitative one-block rate does not propagate to the global law (Remark 16.3). The third factorial moment, which requires finer information on rare tails, is used in none of these proofs.

1.4 Hierarchy of parameters

On first reading, it suffices to keep this in mind: all constants are fixed before N, in an order that excludes any circularity, and the table below serves only as a reference during fine checks. The parameters are chosen once and for all in the following order; no constant chosen at a late stage depends on N.

The formal order of choices is

$$C \longrightarrow (A, \varepsilon_0, \alpha) \longrightarrow (K_\alpha, C_\alpha) \longrightarrow C_{\mathrm{comp}} \longrightarrow C_{\mathrm{term}} \longrightarrow K \longrightarrow N_0.$$

The arrows indicate an order of fixing, not a mandatory dependence: $A = 3$, $\varepsilon_0 = 1/16$ and $\alpha = 3/16$ are absolute. K_α and C_α depend only on α; C_α is chosen in the proof of Theorem 8.1. The constants gathered in C_{comp} depend at most on C and K_α. The terminal constant C_{term} depends at most on $A, C, K_\alpha, C_{\mathrm{comp}}$, and then K is fixed with

$$K > 2C_{\mathrm{term}}/\log 2.$$

Finally N_0 depends on all the previous parameters and makes the asymptotic inequalities simultaneously valid. The absolute constant C_1 of the Runge upper bound (3.3) is distinct from C_{term}.

After these choices, the notations $o_C(1)$ and $N^{o_C(1)}$ allow a dependence on the fixed constants of the table, but never an additional dependence on N. When the bounded size of a component intervenes locally, we write $o_{C,K_0}(1)$ until K_0 has been fixed. This convention makes the order of choices explicit in the final assembly.

1.5 Glossary of structural objects

The abbreviated terms used in the arithmetic sections have the following meaning.

Term	Definition used in the article
start	integer $x \in I_N$ indexing the affine event $J_{x,L}$ and its tree T_x
channel (a, b, h)	for $h = by - ax$ and $(a, b) = 1$, the set of cells (i, j) satisfying $ai - bj = h$, equivalently $a(x + i) = b(y + j)$; these cells are the exact units
host	*ordered* pair $(x, y) \in I_N^2$ whose graph or system has the stated property; all two-block sums keep this convention
residual component	nontrivial unpinned component remaining after removal of the exact components of the rational code; the object is formally defined by $\mathcal{C}_{\mathrm{res}}$ in Section 6
shallow core	sector where the corrected number $c^\#$ of residual components is at most $\alpha B = 3B/16$
deep core	complementary sector $c^\# > \alpha B$, subsequently split into aligned and non-aligned branches
terminal matching	family of isolated residual components of size two obtained after the terminal reduction; its full kernels are pairwise coprime

The symbol $D^\#$ is deliberately absent from the geometric definition of a component: it represents a dimensional contribution of the defective directions modulo the rational code, not a number of vertices of a contracted graph.

Remark 1.5 (Two quantities not to be confused)**.** The threshold separating the cores is αB with $\alpha = \frac{1}{4} - \varepsilon_0 = \frac{3}{16}$. The neighbouring quantity $\frac{1}{2} - 2\varepsilon_0 = 2\alpha = \frac{3}{8}$ is not a threshold on $c^\#$: it is the *exponent of* N obtained at the threshold, since $4^{c^\#} \le 4^{\alpha B} = 2^{2\alpha B} = N^{2\alpha+o(1)} = N^{1/2-2\varepsilon_0+o(1)}$ (proposition 7.5).

Formal verification. The internal deductions corresponding to Sections 2 to 17 — including the canonical endpoints for theorems 1.1, 1.2 and 1.4 — have been formalized in Lean 4 from seven literature statements imported as explicit hypotheses, never as axioms, and subject to two declared restrictions (the instance $\alpha = 3/16$ of Theorem 8.1; for Lemma 13.6, a quantitative bound sufficient for the rate is proved, though not the displayed three-term form); the convergence of the point processes is verified there via a formal surrogate, equivalent under the standard Laplace-functional characterization [21, Theorem 4.11]. Section 18 describes the exact scope, the effective constants obtained, and how to reproduce the audit.

2 Exact affine formulation

For $n \ge 1$, let

$$v(n) = (v_p(n) \bmod 2)_p \in \mathbb{F}_2^{(\mathcal{P})}.$$

Write the prime signs additively: $X = (X_p)_{p\in\mathcal{P}}$ denotes the sequence of independent Bernoulli variables, uniform on $\mathbb{F}_2$, defined by $f(p) = (-1)^{X_p}$. Then

$$f(n) = (-1)^{\langle X, v(n)\rangle}.$$

For a start x, introduce the L rows

$$\begin{aligned} a_{x,0} &= v(x-1) + v(x), & b_{x,0} &= 1, \\ a_{x,j} &= v(x) + v(x+j), & b_{x,j} &= 0, \qquad 1 \le j < L. \end{aligned}$$

Let A_x be the matrix with rows $a_{x,j}$ and $b_x = (1, 0, \dots, 0)$. Then

$$J_{x,L} = \mathbf{1}_{\{A_x X = b_x\}}.$$

For a tuple $\mathbf{x} = (x_1, \dots, x_k)$, stack the matrices and the right-hand sides. Set

$$\mathcal{R}(\mathbf{x}) = \ker(A_{\mathbf{x}}^{\mathsf{T}}), \qquad \rho(\mathbf{x}) = \dim \mathcal{R}(\mathbf{x}),$$

and

$$q_{\mathbf{x}}(c) = \langle c, b_{\mathbf{x}}\rangle.$$

Lemma 2.1 (Affine Fourier identity)**.** *For every tuple* $\mathbf{x} = (x_1, \dots, x_k)$*,*

$$2^{kL}\mathbb{P}(J_{x_1,L} = \cdots = J_{x_k,L} = 1) = \sum_{c\in\mathcal{R}(\mathbf{x})} (-1)^{q_{\mathbf{x}}(c)}.$$

Consequently,

$$\mathbb{P}(J_{x_1,L} = \cdots = J_{x_k,L} = 1) = 2^{-kL}\eta(\mathbf{x})2^{\rho(\mathbf{x})},$$

where $\eta(\mathbf{x}) = 1$ *if* $q_{\mathbf{x}}$ *vanishes on* $\mathcal{R}(\mathbf{x})$*, and* 0 *otherwise. Finally,*

$$\left|\eta(\mathbf{x})2^{\rho(\mathbf{x})} - 1\right| \le 2^{\rho(\mathbf{x})} - 1. \tag{2.1}$$

Proof. For a matrix with $m = kL$ rows,

$$\mathbf{1}_{\{AX=b\}} = 2^{-m} \sum_{c\in\mathbb{F}_2^m} (-1)^{\langle c, AX-b\rangle}.$$

Taking expectations, the terms with $A^{\mathsf{T}}c \ne 0$ vanish. A character sum over a vector space equals the cardinality of that space if the character is trivial, and 0 otherwise. If $\eta = 0$, then q is nontrivial, hence $\rho \ge 1$, and $1 \le 2^{\rho} - 1$. □

2.1 Relations as even subsets

The system of a start is carried by the tree T_x with vertices

$$V_x = \{x-1, x, x+1, \dots, x+L-1\}$$

and edges

$$\{x-1, x\}, \qquad \{x, x+j\} \quad (1 \le j < L).$$

Lemma 2.2 (Multiplicative translation)**.** *For disjoint blocks* $V_{x_1}, \dots, V_{x_k}$*, a relation* $c \in \mathcal{R}(\mathbf{x})$ *is equivalent to a* k*-tuple of even subsets*

$$S_i \subset V_{x_i}$$

such that

$$\prod_{i=1}^{k} \prod_{n \in S_i} n$$

is a square. Under this identification,

$$q_{\mathbf{x}}(c) = \sum_{i=1}^{k} \mathbf{1}_{\{x_i - 1 \in S_i\}} \pmod 2.$$

The same assertion holds for any union of start-trees which remains a tree, in particular for two starts at distance exactly L.

Proof. On a tree, the boundary of a subset of edges is the set of vertices of odd degree; it is even, and every even part is obtained in a unique way. The sum of the corresponding rows is

$$\sum_{i} \sum_{n \in S_i} v(n).$$

It vanishes exactly when the product of the selected occurrences is a square. The unique edge whose right-hand side equals 1 is incident to the leaf $x_i - 1$. □

3 Defective pivots, first moment, and local pairs

For a threshold $H \geq 2$, define

$$\mathcal{K}_H(n) = \prod_{\substack{p > H \\ v_p(n) \text{ odd}}} p.$$

An integer is H-defective if $\mathcal{K}_H(n) = 1$.

3.1 Runge at infinity

The following lemma is a self-contained quantitative specialization of Runge's method. The historical reference is [29]; for general quantitative versions of the method, see [34]. The precise bound $(C_0 dR)^{2d}$ is proved here and is not attributed to these references.

Lemma 3.1 (Runge for a split product)**.** *There is an absolute constant C_0 with the following property. Let $d = 2k \geq 2$, let $\gamma_1, \ldots, \gamma_d$ be distinct integers with $|\gamma_\nu| \leq R$, and*

$$G(T) = \prod_{\nu=1}^{d} (T + \gamma_\nu).$$

If $U \geq 2R$ and $G(U)$ is an integer square, then

$$U \leq (C_0 dR)^{2d}. \tag{3.1}$$

Proof. The γ_ν being distinct and $d \geq 2$, we have $R \geq 1$ and G is not the square of a polynomial. Set

$$F(z) = \prod_{\nu=1}^{d} (1 + \gamma_\nu z)^{1/2} = \sum_{m \geq 0} c_m z^m,$$

where each square root is the analytic branch equal to 1 at 0, and

$$P(T) = \sum_{m=0}^{k} c_m T^{k-m}.$$

For $m \le k$, the identity

$$\binom{1/2}{j} = (-1)^{j-1}\frac{\mathrm{Cat}_{j-1}}{2^{2j-1}} \quad (j \ge 1)$$

implies $2^{2j}\binom{1/2}{j} \in \mathbb{Z}$ and $|\binom{1/2}{j}| \le 1$. More explicitly,

$$c_m = \sum_{j_1+\cdots+j_d=m} \prod_{\nu=1}^{d} \binom{1/2}{j_\nu} \gamma_\nu^{j_\nu}.$$

The number of weak compositions of m into d parts is $\binom{m+d-1}{d-1} \le 2^{m+d}$. Since $m \le k = d/2$ and $|\gamma_\nu| \le R$, we obtain simultaneously

$$2^d c_m \in \mathbb{Z}, \qquad |c_m| \le 2^{m+d} R^m \le (8R)^d, \qquad 0 \le m \le k. \tag{3.2}$$

In particular $2^d P \in \mathbb{Z}[T]$, uniformly as d grows with N.

If $2R \le U \le 4R$, the conclusion is immediate after enlarging C_0. Assume therefore $U > 4R$. On the circle $|z| = (2R)^{-1}$,

$$|F(z)| \le (3/2)^{d/2} =: M_d.$$

Cauchy's formula gives $|c_m| \le M_d(2R)^m$. Since $2R/U < 1/2$,

$$\begin{aligned} |\sqrt{G(U)} - P(U)| &= U^k \left| \sum_{m \ge k+1} c_m U^{-m} \right| \\ &\le U^k M_d \sum_{m \ge k+1} (2R/U)^m \\ &\le \frac{2M_d(2R)^{k+1}}{U} \le \frac{(C_1 dR)^d}{U}. \end{aligned} \tag{3.3}$$

If the difference is nonzero, $\sqrt{G(U)}$ is an integer and, by (3.2), $P(U) \in 2^{-d}\mathbb{Z}$. It is therefore at least 2^{-d} in absolute value. Inequality (3.3) then implies

$$U \le 2^d (C_1 dR)^d \le (C_0 dR)^{2d}.$$

Assume finally $\sqrt{G(U)} = P(U)$. The polynomial

$$Q(T) = 2^{2d} G(T) - (2^d P(T))^2$$

has integer coefficients. It is nonzero, since G is not a square. The expansion at infinity gives $\sqrt{G(T)} - P(T) = O(T^{-1})$, whence

$$\deg Q \le k - 1.$$

The coefficient of T^r in G is a symmetric polynomial containing at most $\binom{d}{r} \le 2^d$ monomials, each of modulus at most R^d after the uniform upper bound $R \ge 1$. By (3.2), every coefficient of $2^d P$ is at most $(16R)^d$ in modulus. A coefficient of $(2^d P)^2$ is the sum of at most $d+1$ products of two such coefficients. Consequently

$$H(2^{2d} G) \le (8R)^{2d}, \qquad H((2^d P)^2) \le (d+1)(16R)^{2d},$$

and therefore, for an absolute constant C_2,

$$H(Q) \le (C_2 dR)^{2d}.$$

This estimate is uniform in d; it hides no step where d would be fixed. An integer zero of a nonzero integer polynomial has modulus at most $1 + H(Q)$, by the elementary Cauchy bound. After enlarging C_0, we again obtain (3.1). □

3.2 Weighted defect mass

Proposition 3.2 (Weighted defect mass)**.** *Fix $0 < c_1 < c_2$. Uniformly for*

$$c_1 \log N \le H \le c_2 \log N,$$

every interval of $H+1$ integers contained in $[N/2, 3N]$ contains at most

$$O_{c_1,c_2}\Big(\frac{\log N}{\log\log N}\Big)$$

H-defective integers. If m_u denotes the number of defects in $[u, u+H]$, then, over the distinct translates $N \le u < 2N$,

$$\sum_{N\le u<2N} (2^{m_u} - 1) \le N^{1/2+o(1)}. \tag{3.4}$$

Proof. Write every defective integer in the form $n = sa^2$, with s squarefree and supported on the primes $p \le H$. To each of the m defects of an interval, associate the column

$$w(n) = ((v_p(s))_{p\le H}, 1) \in \mathbb{F}_2^{r_0}, \qquad r_0 = \pi(H) + 1.$$

The kernel $\mathcal{C}$ of the matrix of these columns is a binary code of length m and dimension at least $m - r_0$. A nonzero word of weight d selects an even number $d = 2q$ of distinct integers whose product is a square. Translating by the smallest selected integer, Lemma 3.1 gives

$$N/2 \le (CqH)^{4q}.$$

There is therefore $c = c(c_1, c_2) > 0$ such that the minimum distance $\delta(\mathcal{C})$ satisfies

$$\delta(\mathcal{C}) \ge c\frac{\log N}{\log H}.$$

If $\mathcal{C} = \{0\}$, the m columns are linearly independent, hence $m \le r_0 \ll \log N/\log\log N$ and the pointwise conclusion holds. Otherwise, set

$$t_0 = \Big\lfloor \frac{1}{2}\Big(c\,\frac{\log N}{\log H} - 1\Big)\Big\rfloor,$$

so that $2t_0 + 1 \le c\log N/\log H \le \delta(\mathcal{C})$ — and, for N large enough uniformly in H (since $\log H = \log\log N + O_{c_1,c_2}(1)$), $t_0 \ge 1$: the Runge bound bounds $\delta(\mathcal{C})$ *from below*, and it is this direct choice of t_0 — not $\lfloor(\delta-1)/2\rfloor$, which the lower bound alone does not bound from above — that sets the scale. If $m < 2t_0$, then $m \ll_{c_1,c_2} \log N/\log H \asymp \log N/\log\log N$ and the pointwise conclusion holds. Otherwise, the Hamming balls of radius t_0 centred at the words of $\mathcal{C}$ are disjoint, whence

$$2^{m-r_0}\sum_{j=0}^{t_0}\binom{m}{j} \le 2^m, \qquad \sum_{j=0}^{t_0}\binom{m}{j} \le 2^{r_0}, \tag{3.5}$$

then, since $m \ge 2t_0$,

$$\binom{m}{t_0} \ge (m/t_0)^{t_0},$$

and (3.5) implies

$$m \le t_0\, 2^{r_0/t_0}.$$

Now $H \asymp \log N$ gives

$$t_0 \asymp \frac{\log N}{\log\log N}, \qquad r_0 \ll \frac{\log N}{\log\log N},$$

hence $r_0/t_0 = O_{c_1,c_2}(1)$ and

$$m \ll_{c_1,c_2} \frac{\log N}{\log\log N}.$$

For the global mass, every H-defective integer can be written sa^2 with s squarefree and H-smooth. Consequently

$$\#\{n \le 3N : \mathcal{K}_H(n) = 1\} \le \sqrt{3N} \prod_{p \le H} (1 + p^{-1/2}) = N^{1/2+o(1)}.$$

Every integer belongs to at most $H + 1 = N^{o(1)}$ translated intervals. The number of intervals with $m_u > 0$ is therefore $N^{1/2+o(1)}$, while the pointwise bound gives

$$2^{m_u} \le \exp\Big(O\Big(\frac{\log N}{\log\log N}\Big)\Big) = N^{o(1)}.$$

The product of these two estimates proves (3.4). □

3.3 Consolidated first moment

Set

$$U_x = \{x, x+1, \dots, x+L-1\},$$

the set of non-root vertices of the start-tree, and

$$m_x^\circ = \#\{n \in U_x : \mathcal{K}_B(n) = 1\}.$$

Since

$$U_x = [x, x+B-2] \subset [x, x+B],$$

proposition 3.2 applies directly to m_x°.

Corollary 3.3 (First moment)**.** *In the window $|L - \log_2 N| \le C$,*

$$\mathbb{E} Z_{N,L} = N 2^{-L} + O_C\Big(N^{-1/2+o_C(1)}\Big).$$

Proof. Every vertex of V_x that is not B-defective has a prime $p > B$ of odd valuation, and this prime is *private*: two distinct multiples of $p > B$ are at distance at least p, while V_x has diameter B. Such a vertex therefore cannot belong to the support of a relation, otherwise the valuation at p of the selected product would be odd. By Lemma 2.2, a relation is moreover an *even* part of V_x. Let m_x be the number of B-defective vertices of the whole $V_x = \{x-1\} \cup U_x$. The even parts of a set with m_x elements form a space of dimension $\max(m_x - 1, 0)$, hence

$$\rho(x) \le \max(m_x - 1, 0) \le m_x^\circ,$$

the last inequality being an equality when $x-1$ is defective or when $m_x^\circ = 0$, and strict otherwise. It is this parity step that absorbs the root $x-1$, absent from m_x°. Lemma 2.1 gives

$$\Big|\mathbb{P}(J_{x,L} = 1) - 2^{-L}\Big| \le 2^{-L}(2^{m_x^\circ} - 1).$$

The sum of the weights is $N^{1/2+o(1)}$ by (3.4), and $2^{-L} \asymp_C N^{-1}$. ☐

3.4 Overlapping and touching pairs

Lemma 3.4 (Local geometries)**.** *For $x \neq y$:*
(i) if $0 < |x-y| < L$, then $J_{x,L}J_{y,L} = 0$;
(ii) the sum, over ordered pairs with $|x-y| = L$, of the weights $2^{\rho(x,y)} - 1$ is $N^{1+o(1)}$.

Proof. If $y = x + d$ with $1 \le d < L$, the values $f(y-1)$ and $f(y)$ belong to the constant stretch issued from x, while the start at y requires them to be opposite.

If $y = x + L$, the two trees share a vertex and their union is a tree with $2L$ edges on an interval of diameter $2L$. With the threshold $H = 2L$, every non-defective occurrence has a private pivot in the union. Proposition 3.2 gives

$$\rho(x, x+L) \ll \frac{\log N}{\log\log N},$$

hence $2^\rho = N^{o(1)}$. There are $O(N)$ ordered touching pairs. ☐

4 Two separated blocks: hosts and interpolation

Definition 4.1 (Counting terminology)**.** A *population* is a subset of ordered pairs $(x, y) \in I_N^2$. A *relational host* is a pair with $\rho(x, y) > 0$. A *channel* is a reduced triple (a, b, h) describing the exact equalities $a(x + i) = b(y + j)$; its solutions (i, j) are called *exact units.* After contraction of the even code formed by these units, a *residual component* is an unpinned component of the graph of the primes $p > B$. The *full kernel* of an occurrence n is $\mathcal{K}_B(n)$. A family of isolated components of size two, covering all offsets except $o(B)$ of them, is called a *near-perfect matching.*

In what follows, $|x - y| > L$. Set

$$\rho = \rho(x, y), \qquad w(x, y) = 2^{\rho(x,y)} - 1.$$

Lemma 4.2 (Number of pairs carrying a relation)**.** *The number of separated ordered pairs $(x, y) \in I_N^2$ with $\rho(x, y) > 0$ is*

$$H_2(N, L) \le N^{3/2+o_C(1)}. \tag{4.1}$$

Proof. For each host, choose a nonzero relation, then the first selected occurrence in the following order: block x before block y, then increasing offsets. Suppose it is $n = x+i$; the other orientation only contributes a factor 2. Set

$$K = \mathcal{K}_B(n).$$

For every prime $p \mid K$, the total parity of the relation forces an occurrence with odd valuation in the other block. Since $p > B$ and each block has diameter $B - 1$, this occurrence is unique in that block. Forgetting the compatibility constraints between the various primes, there are at most

$$B^{\omega(K)}$$

assignments of the prime factors of K to the offsets of the other block.

A fixed assignment forces on the start y a congruence class modulo each $p \mid K$, hence a unique class modulo K by the Chinese remainder theorem. Since $K \le n \le 3N$, the number of representatives in I_N is at most

$$\frac{N}{K} + 1 \le \frac{4N}{K}.$$

This inequality remains valid for $K=1$. The choice of orientation and offset costs at most $2B$. Thus

$$H_2(N,L) \le 8BN \sum_{n\le 3N} \frac{B^{\omega(\mathcal{K}_B(n))}}{\mathcal{K}_B(n)}. \tag{4.2}$$

Write uniquely

$$n = a^2 ur,$$

where u is squarefree and supported on $p \le B$, and where

$$r = \mathcal{K}_B(n)$$

is squarefree and supported on $p > B$. Summing first over a,

$$\begin{aligned} \sum_{n\le 3N} \frac{B^{\omega(r)}}{r} &\le \sqrt{3N} \sum_u \frac{1}{\sqrt{u}} \sum_r \frac{B^{\omega(r)}}{r^{3/2}} \\ &\le \sqrt{3N} \prod_{p\le B} (1+p^{-1/2}) \prod_{p>B} (1+Bp^{-3/2}). \end{aligned}$$

The logarithms of the two products are respectively

$$O\left(\frac{\sqrt{B}}{\log B}\right) \quad \text{and} \quad O\left(B \sum_{p>B} p^{-3/2}\right) = O\left(\frac{\sqrt{B}}{\log B}\right).$$

Since $B \asymp_C \log N$, their product is $N^{o_C(1)}$. Inserting into (4.2), with $B = N^{o(1)}$, gives (4.1). $\square$

Lemma 4.3 (Relational interpolation)**.** *Let $\mathcal{A}$ be a population of separated pairs, all carrying at least one relation. If $u(x,y) \ge 0$, then*

$$\sum_{\mathcal{A}} u(x,y) \le H_2(N,L)^{1/2} \left(\sum_{\mathcal{A}} u(x,y)^2\right)^{1/2}. \tag{4.3}$$

In particular, a quadratic bound $N^{5/2-\delta+o(1)}$ implies a linear mass $N^{2-\delta/2+o(1)}$.

Proof. This is Cauchy–Schwarz combined with (4.1). $\square$

5 Systematic rational subspaces

Set

$$\mathcal{I}_L = \{-1, 0, \dots, L-1\}.$$

Let $a, b \ge 1$ be coprime and

$$h = by - ax.$$

The exact units of the channel (a,b,h) are

$$M_{a,b}(h) = \{(i,j) \in \mathcal{I}_L^2 : ai - bj = h\}.$$

We set

$$m(a,b,h) = |M_{a,b}(h)|, \qquad \sigma(a,b,h) = \max\{m(a,b,h)-1, 0\}.$$

When the channel is the canonical channel of a pair (x,y), the latter quantity coincides with $\sigma(x,y)$. For $(i,j) \in M_{a,b}(h)$,

$$a(x+i) = b(y+j).$$

Lemma 5.1 (Rational code)**.** *If $m = |M_{a,b}(h)|$, the even subsets of the m units form a subspace*

$$\mathcal{S}_{a,b,h} \subset \mathcal{R}(x,y)$$

of dimension $m-1$ when $m \geq 1$. Its affine character is the restriction of the vector

$$d_{i,j} = \mathbf{1}_{\{i=-1\}} + \mathbf{1}_{\{j=-1\}}.$$

If $m \geq 2$, then

$$\max(a,b) \leq L.$$

Proof. Write $x+i = bt$ and $y+j = at$. For an even set E of units,

$$\prod_{(i,j)\in E} (x+i)(y+j) = (ab)^{|E|} \left(\prod_{(i,j)\in E} t \right)^2$$

is a square, and each block contains $|E|$ occurrences. Two distinct solutions differ by (b,a), whence $a, b \leq L$. ☐

Fix $A \geq 3$. A pair is called *aligned* if there exist coprime integers $a, b \leq B^A$ such that

$$|by - ax| < (a+b)B. \tag{5.1}$$

Lemma 5.2 (Asymptotic uniqueness of the channel)**.** *For N large enough in terms of A, C, a pair has at most one reduced fraction satisfying* (5.1)*. Every channel whose rational code is nonzero is covered by this definition.*

Proof. If (a,b) and (a',b') both qualify,

$$|(a'b - ab')x| < 4B^{2A+1}.$$

Since $x \geq N$, the integer determinant vanishes for N large enough. The second assertion comes from Lemma 5.1. ☐

Choose the canonical channel when it exists. If it contains $m \geq 2$ units, set

$$\mathcal{S}_{\mathrm{rat}}(x,y) = \mathcal{S}_{a,b,h}, \qquad \sigma(x,y) = m-1.$$

Otherwise set $\mathcal{S}_{\mathrm{rat}} = 0$ and $\sigma = 0$. Finally,

$$\tau(x,y) = \rho(x,y) - \sigma(x,y).$$

Remark 5.3 (What survives when $m_{\mathrm{ex}} \leq 1$)**.** The distinction $m \geq 2$ versus $m \leq 1$ governs everything that follows and its consequences are scattered; we gather them here once and for all.

Lemma 5.1 bounds the height, $a, b \leq L$, only when the channel carries *at least two* exact units. For $m_{\mathrm{ex}} \leq 1$ one has neither $a < B$, nor $b < B$, nor any identity of valuations between the two occurrences of a cell; on the other hand

$$\mathcal{S}_{\mathrm{rat}} = 0, \quad \sigma = 0, \quad \tau = \rho, \quad D^{\#} = D, \quad c^{\#} = c,$$

and these equalities are definitional. No contraction is performed, so the objects of the large-prime graph keep their usual meaning.

The restriction is not one of caution. Take $N = 2^{100}$, $L = 100$, $a = 103$, $b = 102$, $t = 12427947061061072563693168687$, then $x = bt$ and $y = at$. Then $x, y \in [N, 2N)$, $y - x = t > L$, the canonical channel is $(a, b, 0)$ so the pair is aligned, and the units $103i = 102j$ meet $\mathcal{I}_L$ only at $(0, 0)$: $m_{\text{ex}} = 1$. But $q = 103 > B = 101$, the height bound fails, and since $t \equiv 100 \pmod{103}$ we have

$$v_{103}(x) = 0, \qquad v_{103}(y) = 1,$$

while no $x + k$, $-1 \leq k \leq 99$, is divisible by 103: the vertex y is pinned alone. The valuation identity is therefore *false* on this pair, and the conclusion of Lemma 6.3 would not apply to it.

And it is not applied to it. A pair with $m_{\text{ex}} \leq 1$ is handled, by the first test that succeeds in Lemma 11.1, by: the two-coordinate CRT if $P^{\#} \leq N$ (branch $\sigma = 0$ of proposition 7.3); proposition 7.4 if the canonical channel has height $q \leq \sqrt{\log B}$, which *re-proves* $q < B$ on the spot and pays a possible exact component with a $+1$; proposition 7.5 if the core is shallow, where $\sigma = 0$ makes the estimate trivial; Theorem 8.1 in the deep aligned branch, which directly removes the at most two components meeting the occurrences of the unique exact unit; finally Sections 9 and 10, which invoke no canonical channel. In particular, a pair with $m_{\text{ex}} \leq 1$ never reaches the one-coordinate CRT branch in t, which assumes $m \geq 2$.

Proposition 5.4 (Systematic rational mass)**.** *Uniformly in the critical window,*

$$\sum_{\substack{x,y \in I_N \\ |x-y| > L}} (2^{\sigma(x,y)} - 1) \leq N^{4/3 + o_C(1)}, \tag{5.2}$$

$$\sum_{\substack{x,y \in I_N \\ |x-y| > L}} (4^{\sigma(x,y)} - 1) \leq N^{5/3 + o_C(1)}. \tag{5.3}$$

Proof. Fix $q = \max(a, b)$. There are $O(q)$ reduced pairs (a, b) of height q. For such a pair, the values

$$h = ai - bj, \qquad (i, j) \in \mathcal{I}_L^2,$$

belong to an interval of length $O(qB)$; there are therefore $O(qB)$ nonempty channels. A unit (i_0, j_0) parametrizes the pairs by

$$x = bt - i_0, \qquad y = at - j_0.$$

The intersection with I_N^2 contains $O(N/q + 1)$ values of t. The units are spaced by (b, a), hence

$$m \leq 1 + B/q, \qquad 2^{\sigma} \leq 2^{B/q}, \qquad 4^{\sigma} \leq 4^{B/q}.$$

For $q \geq 3$, the height-q contribution is thus bounded respectively by

$$O\Big(q \cdot qB \cdot (N/q + 1)\, 2^{B/q}\Big) \quad \text{and} \quad O\Big(q \cdot qB \cdot (N/q + 1)\, 4^{B/q}\Big).$$

Since $m \geq 2$ throughout this sum, Lemma 5.1 gives $q \leq L < B$; since $B = N^{o(1)}$, we may therefore use the uniform upper bound

$$N^{1+o(1)} q^2 2^{B/q} \quad \text{or} \quad N^{1+o(1)} q^2 4^{B/q}.$$

The sum over $3 \leq q \leq B$ is dominated, up to a factor polynomial in B, by $q = 3$. Since $2^B \asymp_C N$,

$$N^{1+o(1)} 2^{B/3} = N^{4/3 + o_C(1)}, \qquad N^{1+o(1)} 4^{B/3} = N^{5/3 + o_C(1)}.$$

There remains $q = 2$. The only reduced ratios are $2 : 1$ and $1 : 2$. For instance, for $(a, b) = (2, 1)$,

$$y = 2x + h, \qquad |h| \ll B.$$

Both starts lying in $[N, 2N)$, x is confined to a band of width $O(B)$ near N and y to a band of width $O(B)$ near $2N$; there are $O(B^2)$ pairs. Even weighted, this branch costs at most

$$B^2 2^{B/2} = N^{1/2+o(1)} \quad \text{and} \quad B^2 4^{B/2} = N^{1+o(1)},$$

which is below the stated bounds. Finally $q = 1$ corresponds to $|x - y| \le L$ and does not belong to the separated sector. □

Lemma 5.5 (Weighted channel sum)**.** *We have*

$$\sum_{\substack{(a,b)=1,\ \max(a,b)\ge 2,\ h\in\mathbb{Z} \\ m(a,b,h)\ge 2}} 4^{\sigma(a,b,h)} \le N^{1+o_C(1)}. \tag{5.4}$$

Here $m(a,b,h) = |\{(i,j) \in \mathcal{I}_L^2 : ai - bj = h\}|$ *and* $\sigma(a,b,h) = \max\{m(a,b,h) - 1, 0\}$*, as in Lemma 5.1: the sum ranges over the geometries* (a,b,h) *themselves, independently of any pair of starts. The exclusion of* $\max(a,b) = 1$ *is necessary and lossless: a channel* $(1,1,h)$ *with* $m \ge 2$ *units forces* $|h| \le L$*, hence carries no separated pair, and its weight can reach* 4^{B-1}*.*

Proof. Set $q = \max(a,b)$. There are $O(q)$ reduced pairs (a,b) of height q. For a fixed pair, the values $h = ai - bj$ with $(i,j) \in \mathcal{I}_L^2$ belong to an interval of length $O(qB)$, so there are $O(qB)$ nonempty channels. The units of a channel are spaced by the vector (b,a), whence

$$m(a,b,h) \le 1 + \frac{B}{q}, \qquad 4^{\sigma(a,b,h)} \le 4^{B/q}.$$

For $q \ge 3$, the total height-q contribution is

$$\ll q^2 B\, 4^{B/q}.$$

The summation ranges over $m(a,b,h) \ge 2$, hence over $q \le L$ by Lemma 5.1. The sum over $q \ge 3$ is dominated, up to a factor polynomial in B, by $q = 3$ and equals $N^{2/3+o_C(1)}$. For $q = 2$, only the ratios $2:1$ and $1:2$ exist, and only $O(B)$ values of h; the contribution is

$$O(B4^{B/2}) = N^{1+o_C(1)}.$$

The case $q = 1$ is excluded from the domain. This proves (5.4). □

6 Residual quotient and the large-prime core

Fix a separated pair (x,y). Its set of occurrences is the disjoint union

$$\mathcal{O} = V_x \sqcup V_y.$$

For a prime $p > B$, let $\mathcal{O}_p$ be the set of occurrences where v_p is odd. Since each block has diameter $B - 1 < p$, we have $|\mathcal{O}_p| \le 2$, with at most one occurrence in each block. The parity equations of the primes $p > B$ therefore take the following forms:

- $|\mathcal{O}_p| = 1$: the occurrence is *pinned* and its variable equals 0;
- $|\mathcal{O}_p| = 2$: the two variables are equal;
- $|\mathcal{O}_p| = 0$: no equation.

We deduplicate parallel equations and build the graph $G_{>B}(x,y)$ whose vertices are the occurrences, whose edges are the pairs $\mathcal{O}_p$ of size two, and whose pinning marks are the sets $\mathcal{O}_p$ of size one.

A vertex carrying no prime $p > B$ with odd valuation is called *defective.* Let D be the number of defective vertices and c the number of nontrivial unpinned connected components of the graph. Let

$$W_{>B}(x, y) \subset \mathbb{F}_2^{\mathcal{O}}$$

be the solution space of the equations associated with the primes $p > B$ alone.

Lemma 6.1 (Resolution of the large-prime graph)**.** *There is a canonical isomorphism, after choosing a representative vertex in each component,*

$$W_{>B}(x, y) \simeq \mathbb{F}_2^D \oplus \mathbb{F}_2^c.$$

In particular,

$$\dim W_{>B} = D + c, \qquad D + 2c \leq 2B.$$

Proof. A pinned component is identically zero. In an unpinned component containing an edge, all variables are equal and provide one free variable. An isolated defective vertex is subject to no equation and also provides a free variable. Nontrivial components consume at least two occurrences, whence $D + 2c \leq 2B$. □

Lemma 6.2 (Square class of a component)**.** *Let $\mathcal{C}$ be a nontrivial unpinned component of $G_{>B}(x, y)$. Then $\mathcal{C}$ meets both blocks and there exist a unique squarefree B-smooth integer $d_{\mathcal{C}} > 0$ and an integer $z_{\mathcal{C}} > 0$ such that*

$$\prod_{n \in \mathcal{C}} n = d_{\mathcal{C}} z_{\mathcal{C}}^2. \tag{6.1}$$

Proof. Every edge is bipartite: for $p > B$, there is at most one occurrence of odd p-valuation in each of the two blocks. A nontrivial component contains an edge and therefore meets both blocks.

Fix a prime $p > B$. If it labels the component, its two occurrences of odd valuation are its two endpoints and belong to $\mathcal{C}$; the other p-valuations in the component are even. If it does not label it, no occurrence of $\mathcal{C}$ has odd valuation at p. The case of a single occurrence is excluded, since it would pin the component. Thus

$$\sum_{n \in \mathcal{C}} v_p(n) \equiv 0 \pmod 2 \qquad (p > B).$$

The squarefree part of the product of its vertices is therefore supported on the primes $p \leq B$. It provides the unique $d_{\mathcal{C}}$; the positivity of the product then provides the unique $z_{\mathcal{C}} > 0$ in (6.1). □

Now assume the canonical rational channel has $m \geq 2$ exact units. Let $u_1, \ldots, u_m \in \mathbb{F}_2^{\mathcal{O}}$ be the vectors selecting the two occurrences of a unit. The rational code is

$$\mathcal{S}_{\text{rat}} = \left\{ \sum_{r=1}^{m} \lambda_r u_r : \sum_{r=1}^{m} \lambda_r = 0 \right\},$$

of dimension $m - 1$. Let d be the number of units whose two occurrences are defective and $n = m - d$ the number of the others. We set

$$D^\# = D - d, \qquad c^\# = c - n.$$

When $m \geq 2$, the n non-defective units determine, by Lemma 6.3, n distinct isolated exact components. Let

$$\mathcal{C}_{\text{res}} = \{\text{nontrivial unpinned components}\} \setminus \{\text{these } n \text{ exact components}\}.$$

Then

$$|\mathcal{C}_{\mathrm{res}}| = c - n = c^{\#}.$$

By contrast, $D^{\#} = D - d$ is a *dimensional correction*; it does not literally count a set of defective vertices after contraction. This distinction will be preserved in all counts.

If $m < 2$, no nontrivial rational quotient is taken: we set $D^{\#} = D$, $c^{\#} = c$ and $\mathcal{C}_{\mathrm{res}}$ equal to the set of the c nontrivial unpinned components. In the case $m = 1$, the one or two components containing the two occurrences of the unique exact unit will be flagged explicitly whenever the non-vanishing of $h + bj - ai$ is required; no valuation identity above B is assumed in this branch.

Lemma 6.3 (Isolation of exact units)**.** *Write an exact unit in the form*

$$x + i = bt, \qquad y + j = at, \qquad (a, b) = 1.$$

Assume explicitly $\max(a, b) < B$. *Then, for every prime* $p > B$*,*

$$v_p(x + i) \equiv v_p(t) \equiv v_p(y + j) \pmod 2.$$

There are exactly two possibilities.

(i) If $\mathcal{K}_B(t) = 1$*, the two occurrences are two distinct defective vertices.*

(ii) If $\mathcal{K}_B(t) > 1$*, they form an isolated unpinned component of size two of* $G_{>B}(x, y)$*.*

Two distinct exact units use disjoint occurrences and therefore give distinct defect pairs or components.

Proof. The valuation congruence comes from $p \nmid ab$. In the second case, the two occurrences carry exactly the same labels $p > B$. Such a label cannot appear at any other position of the same block, since two distinct multiples of p are at distance at least $p > B - 1$. The component is therefore exactly the pair under consideration. Two distinct units have distinct offsets in each block, since the solutions of the channel are spaced by (b, a). □

Let

$$\mathcal{R}(x, y) = \ker A_{x,y}^{\mathsf{T}}$$

be the full space of homogeneous relations. It is obtained from $W_{>B}$ by imposing the equations of the primes $p \leq B$ and the two block parities. We therefore have the inclusions

$$\mathcal{S}_{\mathrm{rat}} \subset \mathcal{R}(x, y) \subset W_{>B}(x, y).$$

The induced map provides an injection

$$\mathcal{R}(x, y)/\mathcal{S}_{\mathrm{rat}} \hookrightarrow W_{>B}(x, y)/\mathcal{S}_{\mathrm{rat}}. \tag{6.2}$$

Let λ_x denote the parity of the number of selected occurrences in the first block.

Lemma 6.4 (Quotient core lemma)**.** *We have*

$$\boxed{\tau(x, y) \leq D^{\#}(x, y) + c^{\#}(x, y).} \tag{6.3}$$

Moreover,

$$D^{\#} = O_C\left(\frac{B}{\log B}\right), \tag{6.4}$$

and

$$D^{\#} + 2c^{\#} \leq 2B. \tag{6.5}$$

Proof. Assume first $m \geq 2$. Lemma 6.3 shows that the vectors $u_1, \ldots, u_m$ belong to $W_{>B}$ and have disjoint supports; they are therefore independent. Thus

$$\dim(W_{>B}/\mathcal{S}_{\mathrm{rat}}) = D + c - (m-1).$$

The form λ_x vanishes on $\mathcal{S}_{\mathrm{rat}}$ and descends to the quotient. It is nonzero there: the class of u_1 is nonzero modulo the even code, and $\lambda_x(u_1) = 1$. Consequently

$$\dim \ker \bar{\lambda}_x = D + c - m.$$

Every full relation has even parity in the first block. The injection (6.2) therefore factors through $\ker \bar{\lambda}_x$. The small-prime equations and the second block parity can only lower the dimension. Whence

$$\tau = \dim(\mathcal{R}/\mathcal{S}_{\mathrm{rat}}) \leq D + c - m = (D-d) + (c-n) = D^{\#} + c^{\#}.$$

If $m < 2$, then $\mathcal{S}_{\mathrm{rat}} = 0$ and $\tau = \rho \leq D + c = D^{\#} + c^{\#}$ directly.

Proposition 3.2, applied separately to the two blocks, gives $D = O_C(B/\log B)$; since $D^{\#} \leq D$, we obtain (6.4). Finally $D^{\#} \leq D$ and $c^{\#} \leq c$, so (6.5) follows from Lemma 6.1. □

Lemma 6.5 (Canonical extraction of residual certificates)**.** *The $c^{\#}$ components of $\mathcal{C}_{\mathrm{res}}$ canonically determine a list*

$$(i_s, j_s, p_s), \qquad 1 \leq s \leq c^{\#}, \qquad p_s > B,$$

such that:

(i) the offsets i_s are pairwise distinct, and so are the offsets j_s;
(ii) the primes p_s are pairwise distinct;
(iii) $p_s \mid x + i_s$ and $p_s \mid y + j_s$ for every s;
(iv) if the rational channel contains $m \geq 2$ units, then the extracted cells are not exact: $h + bj_s - ai_s \neq 0$;
(v) if $m = 1$, only the at most two components containing the occurrences of the unique exact unit may escape this last non-vanishing.

For $c^{\#} = 0$, the list is empty.

Proof. In each component of $\mathcal{C}_{\mathrm{res}}$, choose the smallest prime labelling an edge, then the edge with lexicographically minimal offsets. Distinct components are disjoint, hence cannot reuse an offset in either block. A given prime $p > B$ produces at most one edge in the whole graph; the chosen primes are therefore distinct. The divisibilities in (iii) are the definition of an edge labelled by p_s.

When $m \geq 2$, a cell with $h + bj_s - ai_s = 0$ would be an exact unit. Lemma 6.3 would place its edge in one of the exact components removed from $\mathcal{C}_{\mathrm{res}}$, a contradiction. For $m = 1$, we do not use this valuation argument; only the components meeting one of the two distinguished occurrences are set aside, whence the stated exception. □

Set

$$P^{\#} = \prod_{s=1}^{c^{\#}} p_s,$$

with $P^{\#} = 1$ if $c^{\#} = 0$.

For a population $\mathcal{A}$, define

$$\mathcal{Q}_{\mathrm{res}}(\mathcal{A}) = \sum_{\mathcal{A}} 4^{\sigma}(4^{\tau} - 1), \qquad \mathcal{R}_{\mathrm{res}}(\mathcal{A}) = \sum_{\mathcal{A}} 2^{\sigma}(2^{\tau} - 1).$$

Since

$$[2^{\sigma}(2^{\tau} - 1)]^2 \leq 4^{\sigma}(4^{\tau} - 1),$$

Lemma 4.3 gives

$$\mathcal{R}_{\mathrm{res}}(\mathcal{A}) \le N^{3/4+o(1)} \mathcal{Q}_{\mathrm{res}}(\mathcal{A})^{1/2}. \tag{6.6}$$

7 Closing the moderate sectors

The previous sections reduced the systematic relations to rational channels and described the residual quotient through the large-prime graph. This section closes the sectors where a certificate of product at most N or a channel of small height is available; the saving sought there is a power of N, obtained by CRT and interpolation.

7.1 Certificates of product at most N

Lemma 7.1 (CRT certificate summation)**.** *Let $\mathcal{P}$ be a set of primes. A* cell *is a pair (p, c) consisting of a prime $p \in \mathcal{P}$ and a congruence class c modulo p imposed on a common integer parameter t; for each p, a finite set of at most E_p* admissible *cells is fixed. A* certificate *of size r is a set, with no intrinsic order, of r admissible cells whose primes are pairwise distinct; a parameter t is* counted *by a certificate when it satisfies each of its congruences. If the product P of the primes of every certificate under consideration is at most N, then, for every weight $u \ge 1$, the sum over certificates, weighted by u^r, of the number of parameters t counted in a fixed interval of length at most $C_0 N$ is*

$$\ll_{C_0} N \sum_{r \ge 0} \frac{1}{r!} \left(u \sum_{p \in \mathcal{P}} \frac{E_p}{p} \right)^r. \tag{7.1}$$

In two coordinates, a cell is a triple (p, c_x, c_y) imposing a class on x and a class on y modulo p; when there are at most M admissible cells per prime and (x, y) ranges over the product of two fixed intervals of length at most $C_0 N$, the corresponding sum over the pairs $(\text{certificate}, (x, y))$ is bounded by

$$\ll_{C_0} N^2 \sum_{r \ge 0} \frac{1}{r!} \left(uM \sum_{p \in \mathcal{P}} \frac{1}{p^2} \right)^r. \tag{7.2}$$

Proof. First fix an unordered set of r incidences $(\mathfrak{c}_s, p_s)$, with the p_s distinct. The Chinese remainder theorem gives a unique class of t modulo

$$P = p_1 \cdots p_r.$$

Since $P \le N$, an interval of length $C_0 N$ contains at most

$$C_0 N/P + 1 \le (C_0 + 1) N/P$$

representatives. The weight u^r therefore gives a contribution

$$\ll N\, u^r \prod_{s=1}^{r} \frac{1}{p_s}.$$

To sum the unordered certificates, we order them temporarily. The ordered sum is bounded by

$$\left(\sum_p E_p/p \right)^r.$$

Every certificate with distinct primes has exactly $r!$ orderings, whence the factor $1/r!$ in (7.1).

Dropping the distinctness constraint on the primes only increases the sum.

In two coordinates, the CRT determines one class of x and one class of y modulo P. The number of pairs is

$$\ll N^2/P^2.$$

There are at most M possible cells for each prime; the same argument gives (7.2). □

Lemma 7.2 (Residual cells of a channel). *Fix a channel (a,b,h) with $(a,b)=1$, $m(a,b,h)\geq 2$ and $q=\max(a,b)$. After contraction of its exact units, let $E_p(a,b,h)$ be the number of residual cells $(i,j)\in\mathcal{I}_L^2$ that can be linked by a prime $p>B$. Then*

$$E_p(a,b,h)\ll\left(1+\frac{qB}{p}\right)\left(1+\frac{B}{q}\right), \tag{7.3}$$

and

$$\sum_{p>B}\frac{E_p(a,b,h)}{p}\ll\frac{B}{\log B} \tag{7.4}$$

uniformly in (a,b,h).

Proof. For a residual cell, set

$$\Delta_{i,j}=h+bj-ai.$$

If $p>B$ divides $x+i$ and $y+j$ with odd valuation, then

$$p\mid a(x+i)-b(y+j)=-\Delta_{i,j}.$$

The cell is not an exact unit, so $\Delta_{i,j}\neq 0$, and $|\Delta_{i,j}|\ll qB$.

Write $\Delta_{i,j}=\ell p$. The number of possible values of ℓ is

$$O(1+qB/p).$$

For fixed ℓ, the linear equation

$$ai-bj=h-\ell p$$

has, in the box $\mathcal{I}_L^2$, a progression of solutions with step (b,a) and hence at most $O(1+B/q)$ solutions. This gives (7.3).

Only the primes $B<p\ll qB$ intervene. Expanding (7.3),

$$\sum_{p>B}\frac{E_p}{p}\ll\left(1+\frac{B}{q}\right)\sum_{B<p\ll qB}\frac{1}{p}+qB\left(1+\frac{B}{q}\right)\sum_{p>B}\frac{1}{p^2}.$$

By Mertens and partial summation,

$$\sum_{B<p\ll qB}\frac{1}{p}\ll\frac{\log(2q)}{\log B},\qquad\sum_{p>B}p^{-2}\ll\frac{1}{B\log B}.$$

Since the lemma concerns $m\geq 2$, Lemma 5.1 gives $2\leq q\leq L<B$, and both terms are $O(B/\log B)$. □

Proposition 7.3 (Branch $P^{\#}\leq N$). *For separated pairs with $P^{\#}\leq N$,*

$$\mathcal{Q}_{\mathrm{res}}\leq N^{2+o_C(1)}. \tag{7.5}$$

Their residual linear mass is $N^{7/4+o_C(1)}$.

Proof. We separate the branches $\sigma=0$ and $\sigma>0$.

No systematic code. Fix $r = c^{\#}$. Lemma 6.5 associates to each pair a skeleton consisting of r distinct offsets in each block, a matching, and r distinct primes. Parallel edges and multiple primes of a single component create no extra choice: the minimal rule of the lemma retains only one. The case $r = 0$ is the empty certificate. Forgetting canonicity, hence overcounting, the number of skeletons is

$$\binom{B}{r}^2 r! \le \frac{B^{2r}}{r!}.$$

The components carry distinct primes $p_1, \dots, p_r > B$. For a fixed skeleton and fixed primes, the congruences

$$p_s \mid x + i_s, \qquad p_s \mid y + j_s$$

determine two classes modulo $P = p_1 \cdots p_r$. In the branch $P = P^{\#} \le N$, there are

$$O(N^2/P^2)$$

pairs in I_N^2. By (6.3) and (6.4),

$$4^{\tau} \le 4^{D^{\#}+r} = N^{o_C(1)} 4^r.$$

Lemma 7.1, with $M = B^2$, therefore gives

$$\begin{aligned} \mathcal{Q}_{\mathrm{res}}^{(\sigma=0)} &\ll N^{2+o_C(1)} \sum_{r \ge 0} \frac{1}{r!} \left(4B^2 \sum_{p>B} p^{-2} \right)^r \\ &\ll N^{2+o_C(1)} \exp\left(O\left(\frac{B}{\log B} \right) \right) = N^{2+o_C(1)}. \end{aligned}$$

Nontrivial rational channel. Fix a channel (a, b, h) with $m(a, b, h) \ge 2$. Then $q = \max(a, b) \le L$. A reference unit (i_0, j_0) allows us to write all pairs carried by the channel in the form

$$x = bt - i_0, \qquad y = at - j_0,$$

where t ranges over an interval of length $O(N/q + 1) = O(N)$.

For a residual cell (i, j) and a prime $p > B$, Lemma 7.2 shows that $p \mid \Delta_{i,j} \ne 0$. Since $p \nmid ab$, the congruence $p \mid x + i$ fixes a class of t modulo p, and $p \mid \Delta_{i,j}$ then automatically forces $p \mid y + j$. Thus each cell imposes a single class of t modulo its prime.

Fix r cells and distinct primes of product $P \le N$. The CRT leaves one class of t modulo P, hence $O(N/P)$ values. The weight satisfies

$$4^{\tau} \le N^{o_C(1)} 4^r.$$

Lemma 7.1 and Lemma 7.2 give, for the fixed channel,

$$\begin{aligned} \sum_{\substack{(x,y) \text{ carried by } (a,b,h) \\ P^{\#} \le N}} 4^{\tau} &\ll N^{1+o_C(1)} \sum_{r \ge 0} \frac{1}{r!} \left(4 \sum_{p>B} \frac{E_p(a,b,h)}{p} \right)^r \\ &\le N^{1+o_C(1)} \exp\left(O\left(\frac{B}{\log B} \right) \right) = N^{1+o_C(1)}. \end{aligned}$$

The systematic weight of the channel is $4^{\sigma(a,b,h)}$. The pairs being separated, their channels satisfy

$\max(a, b) \geq 2$, and the sum over these channels is therefore, by Lemma 5.5,

$$\mathcal{Q}_{\mathrm{res}}^{(\sigma>0)} \ll N^{1+o_C(1)} \sum_{\substack{m(a,b,h)\geq 2\\ \max(a,b)\geq 2}} 4^{\sigma(a,b,h)} \leq N^{2+o_C(1)}.$$

This proves (7.5). The interpolation (6.6) and the count (4.1) give

$$\mathcal{R}_{\mathrm{res}} \leq N^{3/4+o(1)} N^{1+o(1)} = N^{7/4+o_C(1)}.$$

□

7.2 Small denominators, including $\sigma = 0$

Proposition 7.4 (Channels of small height)**.** *On the population $P^{\#} > N$ having a canonical channel of height*

$$q \leq \sqrt{\log B},$$

we have

$$\mathcal{Q}_{\mathrm{res}} \leq N^{5/3+o_C(1)}.$$

Its residual linear mass is $N^{19/12+o_C(1)} = o(N^2)$.

Proof. If $\sigma > 0$, all exact units have been contracted. If $\sigma = 0$, the channel may contain a unique exact unit. Here $q \leq \sqrt{\log B} < B$ for N large enough, so the coefficients a, b of the channel are $< B$ and Lemma 6.3 applies directly to this unit, even when $m = 1$: either its two occurrences are defective, or they form an isolated exact component of size two. We remove this possible component; there is thus at most one. Every other component has a canonical edge with

$$p \mid \Delta_{i,j} = h + bj - ai \neq 0.$$

Since $|\Delta_{i,j}| \ll qB$, its canonical prime satisfies $B < p \ll qB$. The components have distinct canonical primes, whence

$$c^{\#} \leq 1 + \pi(CqB) - \pi(B) = o(B)$$

uniformly for $q \leq \sqrt{\log B}$. With (6.4) and (6.3),

$$\tau = o(B), \qquad 4^{\tau} = N^{o(1)}.$$

If $\sigma \geq 1$, then $4^{\sigma} \leq \frac{4}{3}(4^{\sigma} - 1)$ and (5.3) gives

$$\sum 4^{\sigma}(4^{\tau} - 1) \leq N^{o(1)} \sum 4^{\sigma} \leq N^{5/3+o_C(1)}.$$

If $\sigma = 0$ and $\tau = 0$, the residual weight vanishes. If $\sigma = 0$ and $\tau > 0$, the pair is a relational host; Lemma 4.2 counts $N^{3/2+o(1)}$ of them and each weight is $N^{o(1)}$. This contribution is $N^{3/2+o(1)}$. The stated quadratic upper bound follows. Finally (6.6) gives

$$\mathcal{R}_{\mathrm{res}} \leq N^{3/4+o(1)} N^{5/6+o(1)} = N^{19/12+o_C(1)}.$$

□

7.3 Shallow cores

Fix

$$\varepsilon_0 = \frac{1}{16}, \qquad \alpha = \frac{1}{4} - \varepsilon_0 = \frac{3}{16}.$$

Proposition 7.5 (Cores of density below α)**.** *On the remaining population, the pairs satisfying*

$$c^{\#} \leq \alpha B$$

have

$$\mathcal{Q}_{\mathrm{res}} \leq N^{5/2-2\varepsilon_0+o_C(1)}.$$

Their linear mass is $N^{2-\varepsilon_0+o_C(1)}$.

Proof. If a channel exists, its height is $> \sqrt{\log B}$, so $\sigma \leq B/q = o(B)$; otherwise $\sigma = 0$. By (6.4) and (6.3),

$$4^{\sigma}(4^{\tau}-1) \leq 4^{\sigma+D^{\#}+c^{\#}} \leq N^{1/2-2\varepsilon_0+o(1)}.$$

We sum over at most N^2 pairs and apply (6.6). □

After these propositions, only the pairs with

$$P^{\#} > N, \qquad c^{\#} > \alpha B,$$

remain; they are aligned with height $> \sqrt{\log B}$, or non-aligned.

8 Excluding aligned deep cores

Theorem 8.1 (No aligned core of linear density)**.** *Fix $\alpha > 0$. For N large enough in terms of α, A, C, no aligned pair satisfies $c^{\#} \geq \alpha B$.*

Proof. Let m_{ex} be the number of exact units of the canonical channel and separate the three situations.

- If $m_{\mathrm{ex}} \geq 2$, Lemma 5.1 gives $a, b \leq L < B$. Lemma 6.3 applies: the non-defective exact components are precisely those removed in the definition of $\mathcal{C}_{\mathrm{res}}$; the defective directions are accounted for only through the dimensional correction $D^{\#}$.
- If $m_{\mathrm{ex}} = 1$, we assume neither $a < B$ nor $b < B$ and use no valuation identity between the two occurrences. We directly remove the at most two components containing them.
- If $m_{\mathrm{ex}} = 0$, there is no crossed exact collision to remove.

In all three cases, since $c^{\#} \geq \alpha B$, for N large enough at least $\alpha B/2$ nontrivial components containing no exact unit remain. Their total size is at most $2B$; consequently at least

$$m \geq \alpha B/4$$

of them have size at most

$$K_{\alpha} = \left\lceil \frac{8}{\alpha} \right\rceil.$$

To each small component $\mathcal{C}_t$, associate the column

$$\Phi(\mathcal{C}_t) = \left((v_p(\prod_{n \in \mathcal{C}_t} n) \bmod 2)_{p \leq B}, |\mathcal{C}_t \cap V_x| \bmod 2, |\mathcal{C}_t \cap V_y| \bmod 2 \right) \in \mathbb{F}_2^r,$$

where $r = \pi(B) + 2$. The kernel of the system of these m columns is a binary code $\mathcal{C}$ of length m and codimension at most r.

Set

$$t = \left\lceil C_{\alpha} \frac{B}{\log B \, \log\log B} \right\rceil.$$

Suppose $\mathcal{C}$ has no nonzero word of weight at most $2t$. The Hamming balls of radius t centred at the codewords are then disjoint, and

$$2^{m-r} \sum_{j=0}^{t} \binom{m}{j} \le 2^m.$$

Hence

$$\sum_{j=0}^{t} \binom{m}{j} \le 2^r.$$

For large N, $t < m/2$, and

$$\begin{aligned} \log \binom{m}{t} &\ge t \log(m/t) - O(t) \\ &\ge (C_\alpha + o_\alpha(1)) \frac{B}{\log B}. \end{aligned}$$

Since $r \le C'B/\log B$, the choice $C_\alpha > 2C' \log 2$ gives a contradiction. There is therefore a nonzero word using

$$O_\alpha\Big(\frac{B}{\log B \, \log\log B}\Big)$$

components.

The union of these components gives offset sets I, J. The vanishing of the last two coordinates of Φ forces $|I|$ and $|J|$ to be even; the vanishing of the other coordinates forces even parity at all primes $p \le B$. The large-prime equations are satisfied inside each component. Consequently

$$\prod_{i \in I} (x+i) \prod_{j \in J} (y+j)$$

is a square. Since each chosen component has at most K_α vertices,

$$d := |I| + |J| \ll_\alpha \frac{B}{\log B \, \log\log B}.$$

Write $h = by - ax$ for the canonical channel and set

$$G(T) = \prod_{i \in I} (T + ai) \prod_{j \in J} (T + h + bj).$$

The roots inside each of the two products are distinct. A collision between the two families, $ai = h + bj$, would be an exact unit and was removed before the construction of the code. The polynomial is therefore split with distinct roots. At $T = ax$,

$$G(ax) = a^{|I|} b^{|J|} \prod_{i \in I} (x+i) \prod_{j \in J} (y+j),$$

which is an integer square since $|I|, |J|$ are even. All roots have modulus $O(B^{A+1})$. Lemma 3.1 gives

$$N \le ax \le (CdB^{A+1})^{2d}.$$

But

$$\log\big((CdB^{A+1})^{2d}\big) \ll_\alpha \frac{B}{\log\log B} = o(B),$$

while $\log N = (\log 2)B + O_C(1)$. Contradiction. $\square$

9 Reducing the non-aligned deep core

There remain the non-aligned deep cores: many full components, no channel of small height, and the previous section has just excluded their aligned variant. This section reduces them to a near-perfect matching of full kernels, whose closure occupies the next section; the tools are Diophantine, Evertse–Silverman then Pell.

Lemma 9.1 (Evertse–Silverman input)**.** *Fix $d \geq 3$. Let $h_1, \ldots, h_d$ be distinct integers and e a nonzero squarefree integer. The number of integer pairs (X, Y) satisfying*

$$\prod_{r=1}^{d}(X + h_r) = eY^2$$

is at most

$$\exp\left(O_d\left(1 + \omega(e) + \omega\left(\prod_{r<s}(h_r - h_s)\right)\right)\right). \tag{9.1}$$

In our applications, $|h_r| \leq B$ and $|e| \leq N^{O_d(1)}$; the number of solutions is therefore $N^{o_C(1)}$.

Proof. Set

$$F(T) = e\prod_{r=1}^{d}(T + h_r).$$

If $F(X) = 0$, then $X = -h_r$ for one of the d indices and $Y = 0$; these solutions add at most d pairs. Assume from now on $F(X) \neq 0$. The equation reads

$$F(X) = (eY)^2.$$

The polynomial F splits over $\mathbb{Q}$ and has at least three distinct roots. Apply Theorem 1(b) of Evertse–Silverman [13] with $K = L = \mathbb{Q}$ and $n = 2$; the field L containing three roots can be taken equal to $\mathbb{Q}$ precisely because F splits. Under these hypotheses, the statement provides

$$\#\{(X, Z) \in \mathbb{Z}^2 : \ Z^2 = F(X), \ Z \neq 0\} \leq 2 \cdot 7^{[L:\mathbb{Q}](4+9|S|)} \cdot 4^{\operatorname{rank} \mathrm{Cl}_L[2]}, \tag{9.2}$$

that is, here, the class group of $\mathbb{Q}$ being trivial, $\leq 2 \cdot 7^{4+9|S|}$. This is exactly the form in which de la Bretèche, Wang and Xu [9] use the same theorem for the equation $Z^2 = dP(X)$, with $S = \{p \mid \operatorname{disc}(dP)\} \cup \{\infty\}$: their application and ours differ only in the choice of F. Note that (9.2) does not depend on d; the d-dependence of (9.1) comes only from the $\leq d$ solutions with $Y = 0$ and from passing from $|S|$ to $\omega(\prod_{r<s}(h_r - h_s))$. Set

$$S = \{\infty\} \cup \{p : p \mid 2e \operatorname{disc}(F)\}.$$

After inverting the elements of S, the leading coefficient and the discriminant are units of $\mathbb{Z}_S$; the integer X and the value eY are S-integers. The theorem bounds the number of possible values of X by an exponential in $O_d(|S|)$. For each X, there are at most two values of Y. Now

$$|S| = O_d\left(1 + \omega(e) + \omega\left(\prod_{r<s}(h_r - h_s)\right)\right),$$

which gives (9.1). Finally, for fixed d, $|e| \leq N^{O_d(1)}$ and the differences have size $O(B)$; the bound $\omega(n) \ll \log(2|n|)/\log\log(3|n|)$ gives $N^{o_C(1)}$. $\square$

Lemma 9.2 (Uniform Pell count in a polynomial range)**.** *Fix $K_0 > 0$. Let A, C be positive squarefree integers such that A/C is not a rational square, and let $e \neq 0$. Assume*

$$A, C, |e| \leq N^{K_0}.$$

The number of integer solutions (z, w) of

$$Az^2 - Cw^2 = e, \qquad |z|, |w| \leq N^{K_0},$$

is

$$\exp\left(O_{K_0}\left(\frac{\log N}{\log\log N}\right)\right). \tag{9.3}$$

Proof. Set $g = (A, C)$, $A = gA_0$, $C = gC_0$. Since A, C are squarefree, A_0, C_0 are squarefree, coprime, and their prime supports are disjoint. If a solution exists, $g \mid e$; write $e = ge_0$. After multiplying by A_0 and substituting $X = A_0 z$, the equation becomes

$$X^2 - Dw^2 = M, \qquad D = A_0C_0, \qquad M = A_0e_0. \tag{9.4}$$

The product D is squarefree. The hypothesis on A/C gives $D > 1$.

In the real quadratic field $K_D = \mathbb{Q}(\sqrt{D})$, the element

$$\alpha = X + w\sqrt{D}$$

belongs to the ring of integers $\mathcal{O}_{K_D}$, since $\sqrt{D}$ is an algebraic integer. Its norm is M. Consequently

$$(\alpha)(\bar{\alpha}) = (M),$$

and the integral ideal (α) divides the ideal (M). For each rational prime $p^a \| M$, the number of choices for the part above p is at most $(a+1)^2$: it equals $(a+1)^2$ if p splits, $a+1$ if it is inert, and $2a+1 \leq (a+1)^2$ if it ramifies, since then $(p) = \mathfrak{p}^2$. The number of ideal divisors of (M) is therefore at most

$$\prod_{p^a \| M} (a+1)^2 = \tau(|M|)^2.$$

Only the principal ideals $\mathfrak{a} = (\alpha)$ of absolute norm $|M|$ that admit a generator of norm exactly M can come from a solution of (9.4). Fix such an ideal and a generator α_0 of norm M. Then

$$|\sigma_1(\alpha_0)\sigma_2(\alpha_0)| = |M|.$$

All relevant generators are among

$$\pm\alpha_0\varepsilon_D^n, \qquad n \in \mathbb{Z},$$

where $\varepsilon_D > 1$ generates the free part of the unit group. If a unit of norm -1 exists, the norm-M condition selects only one parity class of n, which cannot increase the number of solutions.

The bounds on A, C, z, w give, for the two real embeddings σ_1, σ_2,

$$|\sigma_j(\alpha)| \leq H := N^{O_{K_0}(1)}.$$

The inequalities

$$|\sigma_1(\alpha_0)|\varepsilon_D^n \leq H, \qquad |\sigma_2(\alpha_0)|\varepsilon_D^{-n} \leq H$$

define an interval of integers n of length at most

$$\frac{2\log H - \log|M|}{\log \varepsilon_D} + O(1) = O_{K_0}(\log N),$$

since every real quadratic fundamental unit satisfies $\varepsilon_D \geq (1+\sqrt{5})/2$. Thus the number of solutions is at most

$$O_{K_0}(\tau(|M|)^2 \log N).$$

Since $|M| \leq N^{O_{K_0}(1)}$, the standard divisor bound gives (9.3); the sign of M does not affect the argument. More precisely, Nicolas and Robin [27, Theorem 1, p. 485] define

$$F(n) = \frac{\log \tau(n)\ \log\log n}{\log 2\ \log n}$$

and prove that its maximum is attained at $n_0 = 6\,983\,776\,800$. We use only the safe exact consequence

$$\log \tau(n)\ \log\log n \leq 2\log 2\ \log n \qquad (n \geq 64). \tag{9.5}$$

The printed value 1.5379 is rounded and is not interpreted as an exact decimal: $\tau(n_0) = 2304$, while the exact integer comparisons $2304^5 < n_0^2$ and $n_0 2^{32} < 5^{32}$ give $F(n_0) < 2$. Equation (9.5) is the Nicolas–Robin proposition imported by the formal verification; the polynomial-height substitution, the square of the divisor factor and the logarithmic unit count are then derived in Lean.

That formal derivation works in $\mathbb{Z}[\sqrt{D}]$ rather than directly in $\mathcal{O}_{K_D}$. Its second external proposition assigns to each norm-M solution a maximal-order ideal divisor and one of four colours (a padding of at most three unit cosets), with equality of both data forcing equality of the corresponding principal ideals in $\mathbb{Z}[\sqrt{D}]$. This record is associated with the conductor and unit-coset results of Halter–Koch [18, Theorem 1.1.6(1)(b), pp. 4–5; Definition 5.1.6 and Theorem 5.1.7(1),(3), pp. 118–119; Theorems 5.2.3(1),(2) and 5.2.5(1), pp. 125–127]. The current literature certificate does not yet establish the full construction from those source results to this record; the interface therefore remains an explicitly open external hypothesis. □

Lemma 9.3 (Normalization of a component)**.** *Let I, J be two nonempty sets of offsets, with $|I| + |J| \leq K_0$, and let d be a positive squarefree B-smooth integer. If*

$$P_I(x)Q_J(y) = dz^2, \qquad P_I(X) = \prod_{i\in I}(X+i), \quad Q_J(Y) = \prod_{j\in J}(Y+j), \tag{9.6}$$

then, after fixing x, I, J, d, there is a positive squarefree integer $e \leq N^{O_{K_0}(1)}$ such that

$$Q_J(y) = ev^2. \tag{9.7}$$

The integer e is the squarefree part of $dP_I(x)$.

Proof. In the group $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$, the equality (9.6) gives

$$[Q_J(y)] = [dP_I(x)].$$

The positive squarefree representative of the right-hand class is e; the squarefree part of $Q_J(y)$ is therefore e. Since $|I| \leq K_0$, we have $dP_I(x) \leq N^{O_{K_0}(1)}$. □

Lemma 9.4 (One-sided count of a component)**.** *Fix K_0. For x, I, J, d fixed as in Lemma 9.3,* *the number of $y \in I_N$ satisfying (9.6) is*

$$\begin{cases} O(N^{1/2}), & |J| = 1, \\ N^{o_{C,K_0}(1)}, & |J| \geq 2. \end{cases}$$

The same assertion holds after exchanging the two blocks.

Proof. Lemma 9.3 gives $Q_J(y) = ev^2$. If $|J| = 1$, the equation $y + j = ev^2$ leaves at most $1 + \sqrt{3N/e} = O(N^{1/2})$ values.

If $|J| = 2$, write

$$(y + j_1)(y + j_2) = ev^2.$$

With $U = 2y + j_1 + j_2$, $V = 2v$ and $\Delta = j_1 - j_2 \neq 0$,

$$U^2 - eV^2 = \Delta^2.$$

For $e > 1$, Lemma 9.2 applies; for $e = 1$,

$$(U - V)(U + V) = \Delta^2$$

and the divisor bound gives $N^{o(1)}$ solutions. If $|J| \geq 3$, Lemma 9.1 applies to Q_J, which is split with distinct roots. $\square$

Lemma 9.5 (Two singleton components)**.** *Fix two offsets i, j. The number of triples (x, y, d), with $x, y \in I_N$, d positive, squarefree and B-smooth, satisfying*

$$(x + i)(y + j) = dz^2$$

is

$$N \exp\left(O\left(\frac{\sqrt{B}}{\log B}\right)\right). \tag{9.8}$$

Proof. Set $X = x + i$, $Y = y + j$. A prime-by-prime analysis of $XY = dz^2$ gives a unique parametrization

$$X = ecu^2, \qquad Y = (d/e)cv^2,$$

where $e \mid d$, where c is squarefree and $(c, d) = 1$. For fixed e, the conditions $X, Y \in [N/2, 3N]$ force

$$c \leq C_e := \frac{3N}{\max(e, d/e)}.$$

Dropping the "squarefree" and "coprime to d" conditions, which only increases the count, the number of triples (u, v, c) is at most

$$\begin{aligned} \sum_{c \leq C_e} \left(1 + \sqrt{\frac{3N}{ec}}\right)\left(1 + \sqrt{\frac{3Ne}{dc}}\right) &\ll C_e + \sqrt{\frac{N}{e}}\sqrt{C_e} + \sqrt{\frac{Ne}{d}}\sqrt{C_e} + \frac{N}{\sqrt{d}} \sum_{c \leq C_e} \frac{1}{c} \\ &\ll \frac{N \log N}{\sqrt{d}}. \end{aligned}$$

Here we used $\max(e, d/e) \geq \sqrt{d}$. After summing over $e \mid d$, the number of solutions for fixed d is

$$\ll \frac{N\tau(d) \log N}{\sqrt{d}}.$$

Finally

$$\sum_{\substack{d \text{ squarefree} \\ p|d \Rightarrow p \le B}} \frac{\tau(d)}{\sqrt{d}} = \prod_{p \le B} \left(1 + \frac{2}{\sqrt{p}}\right) = \exp\left(O\left(\frac{\sqrt{B}}{\log B}\right)\right).$$

The factor $\log N$ is absorbed into the exponential $N^{o(1)}$. □

Lemma 9.6 (Hosts of a bounded component)**.** *Fix K_0. The number of separated pairs (x, y) whose non-aligned residual graph contains a nontrivial unpinned component meeting both blocks and of size at most K_0 is*

$$N^{1+o_{C,K_0}(1)}. \tag{9.9}$$

If this component has exactly two vertices, one in each block, we have the sharper bound

$$N \exp\left(O\left(\frac{\sqrt{B}}{\log B}\right)\right).$$

Proof. Let I, J be the offset sets of the component. Lemma 6.2 provides, with a unique coefficient d, equation (9.6). There are only $B^{O_{K_0}(1)} = N^{o(1)}$ choices of the two offset sets. We separate three branches.

Two degrees at least two. If $|I|, |J| \ge 2$, fix x, the offsets and d. Lemma 9.4 leaves $N^{o_{C,K_0}(1)}$ values of y. Summing over the N values of x and the $2^{\pi(B)} = N^{o(1)}$ coefficients d gives $N^{1+o_{C,K_0}(1)}$ hosts.

A single degree equal to one. If $|I| = 1 < |J|$, the same argument fixes x and uses the mobile side of degree $|J| \ge 2$. If $|J| = 1 < |I|$, we exchange the two blocks. We again obtain $N^{1+o_{C,K_0}(1)}$ after summing over d.

Two singletons. If $|I| = |J| = 1$, Lemma 9.5 applies directly. It already counts the triples (x, y, d) and, in particular, performs the simultaneous summation

$$\sum_{d\ B\text{-smooth, squarefree}} \frac{\tau(d)}{\sqrt{d}}.$$

No factor $2^{\pi(B)}$ must be reintroduced in this branch. After the union over the B^2 offset pairs, absorbed into the exponential, the sharp bound remains

$$N \exp\left(O\left(\frac{\sqrt{B}}{\log B}\right)\right).$$

The three branches give both assertions. □

Corollary 9.7 (Number of deep-core hosts)**.** *For every $\alpha > 0$, the number of non-aligned pairs with $c^\# \ge \alpha B$ is*

$$N^{1+o_{C,\alpha}(1)}.$$

Proof. In the non-aligned branch, $\sigma = 0$ and $c^\# = c$. The nontrivial components use at most $2B$ occurrences. If $c \ge \alpha B$, one of them has size at most $2/\alpha$. Apply Lemma 9.6 with $K_0 = \lceil 2/\alpha \rceil$. □

9.1 Multiple defects

Lemma 9.8 (Two defects in a block)**.** *The number of starts x whose block contains two distinct B-defective vertices is $N^{o_C(1)}$.*

Proof. Write

$$x + i_1 = d_1 u^2, \qquad x + i_2 = d_2 v^2,$$

with d_1, d_2 squarefree and B-smooth. If $d_1 \neq d_2$, the quotient d_1/d_2 is not a square and Lemma 9.2 applies to

$$d_1 u^2 - d_2 v^2 = i_1 - i_2 \neq 0.$$

If $d_1 = d_2 = d$, then $d \mid i_1 - i_2$ and

$$(u - v)(u + v) = \frac{i_1 - i_2}{d},$$

which has $N^{o(1)}$ solutions by the divisor bound. There are $4^{\pi(B)} = N^{o(1)}$ choices of (d_1, d_2) and $B^2 = N^{o(1)}$ choices of offsets. □

Proposition 9.9 (Eliminating $D^\# \geq 3$)**.** *The linear mass of the non-aligned deep-core pairs with $D^\# \geq 3$ is*

$$N^{3/2+o_C(1)}.$$

Proof. One of the two blocks contains at least two defects. By Lemma 9.8, its start has $N^{o_C(1)}$ choices. Since $c^\# > \alpha B$, some nontrivial component has size bounded by a constant depending only on α. Fix the defective block, its start, the offsets of this component and its coefficient d; the cost is $N^{o_C(1)}$.

The second start is the mobile variable. Lemma 9.4 gives $O(N^{1/2})$ choices if the mobile degree is 1, and $N^{o_C(1)}$ if it is at least 2. The total number of hosts is therefore $N^{1/2+o_C(1)}$.

Finally

$$D^\# + 2c^\# \leq 2B, \qquad D^\# = O_C(B/\log B) = o(B).$$

Thus

$$D^\# + c^\# \leq B + D^\#/2 = B + o(B).$$

Lemma 6.4 gives $2^\tau \leq N^{1+o_C(1)}$, and the product with the number of hosts concludes. □

We may now assume $D^\# \leq 2$.

9.2 Exact rank and near-perfect matching

In the non-aligned population, $\sigma = 0$. Solve the large-prime equations and choose one coordinate per defect and per unpinned component. Let $\widetilde{M}$ be the matrix of the remaining equations: primes $p \leq B$ and the two block parities. Set $\widetilde{k} = \operatorname{rank} \widetilde{M}$.

Lemma 9.10 (Exact rank formula)**.** *In the non-aligned core,*

$$\boxed{\tau = D^\# + c^\# - \widetilde{k}.} \tag{9.10}$$

Proof. Lemma 6.1 identifies the solution space of the large primes with $\mathbb{F}_2^{D^\# + c^\#}$ in the non-aligned branch. The only remaining equations are the rows of $\widetilde{M}$. The nullity therefore has the stated dimension. □

Proposition 9.11 (Terminal reduction)**.** *For $K>0$, let $\mathcal{T}_K$ be the set of non-aligned deep-core pairs satisfying*

$$D^{\#} \le 2, \qquad c^{\#} = B - s, \qquad s + \widetilde{k} \le K\frac{\sqrt{B}}{\log B}, \tag{9.11}$$

and having at least

$$M \ge B - 3s \tag{9.12}$$

isolated components of size two. There is a constant $K = K(A,C) > 0$, fixed before N, such that the mass of the non-aligned deep core outside $\mathcal{T}_K$ is $o_C(N^2)$.

Proof. Corollary 9.7 gives $N^{1+o(1)}$ hosts. If $c^{\#} \le 2B/3$, then

$$2^{\tau} \le 2^{D^{\#}+c^{\#}} = N^{2/3+o(1)},$$

and the mass is $N^{5/3+o(1)}$.

Assume $c^{\#} = B - s > 2B/3$. The components use at most $2B$ occurrences; one of them therefore has size two. The two-vertex case of Lemma 9.6, with the union over B^2 offset choices absorbed into $N^{o(1)}$, gives

$$\#\{\text{hosts}\} \le N \exp\left(C_{\text{term}}\frac{\sqrt{B}}{\log B}\right)$$

for a constant $C_{\text{term}} = C_{\text{term}}(A,C)$. By (9.10),

$$2^{\tau} = 2^{D^{\#}+B-s-\widetilde{k}} \ll_C N\,2^{-(s+\widetilde{k})}.$$

Choose once and for all

$$K > \frac{2C_{\text{term}}}{\log 2}.$$

If $s + \widetilde{k} > K\sqrt{B}/\log B$, the product of the two previous estimates is $o(N^2)$.

In the remaining branch, set $e_t = |\mathcal{C}_t| - 2$ for each nontrivial component. Since the sum of the sizes is at most $2B$,

$$\sum_t e_t \le 2B - 2c^{\#} = 2s.$$

There are therefore at most $2s$ components of size greater than two. Among the $B-s$ components, at least $B-3s$ are isolated of size two. ☐

For an isolated component, write $\{x+i_t, y+j_t\}$ and set

$$R_t = \mathcal{K}_B(x+i_t).$$

Lemma 9.12 (Kernels and determinants of the terminal matching)**.** *For every isolated component,*

$$R_t = \mathcal{K}_B(x+i_t) = \mathcal{K}_B(y+j_t) > 1.$$

The R_t associated with distinct components are pairwise coprime. For $s \ne t$,

$$R_sR_t \mid \Delta_{s,t} := (x+i_s)(y+j_t) - (x+i_t)(y+j_s), \tag{9.13}$$

and

$$0 < |\Delta_{s,t}| \le C_0NB. \tag{9.14}$$

Proof. The equality and coprimality of the kernels follow from the fact that an isolated component contains exactly the two occurrences carrying each of its primes $p > B$; a given prime cannot appear at two distinct offsets of one block.

The divisibility (9.13) is immediate. For the non-vanishing, suppose $\Delta_{s,t} = 0$ and write the common ratio in reduced form

$$\frac{x+i_s}{y+j_s} = \frac{x+i_t}{y+j_t} = \frac{b}{a}, \qquad (a,b) = 1.$$

Then $x + i_s = bu_s$, $y + j_s = au_s$, and similarly with u_t. Subtracting,

$$i_s - i_t = b(u_s - u_t), \qquad j_s - j_t = a(u_s - u_t).$$

The offsets are distinct in each block, so $u_s \neq u_t$ and

$$a \le |j_s - j_t| \le L, \qquad b \le |i_s - i_t| \le L.$$

The two offset pairs are therefore two units of an exact channel of height at most L, contradicting the non-aligned branch. Finally

$$|\Delta_{s,t}| \le |x+i_s|\,|j_t - j_s| + |y+j_s|\,|i_s - i_t| \le C_0 NB.$$

□

10 Closing the terminal matching

Theorem 10.1 (Closure of the terminal population)**.** *The population $\mathcal{T}_K$ satisfies*

$$\#\mathcal{T}_K \le N^{3/4+o_C(1)} \tag{10.1}$$

and

$$\sum_{(x,y)\in\mathcal{T}_K} (2^{\tau(x,y)} - 1) \le N^{7/4+o_C(1)} = o(N^2). \tag{10.2}$$

Proof. 1. Almost all kernels are small.

For two distinct isolated components, Lemma 9.12 gives

$$R_s R_t \le C_0 NB.$$

Set $T = (C_0 NB)^{1/2}$. There is at most one index t with $R_t > T$.

2. Counting the first starts.

Define

$$\mathcal{A}_{B,T}(N) = \{n \le 3N : \mathcal{K}_B(n) \le T\}.$$

Every counted n can be written uniquely

$$n = arz^2,$$

where a is squarefree and B-smooth, and where $r = \mathcal{K}_B(n)$ is squarefree, supported on $p > B$ and below T. Overcounting r by all integers $r \le T$,

$$\begin{aligned}\#\mathcal{A}_{B,T}(N) &\le \sqrt{3N}\prod_{p\le B}(1+p^{-1/2})\sum_{r\le T} r^{-1/2}\\ &\ll \sqrt{NT}\prod_{p\le B}(1+p^{-1/2}) = N^{3/4+o_C(1)}.\end{aligned} \tag{10.3}$$

Let $\mathcal{X}$ be the set of first starts appearing in $\mathcal{T}_K$. Relations (9.11)–(9.12) give, for large N,

$$M - 1 \geq B/2.$$

All kernels but at most one being below T, each block of a terminal pair contains at least $B/2$ occurrences in $\mathcal{A}_{B,T}(N)$. Consider

$$\mathcal{J} = \{(x,n) : x \in \mathcal{X},\ n \in V_x \cap \mathcal{A}_{B,T}(N)\}.$$

Every x gives at least $B/2$ incidences. An integer n belongs to at most B blocks V_x. Hence

$$\frac{B}{2}|\mathcal{X}| \leq |\mathcal{J}| \leq B\#\mathcal{A}_{B,T}(N),$$

and (10.3) gives

$$|\mathcal{X}| \leq N^{3/4+o_C(1)}.$$

3. Partners of a fixed start.

Fix $x \in \mathcal{X}$. The bound $M - 1 \geq B/2$ shows in particular that $M \geq 2$ for N large enough. Every terminal pair (x,y) therefore has at least two isolated components. We take the union over the offset quadruples

$$(i,j,u,v) \in \mathcal{I}_L^4, \qquad i \neq u, \quad j \neq v.$$

There are $O(B^4) = N^{o(1)}$ quadruples.

For a fixed quadruple, set

$$R = \mathcal{K}_B(x+i), \qquad S = \mathcal{K}_B(x+u).$$

Lemma 9.12 gives $R, S > 1$, squarefree, distinct and coprime. The kernel equalities in the two components give the unique decompositions

$$y + j = Raz^2, \qquad y + v = Sbw^2,$$

where a, b are squarefree and B-smooth. The coefficients Ra and Sb are squarefree, since their supports above and below B are disjoint. Their quotient is not a rational square: the nonempty disjoint supports of R and S keep odd exponent after reduction. Subtracting,

$$Raz^2 - Sbw^2 = j - v \neq 0.$$

The bounds $y + j, y + v \leq 3N$ give

$$Ra, Sb \leq 3N, \qquad |z|, |w| \leq \sqrt{3N}.$$

Lemma 9.2 therefore applies with an absolute constant K_0. There are $2^{\pi(B)}$ choices of a and as many of b. The number of partners y for fixed x is at most

$$B^4 4^{\pi(B)} \exp\left(O_C\left(\frac{\log N}{\log\log N}\right)\right) = N^{o_C(1)}.$$

Combined with the count of first starts, this gives (10.1).

4. Weighted summation.

In $\mathcal{T}_K$, $D^{\#} \le 2$ and $c^{\#} \le B$. Lemma 6.4 gives

$$\tau \le B + 2, \qquad 2^{\tau} \ll_C N.$$

Consequently

$$\sum_{(x,y)\in\mathcal{T}_K} (2^{\tau} - 1) \ll_C N \#\mathcal{T}_K \le N^{7/4+o_C(1)}.$$

□

11 Assembling the homogeneous sum

All relation populations are now bounded: systematic masses, moderate sectors, deep cores. It remains to assemble these bounds into (1.4), separating the systematic weight from the residual weight. For separated pairs,

$$2^{\rho} - 1 = (2^{\sigma} - 1) + 2^{\sigma}(2^{\tau} - 1). \tag{11.1}$$

Lemma 11.1 (Exhaustive partition of separated pairs)**.** *After extraction of the systematic term $2^{\sigma} - 1$, every separated pair belongs to exactly one of the following residual sectors, the tests being applied in the stated order:*
(1) $P^{\#} \le N$;
(2) $P^{\#} > N$ and the canonical channel has height $q \le \sqrt{\log B}$;
(3) $P^{\#} > N$, no small channel, and $c^{\#} \le \alpha B$;
(4) $P^{\#} > N$, $c^{\#} > \alpha B$, and the pair is aligned;
(5) the pair is non-aligned, $c^{\#} > \alpha B$, and $D^{\#} \ge 3$;
(6) the pair is non-aligned, $c^{\#} > \alpha B$, $D^{\#} \le 2$, and it does not belong to $\mathcal{T}_K$;
(7) the pair belongs to $\mathcal{T}_K$.
These sectors are disjoint by construction and cover all separated pairs.

Proof. We test the listed properties in turn. In the complement of the first sector, the product of the canonical primes is $> N$. We then test the existence and height of the canonical channel; in its absence, we formally set $q = +\infty$. The complement splits according to the depth $c^{\#}$, then according to alignment. In the deep non-aligned branch, we separate $D^{\#} \ge 3$ and $D^{\#} \le 2$. Proposition 9.11 finally partitions this last branch between $\mathcal{T}_K$ and its complement. Each test is performed only after the previous tests have failed, which ensures disjointness. □

	Condition	$\#\mathcal{A}$	$\sup 2^{\sigma}$	$\sup 2^{\tau}$	$\mathcal{Q}_{\mathrm{res}}(\mathcal{A})$	$\mathcal{R}_{\mathrm{res}}(\mathcal{A})$
(1)	$P^{\#} \le N$	$\le H_2 \le N^{3/2+o(1)}$	$\le N^{1/2+o(1)}$	$\le N^{1+o(1)}$	$\le N^{2+o(1)}$ by CRT	$\le N^{7/4+o(1)}$
(2)	$q \le \sqrt{\log B}$	$\le H_2 \le N^{3/2+o(1)}$	$\le N^{1/2+o(1)}$	$= N^{o(1)}$	$\le N^{5/3+o(1)}$	$\le N^{19/12+o(1)}$
(3)	$c^{\#} \le 3B/16$	$\le N^2$	$= N^{o(1)}$	$\le N^{3/16+o(1)}$	$\le N^{19/8+o(1)}$	$\le N^{31/16+o(1)}$
(4)	deep aligned	0 for large N	—	—	0	0
(5)	non-aligned, $D^{\#} \ge 3$	$\le N^{1/2+o(1)}$	1	$\le N^{1+o(1)}$	$\le N^{5/2+o(1)}$ (raw, unused)	$\le N^{3/2+o(1)}$ directly
(6)	non-terminal	$\le N^{1+o(1)}$	1	$\le N^{1+o(1)}$	$\le N^{3+o(1)}$ (raw, unused)	$o(N^2)$ directly
(7)	$\mathcal{T}_K$	$\le N^{3/4+o(1)}$	1	$\le N^{1+o(1)}$	$\le N^{11/4+o(1)}$ (raw, unused)	$\le N^{7/4+o(1)}$ directly

In rows (1)–(3), the quadratic column is the bound actually interpolated through (6.6); the suprema, deliberately displayed, do not replace the structured summations by CRT or by channels. In rows (5)–(7), the quadratic bounds shown are the raw bounds deduced from the two previous columns and are not used: proposition 9.9, proposition 9.11 and Theorem 10.1 sum $\mathcal{R}_{\mathrm{res}}$ directly. The seven rows are respectively established by proposition 7.3, proposition 7.4, proposition 7.5, Theorem 8.1, proposition 9.9, proposition 9.11 and Theorem 10.1.

Proposition 11.2 (Absolute homogeneous two-block sum). *Uniformly for* $|L - \log_2 N| \le C$*,*

$$R_2(N,L) := \sum_{\substack{x,y \in I_N \\ |x-y|>L}} (2^{\rho(x,y)} - 1) = o_C(N^2).$$

Proof. The systematic term of (11.1) is $N^{4/3+o_C(1)}$ by proposition 5.4. For the residual term, Lemma 11.1 and the preceding table give a finite family of bounds, each $o_C(N^2)$. The sum of the sectors and of the systematic term is therefore $o_C(N^2)$. □

Corollary 11.3 (Quantitative form of the homogeneous sum). *There is* $c_R = c_R(A,C) > 0$ *such that, uniformly for* $|L - \log_2 N| \le C$*,*

$$R_2(N,L) \ll_C N^2 \exp\left(-c_R \frac{\sqrt{\log N}}{\log\log N}\right).$$

Proof. Let us make the assembly explicit. *(i) Polynomial sectors.* The systematic term is $N^{4/3+o_C(1)}$ and sectors (1), (2), (3), (5), (7) save a power of N, the weakest being $N^{-1/16+o_C(1)}$ at sector (3): for $N \ge N_1(C)$, each of these six terms is $\ll_C N^2 \exp(-\sqrt{\log N})$, which dominates any scale $\exp(-c\sqrt{\log N}/\log\log N)$. *(ii) Terminal constant.* The constant $C_{\mathrm{term}} = C_{\mathrm{term}}(A,C)$ is that of the two-vertex case of Lemma 9.6 (the exponential of the host count), and K was fixed at rank 6 of the hierarchy with $K > 2C_{\mathrm{term}}/\log 2$. *(iii) Nonterminal sector.* By proposition 9.11, sector (6) is $\ll_C N^2 \exp((C_{\mathrm{term}} - K\log 2)\sqrt{B}/\log B)$ with $C_{\mathrm{term}} - K\log 2 < -C_{\mathrm{term}} < 0$. *(iv) Choice of* c_R*.* Since $B = L+1$ and $|L - \log_2 N| \le C$, we have $\sqrt{B}/\log B \ge c_4(C)\sqrt{\log N}/\log\log N$ for $N \ge N_2(C)$; any $c_R < \min(1,\ C_{\mathrm{term}}\, c_4(C))$ works. *(v) Threshold.* $N_0(C) = \max(N_1, N_2)$ makes the five estimates simultaneously valid; the seven terms sum to $\ll_C N^2 \exp(-c_R\sqrt{\log N}/\log\log N)$. □

12 Deducing the moments

Proof of Theorem 1.4. The first moment is Corollary 3.3, and (1.4) is proposition 11.2.

For the second factorial moment,

$$\mathbb{E}[(Z_{N,L})_2] = \sum_{\substack{x,y \in I_N \\ x \ne y}} \mathbb{P}(J_{x,L} = J_{y,L} = 1).$$

Subtract $(N)_2 2^{-2L}$. If $0 < |x-y| < L$, the joint probability vanishes, and the $O(NL)$ pairs give $O(NL2^{-2L}) = o(1)$. For $|x-y| = L$, Lemma 3.4 and (2.1) give $2^{-2L}N^{1+o(1)} = o(1)$. For $|x-y| > L$, proposition 11.2 and (2.1) give $2^{-2L}o(N^2) = o(1)$. Thus

$$\mathbb{E}[(Z_{N,L})_2] = (N)_2 2^{-2L} + o(1) = \lambda_N^2 + o(1).$$

Finally,

$$\mathrm{Var}(Z) = \mathbb{E}[(Z)_2] + \mathbb{E}Z - (\mathbb{E}Z)^2 = \lambda_N + o(1).$$

□

13 Conditioning on the small primes and the Chen–Stein method

We now deduce Theorem 1.1 from the defect control and proposition 11.2. We keep the notation

$$R_2(N,L) = \sum_{\substack{x,y\in I_N \\ |x-y|>L}} (2^{\rho(x,y)} - 1)$$

and set

$$Y = \lfloor B^2 \log B \rfloor.$$

For $T \geq B$, define

$$\mathcal{K}_T(n) = \prod_{\substack{p>T \\ v_p(n) \text{ odd}}} p, \qquad m_T(x) = \#\{n \in U_x : \mathcal{K}_T(n) = 1\},$$

and

$$\mathcal{D}_T = \{x \in I_N : m_T(x) \geq 1\}.$$

Finally,

$$\mathcal{M}_B = \sum_{x\in I_N} (2^{m_B(x)} - 1).$$

Since $U_x \subset [x, x+B]$, proposition 3.2 gives

$$\mathcal{M}_B \leq N^{1/2+o_C(1)}. \tag{13.1}$$

13.1 Private pivots in a tree

Lemma 13.1 (Rank via private pivots)**.** *Let $\mathcal{T}$ be a tree whose vertices are integers contained in an interval of diameter strictly less than Y. Fix a root r. Assume that, for each vertex $n \neq r$, there is a prime $p_n > Y$ with $v_{p_n}(n)$ odd. Then the vectors*

$$v(a) + v(b), \qquad \{a,b\} \in E(\mathcal{T}),$$

projected onto the prime coordinates $p > Y$, are linearly independent.

Proof. The prime p_n divides no other vertex, since two distinct multiples of $p_n > Y$ are at distance at least p_n, while the diameter is less than Y. In a combination of edges, the coordinate p_n is therefore the degree modulo 2 of the vertex n in the selected subgraph. If the combination is zero, all non-root vertices have even degree; the root then also has even degree. A nonempty forest has a leaf of degree 1, a contradiction. □

Corollary 13.2 (Exact conditional marginal)**.** *Let*

$$\mathcal{F}_Y = \sigma(X_p : p \leq Y).$$

If $x \notin \mathcal{D}_Y$, then, for every realization of $\mathcal{F}_Y$,

$$\boxed{\mathbb{P}(J_{x,L} = 1 \mid \mathcal{F}_Y) = 2^{-L}.} \tag{13.2}$$

Proof. Take $x-1$ as the root of the start-tree. The other vertices are exactly U_x and each has a prime $> Y$. Lemma 13.1 gives a projected rank equal to L. Whatever right-hand side the small primes create, the system in the coordinates $p > Y$ is surjective onto $\mathbb{F}_2^L$. □

13.2 Mass of the removed starts

Lemma 13.3 (Number of Y-defective starts)**.** *We have*

$$|\mathcal{D}_Y| \le N^{1/2+o_C(1)}. \tag{13.3}$$

Proof. If $\mathcal{K}_Y(n) = 1$, then $n = a^2 s$, where s is squarefree with all prime factors at most Y. Thus

$$\#\{n \le 3N : \mathcal{K}_Y(n) = 1\} \le \sqrt{3N} \prod_{p \le Y} (1 + p^{-1/2}).$$

By partial summation and the Chebyshev bound for $\pi(t)$,

$$\sum_{p \le Y} p^{-1/2} \ll \frac{\sqrt{Y}}{\log Y} \ll \frac{B}{\sqrt{\log B}} = o(\log N).$$

The number of integers involved is therefore $N^{1/2+o(1)}$. Every integer belongs to $O(B)$ sets U_x, a factor absorbed into $N^{o(1)}$. □

Lemma 13.4 (Probabilistic mass of the bad starts)**.** *We have*

$$\sum_{x \in \mathcal{D}_Y} \mathbb{P}(J_{x,L} = 1) = o_C(1). \tag{13.4}$$

Proof. Since $Y > B$, $\mathcal{D}_B \subseteq \mathcal{D}_Y$. If $m_B(x) = m$, the $L - m$ non-defective vertices give $L - m$ private pivots, hence

$$\mathbb{P}(J_{x,L} = 1) \le 2^{-L+m}.$$

For $m \ge 1$, $2^m \le 2(2^m - 1)$; with (13.1),

$$\sum_{x \in \mathcal{D}_B} \mathbb{P}(J_{x,L} = 1) \le 2^{1-L} \mathcal{M}_B = o(1).$$

If $x \in \mathcal{D}_Y \setminus \mathcal{D}_B$, the system already has full rank thanks to the primes $p > B$, so its unconditional probability is 2^{-L}. Lemma 13.3 gives

$$\sum_{x \in \mathcal{D}_Y \setminus \mathcal{D}_B} \mathbb{P}(J_{x,L} = 1) \le 2^{-L} |\mathcal{D}_Y| = o(1).$$

□

Set

$$Z_{N,L}^{\text{good}} = \sum_{x \in I_N \setminus \mathcal{D}_Y} J_{x,L}.$$

The natural coupling and Lemma 13.4 give

$$d_{\mathrm{TV}}\Big(\mathcal{L}(Z_{N,L}), \mathcal{L}(Z_{N,L}^{\text{good}})\Big) = o(1). \tag{13.5}$$

13.3 Conditional dependency graph

For a good start, set

$$\Pi_x(Y) = \{p > Y : \exists n \in V_x,\ v_p(n) \text{ odd}\}.$$

Join two distinct good starts x, y when

$$\Pi_x(Y) \cap \Pi_y(Y) \ne \varnothing.$$

Let $\mathcal{E}_Y$ be the number of joined ordered distinct pairs.

Lemma 13.5 (Exact dependency graph)**.** *Conditionally on $\mathcal{F}_Y$, this graph is a dependency graph for the family $(J_{x,L})_{x\notin\mathcal{D}_Y}$.*

Proof. After conditioning, $J_{x,L}$ is a function of the coordinates $(X_p)_{p\in\Pi_x(Y)}$ alone. If x is joined to no member of a family of starts, the coordinate sets are disjoint, so the corresponding variables are independent.

Local pairs of good starts are indeed edges of the graph. If $0<|x-y|<L$, the sets U_x and U_y share an integer, which has a prime $>Y$. If $|x-y|=L$, the common vertex $x+L-1$ belongs to U_x and also has a prime $>Y$. □

Lemma 13.6 (Edge count)**.** *We have*

$$\mathcal{E}_Y = o_C(N^2). \tag{13.6}$$

More precisely,

$$\mathcal{E}_Y \ll \frac{N^2}{\log^2 B} + NB^2\log\log N + \frac{NB^2}{\log N}.$$

Proof. For $p>Y>B$, a support V_x contains at most one multiple of p. The number of starts whose support meets a multiple of p is

$$O\Big(B\left(\frac{N}{p}+1\right)\Big).$$

Overcounting by the shared prime,

$$\mathcal{E}_Y \ll B^2\sum_{Y<p\le 3N}\left(\frac{N}{p}+1\right)^2.$$

Expand the square and use, by partial summation from $\pi(t)\ll t/\log t$,

$$\sum_{p>Y} p^{-2} \ll \frac{1}{Y\log Y}, \qquad \sum_{p\le 3N} p^{-1} \ll \log\log N, \qquad \pi(3N)\ll N/\log N.$$

With $Y=\lfloor B^2\log B\rfloor$ we have $\log Y\ge 2\log B$ for B large enough, whence $B^2N^2/(Y\log Y)\ll N^2/\log^2 B$. The coarser bound $\sum_{p>Y}p^{-2}\ll Y^{-1}$ would only give $N^2/\log B$; the factor $\log Y$ is kept because it is what sets the rate of Theorem 1.1 (Corollary 13.10). □

13.4 The Chen–Stein bound

Theorem 13.7 (Arratia–Goldstein–Gordon)**.** *Let $(I_\alpha)_{\alpha\in\mathcal{I}}$ be a finite family of indicators with a dependency graph, and let $W=\sum_\alpha I_\alpha$. For each α, let B_α be its closed neighbourhood, set $p_\alpha=\mathbb{E}I_\alpha$, $\lambda=\sum_\alpha p_\alpha$, and*

$$b_1=\sum_\alpha\sum_{\beta\in B_\alpha}p_\alpha p_\beta, \qquad b_2=\sum_\alpha\sum_{\substack{\beta\in B_\alpha\\ \beta\ne\alpha}}\mathbb{E}(I_\alpha I_\beta).$$

There is an absolute constant C_0 such that

$$d_{\mathrm{TV}}(\mathcal{L}(W),\mathrm{Pois}(\lambda))\le C_0(b_1+b_2). \tag{13.7}$$

Proof. This is Arratia–Goldstein–Gordon's Theorem 1 [1, p. 11]; the definitions of b_1,b_2,b_3 are on p. 10. The general form also has a term b_3. For an exact dependency graph, the indicators are independent outside their closed neighbourhood, so $b_3 = 0$. Arratia–Goldstein–Gordon use the normalization $|\mu-\nu|_{\mathrm{TV}} = 2\sup_A |\mu(A)-\nu(A)|$, whereas our d_{TV} is the supremum itself; together with $(1\wedge\lambda^{-1}) \le 1$, this gives (13.7) (and the absolute constant C_0 absorbs the convention). See also [7, Theorem 4.1]. $\square$

Conditionally on $\mathcal{F}_Y$, all good marginals equal

$$p_N = 2^{-L}$$

and the total parameter

$$\lambda_N^{\mathrm{good}} = |I_N \setminus \mathcal{D}_Y| 2^{-L}$$

is deterministic.

Lemma 13.8 (Chen–Stein terms)**.** *For the preceding conditional graph,*

$$b_1 = o_C(1), \qquad \mathbb{E}b_2 = o_C(1).$$

Proof. Including each start in its neighbourhood,

$$b_1 \le (N + \mathcal{E}_Y)p_N^2 = o(1)$$

by Lemma 13.6 and $p_N \asymp_C N^{-1}$.

After averaging over $\mathcal{F}_Y$,

$$\mathbb{E}b_2 = \sum_{\substack{x\sim y\\ x\ne y}} \mathbb{P}(J_{x,L}J_{y,L} = 1).$$

If $0 < |x-y| < L$, the probability vanishes by Lemma 3.4. If $|x-y| = L$, the union of the two start-trees is a tree with $2L$ edges. With root $x-1$, its non-root vertices are exactly $U_x \sqcup U_y$; for two good starts, Lemma 13.1, with $Y > 2L$, gives a conditional rank equal to $2L$. The joint probability is therefore 2^{-2L}, and the $O(N)$ ordered touching pairs contribute $o(1)$.

For $|x-y| > L$, Lemma 2.1 gives

$$\mathbb{P}(J_{x,L}J_{y,L} = 1) \le 2^{-2L}2^{\rho(x,y)}.$$

Over the edges of the graph,

$$\sum 2^{\rho(x,y)} \le \mathcal{E}_Y + R_2(N,L).$$

Proposition 11.2 and Lemma 13.6 conclude. $\square$

Proof of Theorem 1.1. Apply Theorem 13.7 conditionally on $\mathcal{F}_Y$. Lemma 13.8 gives

$$d_{\mathrm{TV}}\Big(\mathcal{L}(Z_{N,L}^{\mathrm{good}} \mid \mathcal{F}_Y), \mathrm{Pois}(\lambda_N^{\mathrm{good}})\Big) \le C_0(b_1 + b_2).$$

The parameter $\lambda_N^{\mathrm{good}}$ is deterministic. By convexity of the total variation,

$$\begin{aligned} &d_{\mathrm{TV}}\Big(\mathcal{L}(Z_{N,L}^{\mathrm{good}}), \mathrm{Pois}(\lambda_N^{\mathrm{good}})\Big) \\ &\qquad \le \mathbb{E}\Big[d_{\mathrm{TV}}\Big(\mathcal{L}(Z_{N,L}^{\mathrm{good}} \mid \mathcal{F}_Y), \mathrm{Pois}(\lambda_N^{\mathrm{good}})\Big)\Big] \le C_0(b_1 + \mathbb{E}b_2) = o(1). \end{aligned}$$

Moreover,

$$|\lambda_N - \lambda_N^{\mathrm{good}}| = |\mathcal{D}_Y| 2^{-L} = o(1),$$

and the standard coupling of two Poisson variables gives

$$d_{\mathrm{TV}}(\mathrm{Pois}(\lambda_N^{\mathrm{good}}), \mathrm{Pois}(\lambda_N)) \le |\lambda_N - \lambda_N^{\mathrm{good}}|.$$

The conclusion follows from (13.5) and the triangle inequality. □

Corollary 13.9 (Finite bound)**.** *Up to a constant depending at most on C,*

$$d_{\mathrm{TV}}(\mathcal{L}(Z_{N,L}), \mathrm{Pois}(\lambda_N)) \ll_C 2^{-L}(\mathcal{M}_B + |\mathcal{D}_Y|) + 2^{-2L}(N + \mathcal{E}_Y + R_2(N, L)).$$

All the inputs of this bound are quantitative. It is therefore worth evaluating them, if only to see which of the two ingredients — the arithmetic or the conditioning — actually limits the result.

Corollary 13.10 (Explicit rate)**.** *Uniformly for $|L - \log_2 N| \le C$,*

$$\boxed{d_{\mathrm{TV}}(\mathcal{L}(Z_{N,L}), \mathrm{Pois}(\lambda_N)) \ll_C \frac{1}{(\log\log N)^2}.}$$

More precisely, with $c = c(A, C) > 0$, the three groups of terms of Corollary 13.9 are

$$\begin{aligned} 2^{-L}(\mathcal{M}_B + |\mathcal{D}_Y|) &\ll N^{-1/2+o_C(1)}, \\ 2^{-2L}R_2(N, L) &\ll \exp\left(-c\,\frac{\sqrt{\log N}}{\log\log N}\right), \\ 2^{-2L}(N + \mathcal{E}_Y) &\ll \frac{1}{\log^2 B} \asymp \frac{1}{(\log\log N)^2}. \end{aligned}$$

Proof. The first line combines (13.1), Lemma 13.3 and $2^{-L} \asymp_C N^{-1}$. The second is Corollary 11.3 with $2^{-2L} \asymp_C N^{-2}$, sector (6) being the dominant term. The third is Lemma 13.6 with $2^{-2L} \asymp_C N^{-2}$, the term $N^2/\log^2 B$ dominating $NB^2 \log\log N$. □

Remark 13.11 (Probabilistic bottleneck and heuristic ceiling of the method)**.** The dominant term is the edge count of the conditional dependency graph: it is the *conditioning* that limits the rate, not the arithmetic, which leaves an exponential margin. This ceiling is intrinsic to the method. The threshold Y is constrained on both sides: Lemma 13.3 requires $\sqrt{Y}/\log Y = o(\log N)$, i.e. $Y \lesssim B^2 \log^2 B$; in the other direction, Lemma 13.6 only provides an *upper bound* for $\mathcal{E}_Y$, but the direct count of pairs of windows sharing a multiple of a single prime $p > Y$ suggests $2^{-2L}\mathcal{E}_Y \gg B^2/(Y \log Y) \gg \log^{-3} B$ — a lower bound we do not establish formally here. Under this heuristic, no choice of Y goes below $(\log\log N)^{-3}$. Obtaining a polynomial rate would require leaving the Arratia–Goldstein–Gordon dependency graph, which treats the sharing of a single large prime as total dependence, in favour of a local version with nonzero b_3 or of a coupling.

14 Deterministic masks and point processes

This section deduces the extensions announced in Theorem 1.2 from the same arithmetic inputs as the scalar law. No three-block estimate is used.

14.1 First moment and Poisson law under a mask

For $A_N \subseteq I_N$, recall

$$Z_{N,L}(A_N) = \sum_{x \in A_N} J_{x,L}, \qquad \lambda_N(A_N) = |A_N| 2^{-L}.$$

Proposition 14.1 (Uniformly masked first moment)**.** *Fix $C_\star > 0$. Uniformly over deterministic sets $A_N \subseteq I_N$ and integers L with*

$$|L - \log_2 N| \le C_\star \log\log N,$$

we have

$$\mathbb{E} Z_{N,L}(A_N) = |A_N| 2^{-L} + O\Big(2^{-L} N^{1/2 + o_{C_\star}(1)}\Big) = |A_N| 2^{-L} + o_{C_\star}(1).$$

Proof. If $m_B(x) = 0$, the private pivots give rank exactly L, and hence $\mathbb{P}(J_{x,L} = 1) = 2^{-L}$. For $m_B(x) = m \ge 1$, the same argument with the $L - m$ remaining pivots gives

$$\mathbb{P}(J_{x,L} = 1) \le 2^{-L+m}.$$

Consequently

$$\Big|\mathbb{P}(J_{x,L} = 1) - 2^{-L}\Big| \le 2^{1-L}(2^{m_B(x)} - 1).$$

Restricting to A_N can only lower the positive right-hand sum. The weighted bound of proposition 3.2, valid for $B = L + 1 = O(\log N)$, gives the first estimate. In the stated window, $2^{-L} N^{1/2+o(1)} = N^{-1/2+o(1)}$. □

Proposition 14.2 (Poisson under a mask)**.** *Fix $C > 0$. Uniformly for $|L - \log_2 N| \le C$,*

$$\sup_{A_N \subseteq I_N} d_{\mathrm{TV}}(\mathcal{L}(Z_{N,L}(A_N)), \mathrm{Pois}(\lambda_N(A_N))) = o_C(1).$$

Proof. We resume the conditioning on $\mathcal{F}_Y$ of section 13. After removing $A_N \cap \mathcal{D}_Y$, the relevant graph is the subgraph induced by $A_N \setminus \mathcal{D}_Y$. All its conditional marginals equal exactly 2^{-L}. The sums b_1 and b_2 can only be smaller than the unmasked sums of Lemma 13.8; their expectation is therefore $o_C(1)$, uniformly in A_N.

The coupling removing the bad starts costs at most

$$\sum_{x \in \mathcal{D}_Y} \mathbb{P}(J_{x,L} = 1) = o_C(1),$$

and the difference between the target parameter and the conditional parameter is

$$|A_N \cap \mathcal{D}_Y| 2^{-L} \le |\mathcal{D}_Y| 2^{-L} = o_C(1).$$

Theorem 13.7, then the coupling of Poisson laws, conclude. □

Remark 14.3 (Uniform rate under masks)**.** The preceding proof loses nothing of the rate of Corollary 13.10: the terms b_1 and b_2 of the graph induced by $A_N \setminus \mathcal{D}_Y$ are bounded term by term by their unmasked counterparts, while the coupling cost and the parameter shift are bounded by $2^{-L}(2\mathcal{M}_B + 2|\mathcal{D}_Y|) \ll N^{-1/2+o_C(1)}$, independently of A_N. The three groups of terms of Corollary 13.9 therefore uniformly dominate their masked versions, and

$$\sup_{A_N \subseteq I_N} d_{\mathrm{TV}}(\mathcal{L}(Z_{N,L}(A_N)), \mathrm{Pois}(\lambda_N(A_N))) \ll_C \frac{1}{(\log\log N)^2}.$$

14.2 Spatial process via auxiliary thinning

Let $g\colon [1,2] \to [0,\infty)$ be continuous. Set

$$q_x = 1 - e^{-g(x/N)}$$

and introduce, independently of f, variables $\xi_x \sim \mathrm{Ber}(q_x)$. The thinned count is

$$W_g = \sum_{x \in I_N} \xi_x J_{x,L}.$$

Conditionally on $\mathcal{F}_Y$ and after removing $\mathcal{D}_Y$, the same graph is a dependency graph for $(\xi_x J_{x,L})_x$. Its marginals equal $q_x 2^{-L}$, and each Chen–Stein term acquires a factor q_x or $q_x q_y$, hence remains dominated by the unthinned term. It follows that

$$W_g \Longrightarrow \mathrm{Pois}\Big(\lambda \int_1^2 (1 - e^{-g(t)})\, dt\Big)$$

along any subsequence with $\lambda_N \to \lambda$, since

$$2^{-L} \sum_{x \in I_N} q_x \longrightarrow \lambda \int_1^2 (1 - e^{-g(t)})\, dt.$$

On the other hand, conditionally on the indicators $J_{x,L}$,

$$\mathbb{P}(W_g = 0 \mid (J_{x,L})_x) = \prod_x (1 - q_x)^{J_{x,L}} = \exp\Big(-\int g\, d\Xi_{N,L}\Big).$$

The convergence of the void probabilities therefore gives

$$\mathbb{E} \exp\Big(-\int g\, d\Xi_{N,L}\Big) \longrightarrow \exp\Big(-\lambda \int_1^2 (1 - e^{-g(t)})\, dt\Big),$$

which is the Laplace functional of $\mathrm{PPP}(\lambda\, dt)$. This proves Theorem 1.2(ii).

Remark 14.4 (Grid effect and total variation)**.** The preceding convergence is vague, not in total variation. If S_N is the set of configurations all of whose points belong to the grid $N^{-1}\mathbb{Z}$, then

$$\mathbb{P}(\Xi_{N,L} \in S_N) = 1, \qquad \mathbb{P}(\mathrm{PPP}(\lambda\, dt) \in S_N) = e^{-\lambda}.$$

Indeed, a Poisson process with diffuse intensity almost surely places no point on this grid; only the empty configuration is common to the two supports. The restrictions of the two laws to nonempty configurations are therefore singular, and

$$\liminf_N d_{\mathrm{TV}}(\mathcal{L}(\Xi_{N,L}), \mathcal{L}(\mathrm{PPP}(\lambda\, dt))) \geq 1 - e^{-\lambda}.$$

The full laws are not mutually singular because of their common empty atom.

14.3 Exact-length events and mixed systems

For $e \in \mathbb{N}_0$, set

$$q_e = L + e + 1$$

and define $K_{x,e}$ as the indicator of the system

$$f(x-1) = -f(x), \qquad f(x) = \cdots = f(x + L + e - 1), \qquad f(x + L + e) = -f(x).$$

Lemma 14.5 (Almost sure finiteness of stretches)**.** *Almost surely, no constant stretch of f is infinite: for every integer $x \geq 2$ there is $n \geq x$ with $f(n) \neq f(x)$.*

Proof. Fix x. On the event where f is constant on $[x, \infty)$, every prime $p \geq x$ satisfies $f(p) = f(x)$; the signs $(f(p))_{p \geq x}$ are then all equal. They are independent, uniform on $\{-1, +1\}$, and infinitely many: the event has probability zero. A countable union over x concludes. (Incidentally, the constant could only be $+1$: for a prime $p \geq \sqrt{x}$ we have $p^2 \geq x$ and $f(p^2) = f(p)^2 = 1$. This is the degeneracy specific to the extended Rademacher model, the very one producing the defective integers of Section 3.) ☐

Lemma 14.5 gives $\ell_x < \infty$ almost surely on $\{J_{x,L} = 1\}$, hence, almost surely,

$$J_{x,L} = \sum_{e \geq 0} K_{x,e}, \qquad K_{x,e} = \mathbf{1}_{\{J_{x,L}=1,\ \ell_x - L = e\}}.$$

The system of $K_{x,e}$ has q_e rows. Its homogeneous rows are those of the matrix A_{x,q_e} obtained by replacing L with q_e in Section 2; only the last entry of the right-hand side changes from 0 to 1.

Lemma 14.6 (Mixed-length affine identity)**.** *For $e, f \in \mathbb{N}_0$ and two starts x, y, stack the homogeneous matrices of $K_{x,e}$ and $K_{y,f}$. Write*

$$\rho_{e,f}(x, y) = \dim \ker \begin{pmatrix} A_{x,q_e} \\ A_{y,q_f} \end{pmatrix}^{\mathsf{T}}$$

and let $\eta_{e,f}(x, y) \in \{0, 1\}$ be the compatibility indicator of the mixed right-hand side. Then

$$\boxed{\mathbb{P}(K_{x,e} = K_{y,f} = 1) = 2^{-q_e - q_f} \eta_{e,f}(x, y) 2^{\rho_{e,f}(x,y)}.}$$

If E is fixed and $Q = L + E + 1$, then, for $0 \leq e, f \leq E$,

$$\rho_{e,f}(x, y) \leq \rho_Q(x, y),$$

where ρ_Q denotes the homogeneous defect of two matrices of length Q.

Proof. The first identity is the rank–compatibility formula applied to a system of $q_e + q_f$ rows: if it is compatible, its probability equals $2^{-\operatorname{rank}} = 2^{-q_e - q_f + \rho_{e,f}}$; otherwise it is zero.

For the domination, the homogeneous rows of A_{x,q_e} are a subset of the rows of $A_{x,Q}$: changing the last equality constraint into an opposite-sign constraint does not modify its homogeneous vector. We therefore extend the coefficients of a mixed relation by zero on the missing rows. This map injects the mixed kernel into the kernel of the two-length-Q system. ☐

The second junction needed concerns unions that have a cycle.

Lemma 14.7 (Private pivots and cycle space)**.** *Let G be a finite graph. In each connected component, assume that all vertices except possibly one have a private prime coordinate: this coordinate appears with odd valuation at this vertex and at no other vertex of G. Then the kernel of the edge vectors $v(a) + v(b)$, projected onto these coordinates, is contained in the cycle space of G. In particular,*

$$\operatorname{rank}\{v(a) + v(b) : \{a, b\} \in E(G)\} \geq |E(G)| - \beta_1(G),$$

where $\beta_1(G) = |E(G)| - |V(G)| + c(G)$ is the cyclomatic number.

Proof. A vanishing combination of edges selects a subgraph whose degree is even at every vertex carrying a private coordinate. In each component, the parity of the sum of degrees also forces the degree of the possible remaining vertex to be even. The selected subgraph is therefore Eulerian, i.e. an element of the cycle space. The dimension of this space is $\beta_1(G)$, whence the stated rank. □

14.4 Truncated marked process

Fix $E \geq 0$ and set

$$Q = L + E + 1, \qquad Y_Q = (Q+1)^2 \log(Q+1).$$

Remove the starts for which one of the vertices of $\{x, \dots, x+Q-1\}$ has no private prime $> Y_Q$; call this set $\mathcal{D}^{(Q)}$. The removed mass must be evaluated for the variables $K_{x,e}$ themselves, not for $J_{x,Q}$: the removal criterion concerns the long window $\{x, \dots, x+Q-1\}$, while the system of $K_{x,e}$ has only $q_e = L + e + 1$ rows, with support

$$V_{x,e} = \{x-1, x, \dots, x+L+e\}, \qquad \operatorname{diam} V_{x,e} = L + e + 1 \leq Q.$$

A start can therefore be removed because of a late vertex absent from $V_{x,e}$, and the inclusion $K_{x,e} \leq J_{x,Q}$ is false in general. The next lemma provides the estimate actually required.

Lemma 14.8 (Mass of the removed starts, exact lengths)**.** *Uniformly for $|L - \log_2 N| \leq C$ and fixed E,*

$$\sum_{e=0}^{E} \sum_{x \in \mathcal{D}^{(Q)}} \mathbb{P}(K_{x,e} = 1) = o_{C,E}(1).$$

Proof. Fix $e \leq E$ and set $W = Q + 1$, so that $W > \operatorname{diam} V_{x,e}$. Let $m_W(x,e) = \#\{n \in V_{x,e} : \mathcal{K}_W(n) = 1\}$ be the number of W-defective vertices of the support. We separate according to $m_W(x,e)$.

First case: $m_W(x,e) \geq 1$. Every non-W-defective vertex of $V_{x,e}$ has a prime $> W$ of odd valuation, private since $W > \operatorname{diam} V_{x,e}$; it therefore cannot belong to the support of a relation. As in Corollary 3.3, relations are even parts of the W-defective vertices, whence $\rho \leq \max(m_W(x,e) - 1, 0)$ and

$$\mathbb{P}(K_{x,e} = 1) \leq 2^{-q_e} 2^{m_W(x,e)-1} \leq 2^{-q_e}\big(2^{m_W(x,e)} - 1\big).$$

Proposition 3.2 at threshold W, whose proof is unchanged since $W = L + O_E(1)$, applies to the translates $u = x - 1 \in [N, 2N)$; the only remaining start, $x = N$ (translate $u = N-1$), is covered by the pointwise bound, which gives it $2^{m_W(N,e)} - 1 = N^{o(1)}$. In total $\sum_x \big(2^{m_W(x,e)} - 1\big) \leq N^{1/2 + o_{C,E}(1)}$. Since $2^{-q_e} \asymp_{C,E} N^{-1}$, the contribution is $N^{-1/2 + o_{C,E}(1)}$.

Second case: $m_W(x,e) = 0$. Every vertex of $V_{x,e}$ then has a private prime $> W$, and $V_{x,e}$ equipped with the q_e edges of the system is a tree (a path, a star and a pendant edge). Lemma 13.1, applied with root $x-1$, gives a projected rank equal to q_e, hence $\mathbb{P}(K_{x,e} = 1) = 2^{-q_e}$ exactly. The number of such x is bounded by $|\mathcal{D}^{(Q)}| \leq N^{1/2 + o_{C,E}(1)}$, a variant of Lemma 13.3 at threshold Y_Q and length Q. The contribution is therefore again $N^{-1/2 + o_{C,E}(1)}$.

There are $E + 1$ values of e, a fixed number. □

For every remaining start and every $0 \leq e \leq E$, the conditional rank of the system of $K_{x,e}$ is exactly q_e, hence

$$\mathbb{P}(K_{x,e} = 1 \mid \mathcal{F}_{Y_Q}) = 2^{-q_e}.$$

We join two indices (x, e) and (y, f) if they have the same start, if their supports touch, or if their supports share a prime coordinate $p > Y_Q$. This is an exact dependency graph after conditioning. Events with distinct marks at the same start are incompatible. For the local geometry, assume $x < y$ and set $r_e = L + e$ and $d = y - x$.

- If $d < r_e$, the second start forces a sign change at y inside the constant stretch issued from x; the joint probability is zero.
- If $d = r_e$, the union of the two start-trees has exactly one cycle, the triangle on $x, y-1, y$. Its cyclomatic number is 1.
- If $d = r_e + 1$, the two trees touch at one vertex and their union is a tree.
- If $d > r_e + 1$, their supports are disjoint.

The case $y < x$ is symmetric, with $r_f = L + f$ for the stretch of the left-hand start. In the two compatible non-disjoint cases, the union has diameter $O_E(L) < Y_Q$. The kept starts provide a private coordinate to all vertices except the root of each component. At shift $d = r_e$, then at shift $d = r_e + 1$, the root $y - 1$ of the second tree is a non-root vertex of the first tree and therefore also has a pivot; only $x - 1$ may lack one in the connected component. Lemma 13.1 and Lemma 14.7 therefore give, uniformly,

$$\mathbb{P}(K_{x,e}K_{y,f} = 1 \mid \mathcal{F}_{Y_Q}) \le 2^{1-q_e-q_f}.$$

There are only $O_E(N)$ compatible ordered local pairs, hence liable to contribute to b_2. The $O_E(NL)$ pairs with touching but incompatible supports do not contribute to b_2; their contribution to b_1 is $O_E(NL2^{-2L}) = o(1)$.

For separated pairs, Lemma 14.6 and proposition 11.2 applied to Q give

$$\sum_{\substack{x,y\in I_N\\ |x-y|>Q}} \left(2^{\rho_{e,f}(x,y)} - 1\right) \le R_2(N,Q) = o_{C,E}(N^2).$$

The large-prime edge count is also $o_{C,E}(N^2)$. Since E is fixed and $2^{-q_e} \asymp_{C,E} N^{-1}$, the Chen–Stein terms of the finite family $(K_{x,e})_{x\in I_N, 0\le e\le E}$ satisfy

$$b_1 = o_{C,E}(1), \qquad \mathbb{E}b_2 = o_{C,E}(1).$$

Here the local pairs contribute $O_E(N2^{-2L}) = o(1)$, while the separated pairs are bounded by

$$O_E\Big(2^{-2L}[\mathcal{E}_{Y_Q} + R_2(N,Q)]\Big) = o(1).$$

Now let $g \ge 0$ be continuous with compact support in $[1,2] \times \mathbb{N}_0$. Since $\mathbb{N}_0$ is discrete, there is E such that $g(t,e) = 0$ for $e > E$. Thin each $K_{x,e}$ independently with probability

$$q_{x,e} = 1 - e^{-g(x/N,e)}.$$

The preceding reasoning, the thinned terms being dominated by the unthinned ones, gives a Poisson law with asymptotic parameter

$$\begin{aligned}\sum_{x\in I_N}\sum_{e=0}^{E} q_{x,e}2^{-q_e} &= 2^{-L}\sum_{x\in I_N}\sum_{e=0}^{E} 2^{-(e+1)}(1 - e^{-g(x/N,e)})\\ &\longrightarrow \lambda\int_1^2 \sum_{e\ge 0}(1 - e^{-g(t,e)})\,\nu(\{e\})\,dt.\end{aligned}$$

The void-probability identity then provides the Laplace functional

$$\mathbb{E}\exp\Big(-\int g\,d\widehat{\Xi}_{N,L}\Big) \longrightarrow \exp\left(-\lambda\int_1^2\sum_{e\geq 0}(1-e^{-g(t,e)})\nu(\{e\})\,dt\right).$$

It is that of $\mathrm{PPP}(\lambda\,dt\otimes\nu)$.

To make explicit the passage from the truncated process to the full process, set

$$T_{N,E} = \sum_{x\in I_N} J_{x,L+E+1}.$$

The event that a mark exceeds E implies $T_{N,E}\geq 1$, and proposition 14.1 gives

$$\mathbb{E}T_{N,E} = \lambda_N 2^{-(E+1)} + o_{C,E}(1).$$

Thus

$$\limsup_{N\to\infty}\mathbb{P}\Big(\widehat{\Xi}_{N,L}([1,2]\times\{E+1,E+2,\ldots\})>0\Big)\leq \lambda 2^{-(E+1)},$$

which tends to zero with E. This proves Theorem 1.2(iii), including the uniform tightness of the marks. Part (i) is proposition 14.2. All the preceding marked passages fix E first; no uniformity for a truncation $E=E(N)\to\infty$ is claimed.

Corollary 14.9 (Exact-length counts)**.** *Along any subsequence with $\lambda_N\to\lambda$, for every fixed E, the vector*

$$\left(\sum_{x\in I_N} K_{x,e}\right)_{0\leq e\leq E}$$

converges to independent Poisson variables with means $\lambda 2^{-(e+1)}$. If

$$M_{N,L} = \max\{\ell_x - L : x\in I_N,\ J_{x,L}=1\},$$

with the convention $M_{N,L}=-\infty$ when there is no start, then, for fixed $m\in\mathbb{N}_0$,

$$\mathbb{P}(M_{N,L}\leq m)\longrightarrow \exp\Big(-\lambda 2^{-(m+1)}\Big).$$

This maximum is internal to the dyadic block and does not constitute a law of the global longest run on $[1,M]$.

15 Dyadic gluing: truncation of the deep blocks

This section fully proves the "truncation" half of the gluing of $[1,M]$: the starts of height below $M2^{-j_0}$ have a total first moment tending to zero in the double limit of the gluing (j_0 fixed, $M\to\infty$, then $j_0\to\infty$). The other half — the extension of proposition 11.2 to ratios $\kappa=2^{j_0}=O(1)$ — is the subject of section 16.

Throughout the section, $L=\log_2 M+O_C(1)$, $B=L+1$, and m_x denotes the number of B-defective vertices of $V_x=\{x-1,\ldots,x+L-1\}$, a vertex being B-defective when its squarefree part is B-smooth. As in the proof of Corollary 3.3, every non-defective vertex carries a prime $p>B$ of odd valuation, private in the window of diameter $L<p$; the support of a relation is therefore contained in the defective vertices, and the even parts of a set with m_x elements form a space of dimension $\max(m_x-1,0)$, whence, by Lemma 2.1,

$$\mathbb{P}(J_{x,L}=1)\leq 2^{-L}\,2^{\max(m_x-1,0)}\qquad (x\geq 2). \tag{15.1}$$

15.1 Low zone: intermediate-prime pivots

Lemma 15.1 (Low zone)**.** *Let $0 < \varepsilon < 1$. Uniformly for $2 \le x \le L^{2-\varepsilon}$,*

$$\mathbb{P}(J_{x,L} = 1) \le 2^{-r(x)}, \qquad r(x) = \pi(L) - \pi(\sqrt{x+L}) = (1 - o(1))\,\pi(L).$$

In particular $\sum_{2 \le x \le L^{2-\varepsilon}} \mathbb{P}(J_{x,L} = 1) = o(1)$.

Proof. Consider the primes $q \in (\sqrt{x+L}, L]$, of which there are $r(x)$, and set $A_q = \{n \in V_x : q \mid n\}$. Every interval of $B > q$ consecutive integers contains a multiple of q, so $A_q \ne \emptyset$. Every $n \in A_q$ satisfies $v_q(n) = 1$, since $q^2 > x + L > n$. A vertex cannot belong to two distinct A_q, since $qq' > x + L$: the supports are disjoint. A relation S is an even part of V_x with square product (Lemma 2.2); since the q-valuation of the product is $|S \cap A_q|$, every relation satisfies the $r(x)$ linear conditions $|S \cap A_q| \equiv 0 \pmod 2$.

These conditions are independent on the space E of even parts. Two linear forms $\langle v, \cdot\rangle$, $\langle w, \cdot\rangle$ coincide on $E = \mathbf{1}^\perp$ if and only if $v - w \in \{0, \mathbf{1}\}$; a dependence among the forms $\mathbf{1}_{A_q}$, $q \in T \ne \emptyset$, would therefore require $\sum_{q\in T} \mathbf{1}_{A_q} \in \{0, \mathbf{1}\}$, that is, the supports being disjoint and nonempty, $\bigcup_{q\in T} A_q = V_x$. But V_x contains a perfect square: two consecutive squares near x differ by $2\sqrt{x+L} + O(1) < B$ since $x \le L^{2-\varepsilon}$. This square m^2 belongs to no A_q: $q \mid m^2$ would force $q^2 \mid m^2$ with $q^2 > x + L \ge m^2$. Hence $\bigcup A_q \ne V_x$ and the $r(x)$ conditions are independent on E, whence $\rho(x) \le \dim E - r(x) = L - r(x)$ and, by Lemma 2.1, $\mathbb{P}(J_{x,L} = 1) \le 2^{-L} 2^{L - r(x)} = 2^{-r(x)}$. Finally we use the prime number theorem in the exact form

$$\frac{\pi(t)\log t}{t} \longrightarrow 1 \qquad (t \longrightarrow \infty),$$

as stated by Avigad, Donnelly, Gray and Raff [3, Introduction, p. 1]. Thus $\pi(\sqrt{x+L}) \ll L^{1-\varepsilon/2}/\log L = o(\pi(L))$, uniformly in the stated range, and the sum is at most $L^2\, 2^{-(1-o(1))\pi(L)} = o(1)$. This is exactly the registered external proposition used by the formal verification. $\square$

15.2 Polynomial heights: Laishram–Shorey large primes

Lemma 15.2 (Polynomial band)**.** *There is $c_3 > 0$ such that, uniformly for $B + 1 < x \le 2L^2$, at least $c_3 B/\log B$ vertices of V_x are not B-defective. In particular*

$$\sum_{B+1 < x \le 2L^2} \mathbb{P}(J_{x,L} = 1) \le 2L^2\, 2^{-c_3 B/\log B + 1} = o(1).$$

Proof. Write $V_x = \{m + d : 1 \le d \le B\}$ with $m = x - 2 \ge B$. Laishram and Shorey [23, Corollary 1, equation (10), p. 330; definition of $\delta(k)$, p. 328] show that, after setting their variable $n = m + 1$, the hypothesis $n > k$ is equivalent to $m \ge k$, and the product $\Delta = (m + 1)\cdots(m + k)$ satisfies

$$\omega(\Delta) \ge \min\Big(\pi(k) + \lfloor \tfrac{3}{4}\pi(k)\rfloor - 1 + \delta(k),\ \pi(2k) - 1\Big),$$

where $\delta(k) \ge 0$. With $k = B$, the number of *distinct* primes $q > B$ dividing Δ is therefore at least

$$\min\Big(\pi(B) + \lfloor \tfrac{3}{4}\pi(B)\rfloor - 1,\ \pi(2B) - 1\Big) - \pi(B) \ge \Big(\tfrac{3}{4} - o(1)\Big)\pi(B),$$

the threshold being B and not L: a B-smooth kernel may contain the prime $B = L + 1$ itself. Each such q divides exactly one vertex n (two multiples of $q > B$ differ by at least q, more than the diameter). If $v_q(n)$ is odd, n is not B-defective. If $v_q(n)$ is even, then $q^2 \mid n$ and $n \le 2L^2 + L$, so $n = aq^2$ with $a \le 3$; for each value of a, at most one vertex has this form, two of them differing by at least $a(q'^2 - q^2) \ge 2L + 1 > L$. There remain at least $(\frac{3}{4} - o(1))\pi(B) - 3$ primes with odd valuation. Since a vertex $n \le 2L^2 + L < L^3$ carries at most two distinct primes $> L$, the number of non-defective vertices is at least half of this count, i.e. $\geq c_3 B/\log B$. The conclusion follows from (15.1): $m_x \leq B - c_3 B/\log B$ gives $\mathbb{P}(J_{x,L}=1) \leq 2^{-L+B-c_3B/\log B-1} = 2^{-c_3B/\log B+1}\cdot 2^{-1}$. $\square$

15.3 Higher heights: Pell pairs and the Balasubramanian–Shorey maximum

The next two lemmas together cover all dyadic blocks $[N,2N)$ with $2L^2 \leq N \leq M$: the first bounds the *number* of windows carrying at least two defects, the second bounds the *weight* of each.

Lemma 15.3 (Rarity of two-defect windows)**.** *Uniformly for $2L^2 \leq N \leq M$,*

$$\#\{x \in [N,2N) : m_x \geq 2\} \leq \exp\Big(O_C\Big(\frac{L}{\log L}\Big)\Big).$$

Proof. First note that the number of squarefree B-smooth integers $s \leq 3N$ satisfies, by Rankin's method with $\alpha = 1/\log L$,

$$\Psi^\flat(3N,B) \leq (3N)^\alpha \prod_{p\leq B}(1+p^{-\alpha}) \leq \exp\Big(\frac{\log(3N)}{\log L} + O(\pi(B))\Big) = \exp\Big(O_C\Big(\frac{L}{\log L}\Big)\Big), \tag{15.2}$$

since $p^{-\alpha} \leq 1$ for every p and $\log(3N) \leq (\log 2 + o(1))L$.

Let $n = sa^2$ and $n' = s'b^2$ be two distinct B-defective vertices of a single window V_x, $x \in [N,2N)$, so that $0 < |n-n'| \leq L$ and $n,n' \in [N-1,3N)$. If $s = s'$, then $|n-n'| = s\,|a-b|(a+b) \geq s\,a \geq \sqrt{sn} \geq \sqrt{N-1} > L$, which is excluded: hence $s \neq s'$, and two distinct squarefree integers cannot have a rational square quotient. Lemma 9.2, applied at the parameter $3N$ with $K_0 = 2$, $(A,C,e) = (s,s',n-n')$, $|z| = a$, $|w| = b \leq \sqrt{3N}$, gives at most $\exp(O(\log N/\log\log N)) = \exp(O_C(L/\log L))$ solutions per triple.

Every x with $m_x \geq 2$ provides such a pair; conversely, a triple (s,s',e) and a solution (a,b) determine (n,n'), and n belongs to at most B windows. The number of triples is at most $\Psi^\flat(3N,B)^2 \cdot 2L$. The product of the four factors is $\exp(O_C(L/\log L))$. $\square$

Lemma 15.4 (Pointwise maximum)**.** *Set*

$$g_L = B - \mu_B(\theta_0) = \frac{B}{\log B}(\log\log B - \log\log\log B - \theta_0 + o(1)),$$

where θ_0 is the effective absolute constant of Balasubramanian–Shorey [4] and

$$\mu_k(\theta) = k\Big(1 - \frac{\log\log k}{\log k} + \frac{\log\log\log k}{\log k} + \frac{\theta}{\log k}\Big).$$

Then, for M large enough and every $x > B^2+2$,

$$m_x \leq B - g_L.$$

Proof. Theorem 1 of [4] states: for $k \geq 27$, there exist effective absolute constants θ_0 and C_2 such that, if $(m+d_1)\cdots(m+d_t) = by^2$ with $1 \leq d_1 < \cdots < d_t \leq k$, $P^+(b) \leq k$, $m > k^2$ and $t \geq \mu_k(\theta_0)$, then $k \leq C_2$. The only constraint on b is the smoothness $P^+(b) \leq k$.

Write $V_x = \{m+d : 1 \leq d \leq B\}$, $m = x-2 > B^2$. The product of the m_x B-defective vertices equals $\prod s_i \cdot \big(\prod a_i\big)^2 = by^2$ with $P^+(b) \leq B$. If we had $m_x \geq \mu_B(\theta_0)$, the theorem would give $B \leq C_2$, false for M large enough. $\square$

15.4 The truncation

Proposition 15.5 (Truncation of the deep blocks)**.** *In the window* $|L - \log_2 M| \le C$*,*

$$\lim_{j_0\to\infty} \limsup_{M\to\infty} \sum_{2\le x<M2^{-j_0}} \mathbb{P}(J_{x,L} = 1) = 0.$$

More precisely, the sum is $O_C(2^{-j_0}) + o_M(1)$ *for fixed* j_0*.*

Proof. The ranges $[2, L^{2-\varepsilon}]$ and $(B+1, 2L^2]$ contribute $o_M(1)$ by Lemmas 15.1 and 15.2, and they cover $[2, 2L^2]$. Cut the rest into dyadic blocks $[N, 2N)$, the last one possibly truncated on the right to $[N, M2^{-j_0})$, all with origin $N \ge 2L^2$ and at most $2L$ in number; Lemmas 15.3 and 15.4 apply to any subinterval of a block. Within a block, the windows with $m_x \le 1$ contribute at most $N2^{-L}$ by (15.1). The windows with $m_x \ge 2$ number at most $\exp(O_C(L/\log L))$ (Lemma 15.3, the condition $N \ge 2L^2 > B^2 + 2$ being satisfied), each of weight at most $2^{-L}2^{m_x-1} \le 2^{-L}2^{B-g_L-1} = 2^{-g_L}$ (Lemma 15.4), whence a contribution

$$\exp\Big(O_C\Big(\frac{L}{\log L}\Big) - (\log 2)\, g_L\Big) \le \exp\Big(-\frac{\log 2}{2}\,\frac{L\log\log L}{\log L}\Big)$$

for M large enough, since $g_L \gg L\log\log L/\log L$ dominates the term $O_C(L/\log L)$ by a factor $\log\log L$. Summing,

$$\sum_{2\le x<M2^{-j_0}} \mathbb{P}(J_{x,L} = 1) \le o_M(1) + \sum_{j\ge j_0} M2^{-j-1}2^{-L} + 2L\exp\Big(-\frac{\log 2}{2}\,\frac{L\log\log L}{\log L}\Big) = O_C(2^{-j_0}) + o_M(1),$$

since $M2^{-L} \asymp_C 1$. The double limit concludes. □

Remark 15.6 (Towards the global law)**.** Proposition 15.5 is exactly the first-moment control required to apply Theorem 13.7 *in one go* to the family of starts indexed by $U = [M2^{-j_0}, M)$: the removed starts have vanishing total probability, with no inter-block independence assumption. The required extension of proposition 11.2 to pairs $x, y \in [N, \kappa N)$ with $\kappa = 2^{j_0}$ fixed is proved in section 16 (proposition 16.1), and the interior global law follows (Theorem 16.2).

16 Dyadic gluing II: the bounded-ratio two-block sum and the global law

Proposition 15.5 has dealt with the deep blocks; for the interior global law it remains to control the dependencies between *retained* starts, that is, to extend (1.4) from a dyadic block to a range of bounded ratio. This is the subject of this section, which concludes with the global law (Theorem 16.2).

Proposition 16.1 (Absolute homogeneous sum at bounded ratio)**.** *For* $\kappa \ge 2$ *set* $U_{N,\kappa} = [N, \kappa N) \cap \mathbb{Z}$*. Fix* $C > 0$ *and* $\kappa_0 \ge 2$*. Uniformly for* $2 \le \kappa \le \kappa_0$ *and* $|L - \log_2 N| \le C$*,*

$$R_{2,\kappa}(N, L) := \sum_{\substack{x,y\in U_{N,\kappa}\\ |x-y|>L}} \big(2^{\rho(x,y)} - 1\big) = o_{C,\kappa_0}(N^2).$$

The uniformity is local *in* κ*: the statement does not allow* $\kappa = \kappa(N) \to \infty$*, and the losses due to the widening of the range alone are of size* $\kappa^{O(1)}$*.*

Proof. Set $U = U_{N,\kappa}$. All the definitions of the sections devoted to (1.4) are kept with I_N replaced by U; we have $|U| \ll_{\kappa_0} N$, all vertices lie in $[N-1, c_{\kappa_0}N]$, and $2^B \asymp_{C,\kappa_0} N$. We walk through the proofs and flag every site where the dyadic confinement was consumed; a single one requires a new computation.

Hosts and interpolation. In Lemma 4.2, a class modulo K has in U at most $(\kappa-1)N/K+1 \ll_{\kappa_0} N/K$ representatives (since $K \le n \ll_{\kappa_0} N$), and the heights $n \le 3N$ become $n \le c_{\kappa_0}N$: Lemmas 17.1 and 17.2 give $H_2 \le N^{3/2+o_{C,\kappa_0}(1)}$ and the interpolation (6.6) with the factor $N^{3/4+o_{C,\kappa_0}(1)}$.

Rational channels: the new site. Fix a channel (a,b,h) with $m \ge 2$ exact units and set $q = \max(a,b)$. Lemma 5.1 gives $q \le L$ as in a block: this bound comes from the window length, not from the width of U. A reference unit parametrizes $x = bt - i_0$, $y = at - j_0$, and the *intersection* of the two constraints $x, y \in U$ confines t to an interval of length $\ll_{\kappa_0} N/q + 1$ — the denominator is q, as in the proof of proposition 5.4. For $q \ge 3$, Lemma 17.5 bounds the height-q contribution by $N^{1+o(1)}q^2 2^{B/q}$ (resp. $4^{B/q}$), dominated by $q=3$, i.e. $N^{4/3+o_{C,\kappa_0}(1)}$ and $N^{5/3+o_{C,\kappa_0}(1)}$. The site that changes is $q=2$: in a dyadic block, the ratios 2:1 are attained only at the edges ($O(B^2)$ pairs); in U with $\kappa \ge 2$ they become *volumetric*. The interval of t has length $O_{\kappa_0}(N)$, there are $O(1)$ reduced pairs and $O(B)$ values of h, whence $O_{\kappa_0}(NB)$ pairs, each of weight $2^\sigma \le 2^{B/2}$ (resp. $4^\sigma \le 4^{B/2}$), i.e.

$$O_{\kappa_0}(NB\,2^{B/2}) = N^{3/2+o_{C,\kappa_0}(1)} \qquad \text{and} \qquad O_{\kappa_0}(NB\,4^{B/2}) = N^{2+o_{C,\kappa_0}(1)}.$$

The masses of proposition 5.4 therefore become

$$\sum_{\substack{x,y\in U\\ |x-y|>L}} (2^{\sigma(x,y)}-1) \le N^{3/2+o_{C,\kappa_0}(1)}, \qquad \sum_{\substack{x,y\in U\\ |x-y|>L}} (4^{\sigma(x,y)}-1) \le N^{2+o_{C,\kappa_0}(1)}, \tag{16.1}$$

and these are the only exponents modified in the whole proof. Lemma 17.6 gives the same bound, since its sum ranges over the geometries (a,b,h) alone, with no volume factor in t.

Pointwise defects. Wherever proposition 3.2 is invoked pointwise — the bound $D^\# = O(B/\log B)$ of Lemma 6.4, then consumed by the sectors — its statement requires an interval contained in $[N'/2, 3N']$ at scale N': for a window at $x \in U$, we apply it at the scale of the dyadic sub-block of U containing x. The constants c_1, c_2 of the window $c_1 \log N' \le B \le c_2 \log N'$ can then be chosen positive, bounded and uniform ($B/\log N' = 1/\log 2 + O_{C,\kappa_0}(1/\log N)$), and Lemmas 17.10 and 17.33 uniformly give $D^\# = O_{C,\kappa_0}(B/\log B)$.

Moderate sectors. In proposition 7.3, the branch $\sigma = 0$ gives $O_{\kappa_0}(N^2/P^2)$ pairs by CRT over intervals of length $O_{\kappa_0}(N)$; in the branch $\sigma \ge 1$, the interval of t of length $O_{\kappa_0}(N/q+1)$ contains $O_{\kappa_0}(N/P)$ representatives per class modulo $P \le N$, and Lemmas 17.6 and 17.12 to 17.14 give: $\mathcal{Q}^{(1)}_{\mathrm{res}} \le N^{2+o(1)}$, then $\mathcal{R}^{(1)}_{\mathrm{res}} \le N^{7/4+o(1)}$ by (6.6). Lemmas 17.15 and 17.16 give

$$\mathcal{Q}^{(2)}_{\mathrm{res}} \le N^{2+o_{C,\kappa_0}(1)}, \qquad \mathcal{R}^{(2)}_{\mathrm{res}} \le N^{7/4+o_{C,\kappa_0}(1)},$$

et

$$\mathcal{Q}^{(3)}_{\mathrm{res}} \le N^{19/8+o_{C,\kappa_0}(1)}, \qquad \mathcal{R}^{(3)}_{\mathrm{res}} \le N^{31/16+o_{C,\kappa_0}(1)}.$$

Deep sectors. Lemma 17.17 uniformly excludes the aligned deep sector. Lemmas 17.18, 17.19, 17.25, and 17.26 give the same counts in the polynomial ranges $e \le N^{O_{\kappa_0}(1)}$. Lemmas 17.24 and 17.28 to 17.30 give, with

$$T = (C_{\det}(\kappa_0)NB)^{1/2}, \qquad K > \frac{2C_{\mathrm{term}}(\kappa_0)}{\log 2},$$

the same uniform bounds for the deep sectors.

Assembly. The partition, the identity (11.1) and all the uniform sector bounds are gathered in Lemmas 17.31 and 17.32. The systematic term is $N^{3/2+o_{C,\kappa_0}(1)}$ by (16.1); the residual masses of the sectors are, in order, $N^{7/4}$, $N^{7/4}$, $N^{31/16}$, 0, $N^{3/2}$, $o(N^2)$, $N^{7/4}$, all up to $o_{C,\kappa_0}(1)$ in the exponent. All are $o_{C,\kappa_0}(N^2)$. $\square$

Theorem 16.2 (Interior global Poisson law at the critical scale)**.** *Let $C > 0$. Set $Z_M = \sum_{2\le x<M} J_{x,L}$ and $\Lambda_M = \mathbb{E}Z_M$. Uniformly for $|L - \log_2 M| \le C$,*

$$d_{\mathrm{TV}}(\mathcal{L}(Z_M), \mathrm{Pois}(\Lambda_M)) \longrightarrow 0 \qquad (M \to \infty),$$

and $\Lambda_M = (1+o_C(1))M2^{-L}$; one may replace $\mathrm{Pois}(\Lambda_M)$ *by* $\mathrm{Pois}(M2^{-L})$ *in the limit. In particular $\mathbb{P}(Z_M = 0) = e^{-\Lambda_M} + o_C(1)$: the probability that no interior constant stretch of length $\ge L$ starts in $[1, M]$ follows the Poisson law of parameter Λ_M. The border window $x = 1$ is not included; on the other hand Z_M counts all starts $x < M$, including those whose window V_x overflows beyond M, the function f being defined on all integers.*

Proof. Fix $j_0 \ge 2$ — so that $\kappa = M/N \in [2^{j_0-1}, 2^{j_0}] \subset [2, 2^{j_0}]$ stays in the range of proposition 16.1 —, set $N = \lceil M2^{-j_0}\rceil$, and $U = U_{N,\kappa} = [N, M) \cap \mathbb{Z}$; proposition 16.1 applies with $\kappa_0 = 2^{j_0}$. Set $Z_U = \sum_{x\in U} J_{x,L}$, $\lambda_U = \mathbb{E}Z_U$. Three estimates, at fixed j_0 and $M \to \infty$.

(1) Truncation. By proposition 15.5,

$$d_{\mathrm{TV}}(\mathcal{L}(Z_M), \mathcal{L}(Z_U)) \le \mathbb{P}(Z_M \ne Z_U) \le \sum_{2\le x<M2^{-j_0}} \mathbb{P}(J_{x,L} = 1) = O_C(2^{-j_0}) + o_M(1),$$

and the same sum bounds $|\Lambda_M - \lambda_U|$.

(2) Chen–Stein on U. Section 13 consumes the scale only through four inputs: the cardinality of the range, the exact marginals of Lemma 2.1, the absolute homogeneous sum R_2, and the defect masses and kernel counts at height $O(N)$. On U, in the window $|L - \log_2 N| \le C + j_0 + O(1)$, these inputs become: $|U| < M \asymp_C 2^L$; the same marginals (local to the windows); $R_{2,\kappa} = o_{C,j_0}(M^2\, 2^{-2j_0}) = o_{C,j_0}(2^{2L})$ by proposition 16.1; and the defect masses summed over the j_0 dyadic sub-blocks of U, each $N^{1/2+o(1)}$ by proposition 3.2 applied at its own scale. The edge count of the dependency graph uses only the interval structure of U ($B(N/p + 1)$ becomes $O_{j_0}(B(N/p+1))$). All the bounds of section 13 therefore hold on U with constants $O_{C,j_0}(1)$, and $d_{\mathrm{TV}}(\mathcal{L}(Z_U), \mathrm{Pois}(\lambda_U)) = o_{C,j_0}(1)$. Moreover, the exact marginals and the same defect masses give $\lambda_U = |U|2^{-L} + o_{C,j_0}(1) = (1 - 2^{-j_0} + o_{C,j_0}(1))M2^{-L}$.

(3) Recentring. $d_{\mathrm{TV}}(\mathrm{Pois}(\lambda_U), \mathrm{Pois}(\Lambda_M)) \le |\Lambda_M - \lambda_U| = O_C(2^{-j_0}) + o_M(1)$.

By the triangle inequality, $\limsup_M d_{\mathrm{TV}}(\mathcal{L}(Z_M), \mathrm{Pois}(\Lambda_M)) \le O_C(2^{-j_0})$ for every j_0, whence the convergence. Finally, by (1) and the above asymptotic for λ_U, $\Lambda_M = (1 - 2^{-j_0})M2^{-L} + O_C(2^{-j_0}) + o_{C,j_0}(1)$, and since $M2^{-L} \asymp_C 1$, letting $j_0 \to \infty$ last gives $\Lambda_M = (1 + o_C(1))M2^{-L}$. Finally $|\Lambda_M - M2^{-L}| = o_C(1)$ and the coupling of Poisson laws allow the replacement of the parameter. $\square$

Remark 16.3 (No global rate)**.** The total-variation rate of Theorem 1.1 does not propagate: the proof above is a double limit, and the uniformity of proposition 16.1 is local in κ. A global rate would require an explicit choice $j_0 = j_0(M) \to \infty$, hence a uniformity in growing κ, which we have not established.

Corollary 16.4 (Longest-run law in the finite prefix)**.** *Set*

$$W_{M,L} = \mathbf{1}_{\{f(1)=\cdots=f(L)\}} + \sum_{2\le x\le M-L+1} J_{x,L},$$

and let R_M be the length of the longest constant stretch of the finite sequence $(f(1), \dots, f(M))$. Uniformly for $|L - \log_2 M| \le C$,

$$d_{\mathrm{TV}}(\mathcal{L}(W_{M,L}), \mathrm{Pois}(M2^{-L})) \longrightarrow 0 \qquad (M \to \infty),$$

and in particular

$$\boxed{\mathbb{P}(R_M < L) = \exp(-M2^{-L}) + o_C(1).}$$

Proof. The event $\{W_{M,L} = 0\}$ is exactly $\{R_M < L\}$. A maximal constant stretch of the prefix, of length $\ell \ge L$ and start x, satisfies $x \le M - L + 1$; if $x = 1$ it realizes the border indicator, and if $x \ge 2$ maximality within the prefix gives $f(x-1) = -f(x)$, hence $J_{x,L} = 1$. Conversely each term of $W_{M,L}$ produces a stretch of length at least L contained in the prefix.

Coupling with Z_M. On the same space, $W_{M,L} \ne Z_M$ requires the border indicator or an overflowing start $M - L + 2 \le x < M$. Since $f(1) = 1$, the border event is equivalent to $f(p) = 1$ for every prime $p \le L$, of probability $2^{-\pi(L)} = o(1)$. For the overflowing starts, proposition 14.1, applied to the mask $A = \{M - L + 2, \dots, M - 1\}$ at scale $N' = M - L + 2$ — so that $A \subseteq I_{N'}$ and $|L - \log_2 N'| \le C + o(1)$ —, gives

$$\sum_{M-L+2 \le x < M} \mathbb{P}(J_{x,L} = 1) = O(L2^{-L}) + o_C(1) = o_C(1).$$

Thus $d_{\mathrm{TV}}(\mathcal{L}(W_{M,L}), \mathcal{L}(Z_M)) \le \mathbb{P}(W_{M,L} \ne Z_M) = o_C(1)$.

Conclusion. The recentred form of Theorem 16.2 and the triangle inequality give the convergence; the void law gives the boxed identity. □

Remark 16.5. Corollary 16.4 is the exact multiplicative analogue of the Erdős–Rényi law for the longest runs of an independent Bernoulli sequence [10, 14]: at the critical scale, the border $x = 1$ weighs only $2^{-\pi(L)} = o(1)$ and no border–interior independence is required, a coupling suffices. The discussion in the conclusion about the moderate tail, where the border dominates, is unchanged.

17 Appendix: stability at bounded ratio

Throughout this section, we fix

$$C > 0, \qquad \kappa_0 \ge 2, \qquad 2 \le \kappa \le \kappa_0, \qquad U = U_{N,\kappa} = [N, \kappa N) \cap \mathbb{Z}, \qquad B = L + 1,$$

and assume

$$|L - \log_2 N| \le C.$$

All pairs are ordered and separated, that is, $(x, y) \in U^2$ and $|x - y| > L$. The objects

$$\rho, \sigma, \tau, D^{\#}, c^{\#}, P^{\#}, \mathcal{C}_{\mathrm{res}}, \mathcal{Q}_{\mathrm{res}}, \mathcal{R}_{\mathrm{res}}, \mathcal{T}_K$$

are those of Sections 4 to 11, with I_N replaced by U.

Set once and for all

$$c_{\kappa_0} = \kappa_0 + 2, \qquad j_{\kappa_0} = \lceil \log_2 \kappa_0 \rceil.$$

For N large enough in terms of C and κ_0, every occurrence belonging to a window issued from a start of U belongs to

$$[N/2, c_{\kappa_0} N].$$

17.1 Table of quantifiers

The order of fixing is

$$(C,\kappa_0) \longrightarrow (A,\varepsilon_0,\alpha) \longrightarrow (K_\alpha, C_\alpha) \longrightarrow C_{\mathrm{comp}} \longrightarrow C_{\mathrm{term}} \longrightarrow K \longrightarrow N_0.$$

It is made precise in the following table.

Rank	Fixed parameters	Allowed dependencies	Role
1	C, κ_0	no dependence on N, L or κ	critical window and uniform bound $2 \le \kappa \le \kappa_0$
2	$A = 3$, $\varepsilon_0 = 1/16$, $\alpha = 3/16$	absolute constants	alignment and core threshold
3	K_α, C_α	dependence on α only	bounded component size and aligned argument
4	C_{comp}, together with c_{κ_0}, $C_{\mathrm{crt}}(\kappa_0)$ and $C_{\det}(\kappa_0)$	dependence at most on C,κ_0,K_α	CRT counts, polynomial ranges and determinants
5	C_{term}	dependence at most on $C,\kappa_0,A,K_\alpha,C_{\mathrm{comp}}$	exponential count of terminal hosts
6	K	dependence on all previous parameters, with $$K > \frac{2C_{\mathrm{term}}}{\log 2}$$	uniform saving in the non-terminal branch
7	N_0	dependence on all previous parameters	simultaneous validity of all asymptotic estimates

Once $C, A, \varepsilon_0, \alpha, K_\alpha, C_\alpha$ and C_{comp} are fixed, we abbreviate

$$C_{\mathrm{term}} = C_{\mathrm{term}}(C,\kappa_0,A,K_\alpha,C_{\mathrm{comp}}) = C_{\mathrm{term}}(\kappa_0).$$

The notation $O_{\kappa_0}(X)$ denotes a bound uniform for $2 \le \kappa \le \kappa_0$, whose constant depends neither on N nor on L. A quantity $r_N = o_{C,\kappa_0}(1)$ satisfies

$$\sup_{\substack{2\le\kappa\le\kappa_0\\ |L-\log_2 N|\le C}} |r_N| \longrightarrow 0.$$

The notations $N^{o_{C,\kappa_0}(1)}$ and $\exp(o_{C,\kappa_0}(B))$ have the corresponding uniform meaning. When a local parameter d or K_0 has not yet been fixed, it remains indicated in the subscript.

17.2 List of replacements

The transports below use exclusively the following replacements.

R1.
$$I_N \rightsquigarrow U, \qquad |U| \le \kappa_0 N, \qquad |U|^2 = O_{\kappa_0}(N^2).$$

R2. Every height bound

$$n \le 3N, \qquad X,Y \in [N/2, 3N]$$

is replaced by

$$n \le c_{\kappa_0} N, \qquad X,Y \in [N/2, c_{\kappa_0} N].$$

R3. If $1 \le r \le c_{\kappa_0} N$, a class modulo r has in U at most

$$\frac{(\kappa-1)N}{r} + 1 \le (\kappa_0 - 1 + c_{\kappa_0})\frac{N}{r} = O_{\kappa_0}\left(\frac{N}{r}\right)$$

representatives. This is the replacement of $N/r+1$ by $O_{\kappa_0}(N/r)$.

R4. For a channel (a,b,h) having a unit (i_0,j_0), with $q = \max(a,b)$,

$$x = bt - i_0, \qquad y = at - j_0,$$

and the admissible values of t belong to

$$\left[\frac{N+i_0}{b}, \frac{\kappa N + i_0}{b}\right) \cap \left[\frac{N+j_0}{a}, \frac{\kappa N + j_0}{a}\right).$$

This interval contains at most

$$\frac{(\kappa_0 - 1)N}{q} + 1 \ll_{\kappa_0} \frac{N}{q}$$

integers as soon as $m(a,b,h) \ge 2$, since then $q \le L < N$.

R5. In Lemma 7.1, the intervals of starts or of parameters have length at most $C_{\mathrm{crt}}(\kappa_0)N$. If $P \le N$, a class modulo P has $O_{\kappa_0}(N/P)$ representatives there. Thus the geometric constant C_0 of that lemma becomes $C_{\mathrm{crt}}(\kappa_0) = O_{\kappa_0}(1)$.

R6. Every raw count over I_N^2 is replaced by $O_{\kappa_0}(N^2)$; every free choice of a start is replaced by $O_{\kappa_0}(N)$.

R7. The Diophantine ranges

$$e \le N^{O(1)}, \qquad |z|, |w| \le N^{O(1)}$$

become

$$e \le N^{O_{\kappa_0}(1)}, \qquad |z|, |w| \le N^{O_{\kappa_0}(1)}.$$

The exponents remain fixed before N.

R8. In the terminal sector,

$$|\Delta_{s,t}| \le C_{\det}(\kappa_0)NB, \qquad C_{\det}(\kappa_0) = 2c_{\kappa_0},$$

et

$$T = \left(C_{\det}(\kappa_0)NB\right)^{1/2}.$$

R9. In the global application, if $\kappa_0 = 2^{j_0}$ and $N = \lceil M 2^{-j_0} \rceil$, then

$$|L - \log_2 M| \le C \quad \Longrightarrow \quad |L - \log_2 N| \le C + j_0 + O(1).$$

More generally, one may replace C by the fixed number

$$C^\flat = C + \lceil \log_2 \kappa_0 \rceil + 2.$$

We keep

$$2^B \asymp_{C,\kappa_0} N.$$

R10. The range U is covered by the at most $j_{\kappa_0} + 1$ sub-blocks

$$[2^r N, 2^{r+1} N), \qquad 0 \le r \le j_{\kappa_0}.$$

At scale $N_r = 2^r N$, the windows are contained in $[N_r/2, 3N_r]$ and

$$\frac{B}{\log N_r} = \frac{1}{\log 2} + O_{C,\kappa_0}\left(\frac{1}{\log N}\right).$$

17.3 Hosts, interpolation and rational code

Lemma 17.1 (Hosts at bounded ratio)**.** *Let $H_{2,U}(N,L)$ be the number of separated ordered pairs $(x,y) \in U^2$ with $\rho(x,y) > 0$. Then*

$$H_{2,U}(N,L) \le N^{3/2+o_{C,\kappa_0}(1)}.$$

Proof. The proof of Lemma 4.2 applies after R1–R3: the sum ranges over $n \le c_{\kappa_0} N$ and the factor $N/\mathcal{K}_B(n) + 1$ becomes $O_{\kappa_0}(N/\mathcal{K}_B(n))$. The same Euler product is $N^{o_{C,\kappa_0}(1)}$. □

Lemma 17.2 (Interpolation at bounded ratio)**.** *Let $\mathcal{A} \subseteq U^2$ be a population of separated pairs carrying at least one relation. For every $u(x,y) \ge 0$,*

$$\sum_{\mathcal{A}} u(x,y) \le H_{2,U}(N,L)^{1/2} \left(\sum_{\mathcal{A}} u(x,y)^2\right)^{1/2}.$$

In particular,

$$\mathcal{R}_{\mathrm{res}}(\mathcal{A}) \le N^{3/4+o_{C,\kappa_0}(1)} \mathcal{Q}_{\mathrm{res}}(\mathcal{A})^{1/2}.$$

Proof. This is Cauchy–Schwarz, Lemma 17.1 and $[2^\sigma(2^\tau - 1)]^2 \le 4^\sigma(4^\tau - 1)$. □

Lemma 17.3 (Rational code at bounded ratio)**.** *For a separated pair $(x,y) \in U^2$, $(a,b) = 1$ and $h = by - ax$, set*

$$M_{a,b}(h) = \{(i,j) \in \mathcal{I}_L^2 : ai - bj = h\}, \qquad m = |M_{a,b}(h)|,$$

the even subsets of the units form a subspace of $\mathcal{R}(x,y)$ of dimension $m-1$ when $m \ge 1$. If $m \ge 2$, then

$$q := \max(a,b) \le L.$$

Proof. The proof of Lemma 5.1 is local to the offsets: two distinct units differ by (b,a), whence $a, b \le L$; no metric replacement intervenes. □

Lemma 17.4 (Channel uniqueness at bounded ratio)**.** *For $N \ge N_0(A, C, \kappa_0)$, a pair $(x,y) \in U^2$ has at most one reduced fraction (a,b), with $a, b \le B^A$, satisfying*

$$|by - ax| < (a+b)B.$$

Every channel whose rational code is nonzero is thereby covered.

Proof. The proof of Lemma 5.2 uses only $x \ge N$: two admissible fractions would give $|(a'b - ab')x| < 4B^{2A+1}$, hence $a'b - ab' = 0$ for N large enough. □

Lemma 17.5 (Rational masses at bounded ratio)**.** *The sums range over separated pairs of U^2 whose canonical channel, with $m \ge 2$ exact units, has the indicated height. The contribution of heights $q \ge 3$ satisfies*

$$\sum_{q \ge 3} (2^\sigma - 1) \le N^{4/3+o_{C,\kappa_0}(1)}, \qquad \sum_{q \ge 3} (4^\sigma - 1) \le N^{5/3+o_{C,\kappa_0}(1)}.$$

The height-$q = 2$ contribution satisfies

$$\sum_{q=2}(2^{\sigma} - 1) \leq N^{3/2+o_{C,\kappa_0}(1)}, \qquad \sum_{q=2}(4^{\sigma} - 1) \leq N^{2+o_{C,\kappa_0}(1)}.$$

Consequently,

$$\sum_{\substack{x,y\in U \\ |x-y|>L}} (2^{\sigma(x,y)} - 1) \leq N^{3/2+o_{C,\kappa_0}(1)}$$

and

$$\sum_{\substack{x,y\in U \\ |x-y|>L}} (4^{\sigma(x,y)} - 1) \leq N^{2+o_{C,\kappa_0}(1)}.$$

Proof. For $q \geq 3$, the proof of proposition 5.4 applies after R1 and R4; the contributions $N^{1+o(1)}q^2 2^{B/q}$ and $N^{1+o(1)}q^2 4^{B/q}$ are still dominated by $q = 3$. The case $q = 2$ is exactly the volumetric computation preceding (16.1); the case $q = 1$ forces $|x - y| \leq L$. □

Lemma 17.6 (Sum of channel geometries)**.** *With the intrinsic domain of Lemma 5.5,*

$$\sum_{\substack{(a,b)=1,\ \max(a,b)\geq 2,\ h\in\mathbb{Z} \\ m(a,b,h)\geq 2}} 4^{\sigma(a,b,h)} \leq N^{1+o_{C,\kappa_0}(1)}.$$

Proof. The proof of Lemma 5.5 does not sum the translation parameter t; it uses only $q \leq L$, the $O(qB)$ values of h and $4^{\sigma} \leq 4^{B/q}$. □

17.4 Residual quotient

Lemma 17.7 (Large-prime graph on U)**.** *For every separated pair $(x, y) \in U^2$,*

$$W_{>B}(x, y) \simeq \mathbb{F}_2^D \oplus \mathbb{F}_2^c, \qquad \dim W_{>B} = D + c, \qquad D + 2c \leq 2B.$$

Proof. The proof of Lemma 6.1 is local to the two windows, whose diameter is $B - 1 < p$ for every $p > B$. □

Lemma 17.8 (Square class of components on U)**.** *Every nontrivial unpinned component $\mathcal{C}$ meets both blocks and has unique integers $d_{\mathcal{C}}, z_{\mathcal{C}} > 0$ such that*

$$\prod_{n\in\mathcal{C}} n = d_{\mathcal{C}} z_{\mathcal{C}}^2,$$

where $d_{\mathcal{C}}$ is squarefree and B-smooth.

Proof. The proof of Lemma 6.2 uses only the uniqueness, in each window, of an occurrence of odd valuation for a prime $p > B$. □

Lemma 17.9 (Exact isolation on U)**.** *Consider an exact unit*

$$x + i = bt, \qquad y + j = at, \qquad (a, b) = 1, \qquad \max(a, b) < B.$$

If $\mathcal{K}_B(t) = 1$, its two occurrences are defective and distinct; if $\mathcal{K}_B(t) > 1$, they form an isolated unpinned component of size two. Two distinct units give distinct objects.

Proof. The proof of Lemma 6.3 is unchanged, since $p > B > \max(a, b)$ and two distinct multiples of p cannot belong to the same window. □

Lemma 17.10 (Quotient core on U)**.** *We have*

$$\tau(x,y) \le D^{\#}(x,y) + c^{\#}(x,y), \qquad D^{\#} = O_{C,\kappa_0}\left(\frac{B}{\log B}\right), \qquad D^{\#} + 2c^{\#} \le 2B.$$

Proof. The dimension arguments of Lemma 6.4 are given by Lemmas 17.7 and 17.9. The bound on $D^{\#}$ comes from proposition 3.2 applied according to R10. □

Lemma 17.11 (Residual certificates on U)**.** *The $c^{\#}$ residual components canonically determine triples*

$$(i_s, j_s, p_s), \qquad 1 \le s \le c^{\#},$$

whose offsets are distinct in each block, whose primes $p_s > B$ are distinct, and

$$p_s \mid x + i_s, \qquad p_s \mid y + j_s.$$

If the channel contains at least two units, then $h + bj_s - ai_s \ne 0$; if the channel contains one unit, only the at most two components meeting its occurrences may be exceptions.

Proof. The canonical selection of Lemma 6.5 depends only on the local graph and on Lemma 17.9. □

17.5 Moderate sectors

Lemma 17.12 (CRT summation on U)**.** *Under the hypotheses of Lemma 7.1, if the product P of the primes of the certificate satisfies $P \le N$, the weighted sum in one coordinate is*

$$\ll_{\kappa_0} N \sum_{r\ge 0} \frac{1}{r!}\left(u \sum_{p\in\mathcal{P}} \frac{E_p}{p}\right)^r,$$

and the sum in two coordinates is

$$\ll_{\kappa_0} N^2 \sum_{r\ge 0} \frac{1}{r!}\left(uM \sum_{p\in\mathcal{P}} \frac{1}{p^2}\right)^r.$$

Proof. The Chinese remainder theorem determines one class modulo P in each coordinate; R3–R5 replace the counts $N/P + 1$ and N^2/P^2 by their versions $O_{\kappa_0}(N/P)$ and $O_{\kappa_0}(N^2/P^2)$. □

Lemma 17.13 (Cells of a channel on U)**.** *For a channel (a,b,h) with $m(a,b,h) \ge 2$ and $q = \max(a,b)$,*

$$E_p(a,b,h) \ll \left(1 + \frac{qB}{p}\right)\left(1 + \frac{B}{q}\right),$$

et

$$\sum_{p>B} \frac{E_p(a,b,h)}{p} \ll \frac{B}{\log B},$$

uniformly in (a,b,h) and in $2 \le \kappa \le \kappa_0$.

Proof. The proof of Lemma 7.2 concerns only $\Delta_{i,j} = h + bj - ai$, the offsets and the primes $B < p \ll qB$; no bound on x or y intervenes. □

Lemma 17.14 (Sector $P^{\#} \le N$ on U)**.** *On the population $P^{\#} \le N$,*

$$\mathcal{Q}^{(1)}_{\mathrm{res}} \le N^{2+o_{C,\kappa_0}(1)}, \qquad \mathcal{R}^{(1)}_{\mathrm{res}} \le N^{7/4+o_{C,\kappa_0}(1)}.$$

Proof. In the branch $\sigma = 0$, the proof of proposition 7.3 uses R5 and Lemma 17.12; in the branch $\sigma > 0$, it uses R4 and Lemmas 17.6, 17.12, and 17.13. Lemma 17.2 gives the second bound. $\square$

Lemma 17.15 (Small channels on U)**.** *On the population $P^{\#} > N$ having a canonical channel of height*

$$q \leq \sqrt{\log B},$$

we have

$$\mathcal{Q}^{(2)}_{\text{res}} \leq N^{2+o_{C,\kappa_0}(1)}, \qquad \mathcal{R}^{(2)}_{\text{res}} \leq N^{7/4+o_{C,\kappa_0}(1)}.$$

Proof. The proof of proposition 7.4 still gives $4^{\tau} = N^{o_{C,\kappa_0}(1)}$; for $\sigma \geq 1$ we use the quadratic mass of Lemma 17.5, and for $\sigma = 0$ Lemma 17.1. Lemma 17.2 gives the stated linear mass. $\square$

Lemma 17.16 (Shallow cores on U)**.** *On the remaining population with*

$$c^{\#} \leq \alpha B, \qquad \alpha = \frac{3}{16},$$

we have

$$\mathcal{Q}^{(3)}_{\text{res}} \leq N^{19/8+o_{C,\kappa_0}(1)}, \qquad \mathcal{R}^{(3)}_{\text{res}} \leq N^{31/16+o_{C,\kappa_0}(1)}.$$

Proof. The pointwise bound of proposition 7.5,

$$4^{\sigma}(4^{\tau} - 1) \leq N^{1/2-2\varepsilon_0+o_{C,\kappa_0}(1)},$$

is local to the windows; R6 replaces the number of pairs by $O_{\kappa_0}(N^2)$. Lemma 17.2 concludes. $\square$

17.6 Deep cores

Lemma 17.17 (Aligned exclusion on U)**.** *Fix $\alpha > 0$. For $N \geq N_0(\alpha, A, C, \kappa_0)$, no aligned pair of U^2 satisfies*

$$c^{\#} \geq \alpha B.$$

Proof. The proof of Theorem 8.1 uses $x \geq N$, $a, b \leq B^A$, $|by - ax| < (a+b)B$ and objects local to the windows. The Runge bound remains $\exp(o_{\alpha}(B))$, while $\log N = (\log 2)B + O_C(1)$. $\square$

Lemma 17.18 (Evertse–Silverman input on U)**.** *Fix $d \geq 3$. For distinct integers $h_1, \ldots, h_d$ and a nonzero squarefree integer e, the number of integer pairs (X, Y) satisfying*

$$\prod_{r=1}^{d} (X + h_r) = eY^2$$

is at most

$$\exp\left(O_d\left(1 + \omega(e) + \omega\left(\prod_{r<s}(h_r - h_s)\right)\right)\right).$$

When $|h_r| \leq B$ and $|e| \leq N^{O_{\kappa_0,d}(1)}$, this number is $N^{o_{C,\kappa_0,d}(1)}$.

Proof. The statement of Lemma 9.1 is independent of the interval of starts; R7 keeps $\omega(e) = O_{\kappa_0,d}(\log N/\log\log N)$. $\square$

Lemma 17.19 (Pell in the ranges issued from U)**.** *Fix $K_0 > 0$. If A, C are positive squarefree integers, $A/C \notin \mathbb{Q}^{\times 2}$, if e is a nonzero integer, and if*

$$A, C, |e| \leq N^{K_0},$$

then the number of integer solutions (z,w) *with* $|z|,|w| \le N^{K_0}$ *of*

$$Az^2 - Cw^2 = e$$

est

$$\exp\Big(O_{K_0}\Big(\frac{\log N}{\log\log N}\Big)\Big).$$

In particular, the same conclusion holds when K_0 *is a fixed constant depending on* κ_0*.*

Proof. This is Lemma 9.2; replacement R7 only modifies the fixed value of K_0. □

Lemma 17.20 (Normalization of a component on U)**.** *Let* I, J *be nonempty,* $|I| + |J| \le K_0$*, and* d *positive, squarefree and* B*-smooth. From*

$$P_I(x)Q_J(y) = dz^2$$

we deduce, after fixing x, I, J, d*,*

$$Q_J(y) = ev^2,$$

where e *is positive, squarefree, and*

$$e \le N^{O_{C,\kappa_0,K_0}(1)}.$$

Proof. As in Lemma 9.3, e is the squarefree part of $dP_I(x)$; R2 and R7 give the stated height. □

Lemma 17.21 (One-sided count on U)**.** *Under the hypotheses of Lemma 17.20, the number of admissible* $y \in U$ *is*

$$\begin{cases} O_{\kappa_0}(N^{1/2}), & |J| = 1,\\ N^{o_{C,\kappa_0,K_0}(1)}, & |J| \ge 2.\end{cases}$$

The same assertion holds after exchanging the two blocks.

Proof. For $|J| = 1$, R2 gives $O_{\kappa_0}(\sqrt{N/e} + 1)$ possibilities. For $|J| = 2$ and $e \ge 2$ we apply Lemma 17.19; for $|J| = 2$ and $e = 1$, the ratio $A/C = 1$ is a square and we follow the factorization $(U - V)(U + V) = \Delta^2$ of the proof of Lemma 9.4, which consumes only divisor counts at height $c_{\kappa_0}N$. For $|J| \ge 3$, Lemma 17.18 concludes. □

Lemma 17.22 (Two singletons on U)**.** *For two fixed offsets* i, j*, the number of triples* (x, y, d) *with* $x, y \in U$*,* d *positive, squarefree and* B*-smooth, and*

$$(x + i)(y + j) = dz^2$$

is at most

$$C_{\kappa_0} N \log N \prod_{p \le B}\Big(1 + \frac{2}{\sqrt{p}}\Big) = N \exp\Big(O_{\kappa_0}\Big(\frac{\sqrt{B}}{\log B}\Big)\Big).$$

Proof. The parametrization $X = ecu^2$, $Y = (d/e)cv^2$ of Lemma 9.5 remains unique; R2 replaces $3N$ by $c_{\kappa_0}N$. Summing over $e \mid d$, then over d, gives the displayed Euler product. □

Lemma 17.23 (Hosts of a bounded component on U)**.** *Fix* K_0*. The number of separated pairs of* U^2 *whose non-aligned residual graph contains a component meeting both blocks and of size at most* K_0 *is*

$$N^{1+o_{C,\kappa_0,K_0}(1)}.$$

If a component of size two is present, the explicit recount gives

$$\#\{hosts\} \le C_{\kappa_0} N B^2 \log N \prod_{p \le B} \left(1 + \frac{2}{\sqrt{p}}\right). \tag{17.1}$$

We fix $C_{\mathrm{term}}(\kappa_0)$ *so that, for* $N \ge N_0$*,*

$$C_{\kappa_0} B^2 \log N \prod_{p \le B} \left(1 + \frac{2}{\sqrt{p}}\right) \le \exp\left(C_{\mathrm{term}}(\kappa_0) \frac{\sqrt{B}}{\log B}\right). \tag{17.2}$$

Thus

$$\#\{hosts\ with\ a\ size\text{-}two\ component\} \le N \exp\left(C_{\mathrm{term}}(\kappa_0) \frac{\sqrt{B}}{\log B}\right).$$

Proof. The proof of Lemma 9.6 applies after R2, R6 and Lemmas 17.21 and 17.22; the offset choices cost $B^{O_{\kappa_0}(1)}$. For a size-two component, the union over the B^2 offset pairs gives (17.1), then $\log(B^2 \log N) = o(\sqrt{B}/\log B)$ gives (17.2). □

Lemma 17.24 (Deep-core hosts on U)**.** *For every* $\alpha > 0$*, the number of non-aligned pairs with* $c^\# \ge \alpha B$ *is*

$$N^{1+o_{C,\kappa_0,\alpha}(1)}.$$

More explicitly, a component of size at most $\lceil 2/\alpha \rceil \le K_\alpha$ *exists and all occurrences* $x+i$*,* $y+j$ *of its equation belong to* $[N/2, c_{\kappa_0} N]$*.*

Proof. The components use at most $2B$ occurrences, so one of them has size at most $2/\alpha$; Lemma 17.23 then gives the stated count. The free choice of a first start costs $O_{\kappa_0}(N)$, the offsets $B^{O(K_\alpha)}$, and the partner $N^{o_{C,\kappa_0,K_\alpha}(1)}$. □

Lemma 17.25 (Two defects in a block of U)**.** *The number of starts* $x \in U$ *whose window contains two distinct* B*-defective vertices is*

$$N^{o_{C,\kappa_0}(1)}.$$

Proof. The proof of Lemma 9.8 separates $d_1 = d_2$, handled by divisor factorization at height $c_{\kappa_0} N$, from $d_1 \ne d_2$, which leads to a Pell equation in the range R2; Lemma 17.19 applies, and the $4^{\pi(B)} B^2$ auxiliary choices are $N^{o_C(1)}$. □

Lemma 17.26 (Eliminating multiple defects on U)**.** *The linear mass of the non-aligned deep-core pairs with* $D^\# \ge 3$ *is*

$$N^{3/2+o_{C,\kappa_0}(1)}.$$

Proof. As in the proof of proposition 9.9, $c^\# > \alpha B$ provides a component of size at most $\lceil 2/\alpha \rceil$, and the sums over its offsets and its coefficient d cost $B^{O_\alpha(1)} = N^{o_C(1)}$. Lemma 17.25 leaves $N^{o_{C,\kappa_0}(1)}$ choices for one of the starts, then Lemma 17.21 leaves at most $O_{\kappa_0}(N^{1/2})$ choices for the other. Lemma 17.10 gives $2^\tau \le N^{1+o_{C,\kappa_0}(1)}$. □

Lemma 17.27 (Exact rank on U)**.** *In the non-aligned core,*

$$\tau = D^\# + c^\# - \widetilde{k}.$$

Proof. The proof of Lemma 9.10 is a dimension identity over $\mathbb{F}_2$ and depends on no metric bound. □

Lemma 17.28 (Uniform terminal reduction)**.** *Fix*

$$K > \frac{2C_{\mathrm{term}}(\kappa_0)}{\log 2}.$$

The mass of the non-aligned deep core with $D^\# \le 2$ — the pairs with $D^\# \ge 3$ being eliminated by Lemma 17.26 — is $o_{C,\kappa_0}(N^2)$ outside the set $\mathcal{T}_K$ of non-aligned deep-core pairs of U^2 satisfying

$$D^\# \le 2, \qquad c^\# = B - s, \qquad s + \widetilde{k} \le K\frac{\sqrt{B}}{\log B},$$

and having at least

$$M \ge B - 3s$$

isolated components of size two, as in proposition 9.11.

Proof. Lemma 17.24 handles $c^\# \le 2B/3$ with a mass $N^{5/3+o_{C,\kappa_0}(1)}$; in the branch $c^\# = B - s > 2B/3$, Lemma 17.23 and Lemma 17.27 give

$$\mathcal{R}_{\mathrm{res}} \ll_{C,\kappa_0} N^2 \exp\left((C_{\mathrm{term}}(\kappa_0) - K\log 2)\, \frac{\sqrt{B}}{\log B} \right) = o_{C,\kappa_0}(N^2)$$

when $s + \widetilde{k} > K\sqrt{B}/\log B$. In the complement, $\sum_t (|\mathcal{C}_t| - 2) \le 2s$, so at least $B - 3s$ components have size two. □

Lemma 17.29 (Terminal determinants on U)**.** *For an isolated terminal component,*

$$R_t = \mathcal{K}_B(x + i_t) = \mathcal{K}_B(y + j_t) > 1,$$

and the kernels of distinct components are pairwise coprime. For $s \ne t$,

$$R_s R_t \mid \Delta_{s,t}, \qquad 0 < |\Delta_{s,t}| \le C_{\det}(\kappa_0) NB, \qquad C_{\det}(\kappa_0) = 2c_{\kappa_0}.$$

Proof. The kernel equality, their coprimality, the divisibility and the non-vanishing are those of Lemma 9.12. By R2,

$$|\Delta_{s,t}| \le |x + i_s|\,|j_t - j_s| + |y + j_s|\,|i_s - i_t| \le 2c_{\kappa_0} NB.$$

□

Lemma 17.30 (Terminal closure on U)**.** *With*

$$T = \left(C_{\det}(\kappa_0) NB\right)^{1/2},$$

the terminal population satisfies

$$\#\mathcal{T}_K \le N^{3/4 + o_{C,\kappa_0}(1)}$$

et

$$\sum_{(x,y)\in\mathcal{T}_K} (2^{\tau(x,y)} - 1) \le N^{7/4 + o_{C,\kappa_0}(1)}.$$

Proof. Lemma 17.29 shows that all kernels but at most one are below T, and

$$\#\{n \le c_{\kappa_0} N : \mathcal{K}_B(n) \le T\} \ll_{\kappa_0} \sqrt{NT} \prod_{p \le B} (1 + p^{-1/2}) = N^{3/4 + o_{C,\kappa_0}(1)}.$$

The incidence count gives the same bound for the first starts, then Lemma 17.19 leaves $N^{o_{C,\kappa_0}(1)}$ partners per start; finally $\tau \le B+2$ gives $2^\tau = O_C(N)$. □

17.7 Partition and assembly

Lemma 17.31 (Partition at bounded ratio)**.** *For every separated pair of U^2,*

$$2^\rho - 1 = (2^\sigma - 1) + 2^\sigma(2^\tau - 1).$$

After extraction of the first term, the successive tests
(1) $P^\# \le N$;
(2) $P^\# > N$ and $q \le \sqrt{\log B}$;
(3) $P^\# > N$, no small channel, and $c^\# \le \alpha B$;
(4) $P^\# > N$, $c^\# > \alpha B$, and aligned pair;
(5) deep non-aligned pair with $D^\# \ge 3$;
(6) deep non-aligned pair, $D^\# \le 2$, outside $\mathcal{T}_K$;
(7) pair belonging to $\mathcal{T}_K$
define an exhaustive disjoint partition. The uniform bounds are

sector	$\mathcal{Q}_{\mathrm{res}}$	$\mathcal{R}_{\mathrm{res}}$
(1)	$N^{2+o_{C,\kappa_0}(1)}$	$N^{7/4+o_{C,\kappa_0}(1)}$
(2)	$N^{2+o_{C,\kappa_0}(1)}$	$N^{7/4+o_{C,\kappa_0}(1)}$
(3)	$N^{19/8+o_{C,\kappa_0}(1)}$	$N^{31/16+o_{C,\kappa_0}(1)}$
(4)	0	0
(5)	*unused*	$N^{3/2+o_{C,\kappa_0}(1)}$
(6)	*unused*	$o_{C,\kappa_0}(N^2)$
(7)	*unused*	$N^{7/4+o_{C,\kappa_0}(1)}$.

Proof. The weight identity is algebraic and the partition is that of Lemma 11.1. The seven rows are respectively Lemmas 17.14 to 17.17, 17.26, 17.28, and 17.30. □

Lemma 17.32 (Uniform assembly)**.** *We have*

$$\sum_{\substack{x,y\in U\\ |x-y|>L}} (2^{\rho(x,y)} - 1) = o_{C,\kappa_0}(N^2).$$

Proof. The systematic term is $N^{3/2+o_{C,\kappa_0}(1)}$ by Lemma 17.5. Lemma 17.31 gives for each of the seven sectors a residual mass $o_{C,\kappa_0}(N^2)$. □

17.8 Chen–Stein inputs on U

We keep

$$Y = \lfloor B^2 \log B \rfloor, \qquad m_T(x) = \#\{n \in U_x : \mathcal{K}_T(n) = 1\},$$

and set

$$\mathcal{M}_{B,U} = \sum_{x\in U}(2^{m_B(x)} - 1), \qquad \mathcal{D}_{Y,U} = \{x \in U : m_Y(x) \ge 1\}.$$

Lemma 17.33 (Pointwise defects and sub-blocks)**.** *Uniformly for $x \in U$,*

$$m_B(x) = O_{C,\kappa_0}\left(\frac{B}{\log B}\right),$$

et

$$\mathcal{M}_{B,U} \le N^{1/2+o_{C,\kappa_0}(1)}.$$

Proof. We apply proposition 3.2 to the at most $j_{\kappa_0}+1$ sub-blocks of R10; this is the cutting of the paragraph *Pointwise defects* in the proof of proposition 16.1. The sum of an $O_{\kappa_0}(1)$ number of bounds $(2^r N)^{1/2+o_{C,\kappa_0}(1)}$ gives the result. □

Lemma 17.34 (Y-defective starts on U)**.** *We have*

$$|\mathcal{D}_{Y,U}| \le N^{1/2+o_{C,\kappa_0}(1)}$$

et

$$\sum_{x\in\mathcal{D}_{Y,U}} \mathbb{P}(J_{x,L}=1) = o_{C,\kappa_0}(1).$$

Proof. After R2,

$$\#\{n \le c_{\kappa_0}N : \mathcal{K}_Y(n)=1\} \le \sqrt{c_{\kappa_0}N}\prod_{p\le Y}(1+p^{-1/2}) = N^{1/2+o_{C,\kappa_0}(1)},$$

and every integer belongs to $O(B)$ windows. The proof of Lemma 13.4, with Lemma 17.33, gives the second assertion. □

Lemma 17.35 (Marginals and conditional graph on U)**.** *Let*

$$\mathcal{F}_Y = \sigma(X_p : p \le Y).$$

For every $x \in U \setminus \mathcal{D}_{Y,U}$,

$$\mathbb{P}(J_{x,L}=1 \mid \mathcal{F}_Y) = 2^{-L}.$$

The graph joining two good starts when their prime supports above Y meet is, conditionally on $\mathcal{F}_Y$, an exact dependency graph. Its marginal parameter is the deterministic number

$$\lambda_U^{\text{good}} = |U \setminus \mathcal{D}_{Y,U}|\,2^{-L},$$

et

$$\left|\lambda_U^{\text{good}} - |U|2^{-L}\right| \le N^{-1/2+o_{C,\kappa_0}(1)}.$$

Proof. The proofs of Lemmas 13.1 and 13.5 and Corollary 13.2 are local to the windows and do not depend on their position. The last bound is Lemma 17.34 multiplied by $2^{-L} \asymp_C N^{-1}$. □

Lemma 17.36 (Graph edges on U)**.** *If $\mathcal{E}_{Y,U}$ denotes the number of joined ordered distinct pairs, then*

$$\mathcal{E}_{Y,U} \ll_{\kappa_0} \frac{N^2}{\log^2 B} + NB^2\log\log N + \frac{NB^2}{\log N} = o_{C,\kappa_0}(N^2).$$

Proof. For $p > Y$, the number of starts of U whose support meets a multiple of p is $O_{\kappa_0}(B(N/p+1))$; the proof of Lemma 13.6 therefore applies with the sum $p \le c_{\kappa_0}N$. The estimates of $\sum p^{-2}$, $\sum p^{-1}$ and $\pi(c_{\kappa_0}N)$ give the displayed bound. □

Lemma 17.37 (Chen–Stein terms on U)**.** *For the good starts of U,*

$$b_1 = o_{C,\kappa_0}(1), \qquad \mathbb{E}b_2 = o_{C,\kappa_0}(1).$$

Consequently,

$$d_{\mathrm{TV}}\left(\mathcal{L}\left(\sum_{x\in U} J_{x,L}\right), \mathrm{Pois}(\lambda_U^{\text{good}})\right) = o_{C,\kappa_0}(1).$$

Proof. Lemmas 17.35 and 17.36 give

$$b_1 \le (|U| + \mathcal{E}_{Y,U})2^{-2L} = o_{C,\kappa_0}(1),$$

while the touching pairs cost $O_{\kappa_0}(N)2^{-2L}$ and the separated pairs at most

$$2^{-2L}(\mathcal{E}_{Y,U} + R_{2,\kappa}(N,L)) = o_{C,\kappa_0}(1)$$

by Lemma 17.32. Theorem 13.7 and Lemma 17.34 conclude by coupling of the bad starts. □

18 Formal verification in Lean 4

Purpose and audit boundary. The formalization is intended to expose the logical architecture and trust boundary of the manuscript, not to replace mathematical exposition by the size of a machine-checked object. For each canonical endpoint, the audit records the literature interfaces actually consumed. The Lean kernel checks the deduction from those interfaces; it does not by itself establish that a cited source has been transcribed faithfully or that the source proves the imported statement. Validation of those interfaces remains part of external mathematical review.

Subject to the restrictions stated below, the internal deductions from the registered literature interfaces to the canonical endpoints corresponding to Sections 2 to 17 have been formalized in Lean 4, using toolchain `v4.32.2` and a manifest-pinned version of Mathlib. At the freeze of the present version, the self-contained development contains no `sorry` or added axiom. A deterministic audit regenerates one `#print axioms` report per public declaration and checks the hypothesis register, the generated audit file and the manifest; the standalone repository replays the build and audit in continuous integration. The versioned manifest, rather than the prose article, records the exact module and declaration counts.

Scope and hypotheses. The formalization is *conditional*: the Lean kernel verifies the deduction of the canonical endpoints from seven explicitly stated literature-facing propositions associated with published sources. They are imported as ordinary *hypotheses* — never as axioms — and, as such, are invisible to `#print axioms`. The kernel verifies their downstream use, not the implications from the cited sources to these propositions. The seven statements are recorded in a register that carries, for each, the source citation and the fingerprint of its Lean statement (the full register has thirteen entries, eight of external provenance: the eighth, a general form of the divisor bound, is *discharged* — replaced by a proof — so that only seven remain open; the five historical internal entries are all discharged and kept with their discharge theorems): the Chen–Stein theorem [1], the Evertse–Silverman theorem [13], the bounds of [4] and [23], the prime number theorem in the form $\pi(t)\log t/t \to 1$, where $\pi(t)$ is exactly the Lean count of primes at most t, [3, Introduction, p. 1], the conductor-fibre proposition associated with Halter–Koch [18, Theorem 1.1.6(1)(b), pp. 4–5; Definition 5.1.6 and Theorem 5.1.7(1),(3), pp. 118–119; Theorems 5.2.3(1),(2) and 5.2.5(1), pp. 125–127], and the Nicolas–Robin inequality (9.5) [27, Theorem 1, p. 485].

The last interface is exactly $\log\tau(n)\log\log n \le 2\log 2\log n$ for $n \ge 64$; the passage from the source's rounded maximum to the exact constant 2 is given in the proof of Lemma 9.2. For squarefree D, the cited Halter–Koch results identify $\mathbb{Z}[\sqrt{D}]$ with the order of discriminant $4D$: its conductor is 2 when $D \equiv 1 \pmod 4$ and 1 otherwise, and the further cited results provide the intended unit-index and fixed-principal-ideal-generator ingredients. The Lean proposition packages those ingredients as the ideal-divisor/four-colour record (padding at most three unit cosets) described in Lemma 9.2. Its current literature certificate explicitly leaves the complete source-to-record construction unestablished. Thus this conductor-fibre proposition remains an open external premise, rather than a source transcription certified by the kernel.

The main theorems are LEAN theorems whose hypothesis lists are *exactly*: for Theorem 1.1 (with its rate), Theorem 1.2(i) and parts (ii) and (iii) of Theorem 1.2, Chen–Stein, Evertse–Silverman, conductor and divisors; for Theorem 1.4, the last three only. Parts (ii) and (iii) of Theorem 1.2 are verified in the form of a *formal surrogate*: the LEAN theorem actually proved is the convergence of the expectations of the exponential functionals for every continuous nonnegative test with compact support; the manuscript's statement — vague convergence to a Poisson point process — is equivalent to it by the standard Laplace-functional characterization [21, Theorem 4.11], which is not formalized, the general theory of point processes not being available in Mathlib; the geometric law of the marks is exact in the development. The Rademacher product measure on the set of primes is constructed in the development, and the transfer identities between the infinite law and its finite cylinders are proved exactly.

Table of external dependencies per statement. The following lists are extracted from the audit manifest; they give, for each public statement, the literature hypotheses actually present in the signature of its canonical LEAN theorem. Abbreviations: AGG [1], ES [13], HK [18], NR [27], LS [23], BS [4], PNT = prime number theorem [3, Introduction, p. 1].

Public statement	External hypotheses consumed
Theorem 1.1 (with rate)	AGG, ES, HK, NR
Theorem 1.2(i)	AGG, ES, HK, NR
Theorem 1.2(ii)–(iii), Laplace form	AGG, ES, HK, NR
Theorem 1.4	ES, HK, NR
proposition 11.2 ($R_2 = o(N^2)$)	ES, HK, NR
proposition 15.5	PNT, BS, LS (Pell discharged internally)
proposition 16.1	ES, HK, NR
Theorem 16.2	PNT, AGG, BS, ES, HK, LS, NR
Corollary 16.4	PNT, AGG, BS, ES, HK, LS, NR

Theorem 16.2 consumes all seven bridges: the truncation of Section 15 brings in [23] and [4], and the Pell count is instantiated there by the repository's own discharge theorem (ideals and units, Section 9), which leaves only HK and NR.

Effective constants. The formalization delivers effective versions of several statements that the text leaves with anonymous constants; the following values are admissible, not optimized: $C_0 = 128$ in Lemma 3.1; the explicit threshold $x > 4B^{2A+1}$ in Lemma 5.2; the finite bound $46(L+1)^4\, 4^{\lfloor L/2 \rfloor}$ in Lemma 5.5; and the elementary Chebyshev-type bound $\lfloor \log_2 H \rfloor\, \pi(H) \le 7H$ (for $H \ge 4$), consumed by proposition 3.2.

Restrictions and encoding gaps. Two declared scope restrictions: Theorem 8.1 is mechanized in the instance $\alpha = 3/16$ — the one consumed by the partition of Section 7 — under the explicit hypotheses $3B < 16c^{\#}$ and $B \ge 32$, with extraction of at least $\lfloor B/16 \rfloor$ exact-free components of size at most 43; and, of Lemma 13.6, the formalization establishes an explicit harmonic edge estimate — distinct from the displayed three-term form, but sufficient after multiplication by 2^{-2L} for the rate of Corollary 13.10, and actually consumed by the LEAN rate theorem. The exponential form of Corollary 11.3 is exposed as a final assembled theorem (a dedicated canonical endpoint, with one constant uniform over the seven sectors); Corollary 16.4, the rate version of Remark 14.3, and the reformulations of the main theorems on the infinite product law are mechanized as well. Three encoding gaps, with no effect on the consumers: in Lemma 14.8, the machine-checked proof goes through a direct variant whose constant is twice that of the text, the term remaining $o_{C,E}(1)$; the conditioning of Section 13 is represented extensionally by the uniform law on the remaining cylinder, without a measure-theoretic theory of conditional probability; and, outside the windows entirely covered by the cutoff, the start probability is by convention that of the truncated cylinder. The formalization moreover follows paths sometimes shorter than ours: a direct count of relational hosts at the upper cutoff of the window, without dyadic sub-block covering (appendix, section 17); the exact identity $R_2(N, L) = R_{2,2}(N, L)$ — the instance $\kappa = 2$ of proposition 16.1 — which reduces the original-window mass to the bounded-ratio case; and the pointwise saving $(2^\tau - 1)\, 2^{R_K(B)+1} \le 4 \cdot 2^B$ in the terminal population.

Reproduction. Version-specific public deposits of the English and French manuscripts are maintained under the Zenodo paper concept record doi:10.5281/zenodo.21736676. The public Lean formalization repository provides the companion Lean 4 development; version-specific archival deposits are maintained under its Zenodo software concept record doi:10.5281/zenodo.21735481. For each deposited version, its audit and checksum manifests identify the source revision, toolchain, hypothesis statuses, evidence scope, and exact files. The versioned Cambridge Open Engage record doi:10.33774/coe-2026-z3l74 preserves the public manuscript history.

19 Conclusion and perspectives

Complete multiplicativity creates dependence at every scale, so the usual starting point for a Poisson approximation — an a priori sparse dependency graph — is unavailable. The proof isolates a way around this obstruction. Dyadic localization removes the dominant exact dilation from the bulk. The remaining long-range couplings are encoded arithmetically, and the absolute homogeneous two-block estimate (1.4) controls their total mass. Conditioning on the small primes then converts this arithmetic statement into a probabilistic one: outside a negligible set, the start events have exact conditional marginal 2^{-L} and their shared large-prime coordinates form an exact sparse dependency graph. Chen–Stein applies precisely after, rather than before, the arithmetic classification.

The structural output is the conversion

$$\begin{gathered}\text{long-range multiplicative dependence} \longrightarrow \text{absolute two-block summability}\\ \longrightarrow \text{exact conditional dependency graph} \longrightarrow \text{Poisson approximation.}\end{gathered}$$

The same interface is uniform enough to yield deterministic masks, the spatial Poisson point process and the excess-length process with geometric marks. In this sense, (1.4), rather than the second-moment asymptotic alone, is the central reusable object of the paper.

The dyadic block is structural, not merely a convenient normalization. It keeps the exact pair $(x, 2x)$ out of the bulk and turns the residual relations into a countable family. Passing from a single block to $[1, M]$ requires two additional operations. First, one applies Theorem 13.7 once to the union of starts, rather than applying the one-block theorem separately and presuming inter-block independence. Second, the ratio-2 channel changes from an edge effect into a volumetric family. Proposition 16.1 shows that the arithmetic partition survives this change without worsening the final exponent in the sectors with polynomial saving.

The truncation of the deep blocks also records an instructive failed route. Expanding an exponential moment of large square divisors creates many divisor witnesses even when the target quantity remains bounded; for $x = m^2$, the discrepancy is already measured by the growth of $\tau(m)$. Passing instead through pairs of defects exposes Pell rigidity and avoids this overcounting. This illustrates a principle used throughout the proof: the useful object is the representation that preserves the geometry of the dependence, not the largest readily available arithmetic moment.

The scope remains precise. Higher factorial moments are outside this article and are used in none of its proofs. The refined border–interior joint and the relative moderate-tail asymptotics are not established. At the critical scale the border has probability $2^{-\pi(L)} = o(1)$, so the whole finite-prefix longest-run law follows by coupling, without a border–interior independence statement. In the moderate tail the border eventually dominates the interior Poisson intensity; a natural continuation is a relative truncation together with a joint analysis, leading to the expected correction

$$1 - (1 - 2^{-\pi(L)})e^{-\Lambda}$$

in place of a pure Poisson tail.

Declaration on artificial-intelligence assistance

The research was organized by the author with substantial assistance from artificial-intelligence systems — chiefly Claude Fable 5 and Claude Opus 4.8 (Anthropic) together with GPT 5.6 Sol (OpenAI) — used to explore lemmas, draft intermediate versions, propose additions integrated into the text, develop the Lean formalization described in section 18, generate reconnaissance programs, and carry out separate counter-audits by means of distinct instances and of models from different providers. These procedures constitute neither an independent scientific validation nor a human certification of the proof. The author alone assumes responsibility for the manuscript, its attributions, its possible errors and its limits.

Code and artifact availability

Version-specific public deposits of the English and French manuscripts are maintained under the Zenodo paper concept record doi:10.5281/zenodo.21736676. The public Lean formalization repository provides the companion Lean 4 development; version-specific archival deposits are maintained under its Zenodo software concept record doi:10.5281/zenodo.21735481. The versioned Cambridge Open Engage record doi:10.33774/coe-2026-z3l74 preserves the public manuscript history. For each deposited version, its audit and checksum manifests identify the source revision, evidence scope, and exact files. No numerical output is used in the proof: the programs served only to detect edge cases and to test finite identities, and their outputs are explicitly labelled as numerical diagnostics rather than as asymptotic evidence.

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